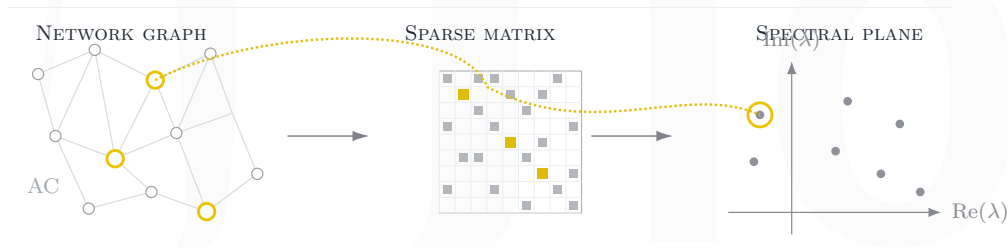


Spectral Grid Dynamics for Converter-Rich Power Systems

Operator-Valued Grid Strength,
Rational Self-Energy, and Structural Fragility



SPECTRAL GRAPHS · OPERATOR-VALUED STRENGTH · SELF-ENERGY ·
STRUCTURAL FRAGILITY

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Abstract

This dissertation develops an operator-valued theory of dynamic grid strength for converter-rich power systems. The central claim is that grid strength is not a scalar such as short-circuit ratio, algebraic connectivity, or a single damping ratio. It is a port-dependent dynamic operator whose static limit is encoded by graph stiffness, whose oscillatory behavior is carried by the Green function of a quadratic or rational eigenvalue problem, whose controller-mediated behavior is represented by rational self-energy, whose fragility is measured by structured pseudospectra, and whose nonlinear extension appears as memory. In the classical swing-equation model,

$$M\ddot{\theta} + D\dot{\theta} + L\theta = p$$

the damping matrix D is fixed. Exact graph-mode decoupling occurs when the inertia-normalized stiffness and damping matrices commute. The special case $D = \gamma M$ is therefore not the whole decouplable class; it is the uniform damping-to-inertia case in which all underdamped modes share the same real decay rate. Rayleigh damping $D = \alpha M + \beta L$, and more generally commuting damping operators, define a broader constant- D floor. If D is fixed but non-commuting, the modal damping matrix $Q^T M^{-1/2} D M^{-1/2} Q$ is not diagonal; graph eigenvectors still organize stiffness, but they no longer diagonalize damping. If converter, phase-locked loop, exciter, governor, stabilizer, or auxiliary control states are condensed, the reduced object is not a fixed-coefficient quadratic eigenvalue problem. It is a rational nonlinear eigenvalue problem,

$$T(s) = s^2 M + s D_0 + L + \Pi(s).$$

Only after the rational correction is separated into $\Pi(s) = \Pi_K(s) + s\Pi_D(s)$ does the damping-channel specialization $\Sigma(s) = D_0 + \Pi_D(s)$ appear.

The original contribution is not a new Laplacian metric or a new eigenvalue solver. It is a mechanism-resolved operator framework showing when graph-theoretic strength is a valid damping certificate, when damping-to-inertia roughness creates modal coupling, when topology actions must be evaluated through spectral effective resistance, and when controller self-energy can reverse the dynamic effect predicted by static graph screens.

The thesis is deliberately theoretical and is organized as a doctoral monograph rather than as an extended article. It develops the physical framework, mathematical foundations, state of the art, operational graph dynamics, constant-damping theory, rational self-energy, nonlinear memory, and applications as separable modules. Its unifying object is the dynamic Green function $G_T(s) = T(s)^{-1}$. For a line $e = (i, j)$, the scalar $\mathcal{R}_e(s) = a_e^T G_T(s) a_e$, with $a_e = e_i - e_j$, is a dynamic effective resistance: at $s = 0$ it recovers classical graph effective resistance, while away from zero it records inertia, damping, stiffness, controller poles, residues, non-normality, and modal coupling. A finite line action then satisfies the exact scalar equation $1 + \Delta b_e \mathcal{R}_e(s) = 0$; general low-rank actions satisfy the operator determinant law $\det T_u(s) = \det T(s) \det[I + K_u(s) V^* T(s)^{-1} U]$. These identities turn local

eigenvalue sensitivities into first-order limits of exact finite-action laws. The remaining modules add port-equivalence of self-energy, model-adequacy certificates, structured fragility radii, directional passivity, data-identified self-energy, and nonlinear memory. Simulation cases and internal project drafts are treated as validation gates and traceability records, not as universal theorems. The main contribution is a coherent mathematical language for deciding when spectral graph intuition is a certificate, when it is only a screen, when a network improvement can reduce a dynamic margin, and when controller-aware rational self-energy or nonlinear memory is required.

Keywords: operator-valued grid strength, operational graph, graph-time resonance, dynamic effective resistance, Green function, damping ratio, inverter-based resources, quadratic eigenvalue problem, Schur complement, nonlinear eigenvalue problem, rational self-energy, structured pseudospectrum, dynamic hosting capacity, virtual inertia, Braess-like reversal, weak nodes, weak links, hidden damping margin, risk vector, ANDES, IEEE 39-bus system.

Scope and Reproducibility Statement

This document treats the current PDF as version 0 of the intellectual map, not as a manuscript to be enlarged linearly. It is a doctoral theoretical draft, not an experimental report. It does not present a final benchmark, a calibrated IEEE test case, or a claim of empirical closure. Its purpose is to establish the modular mathematical structure that later numerical experiments must respect. Claims about constant damping, non-commuting modal coupling, Schur condensation, rational self-energy, and nonlinear memory are presented as theoretical statements, derivations, or chapter contracts. Claims about practical planning are formulated as requirements for future validation.

All source files for this manuscript are kept under `reports/thesis/`. The compiled PDF is copied to `outputs/pdf/thesis_draft.pdf`. Scratch material in `temp/` is preserved as historical context and is not part of the theoretical thesis source. The integrated claim ledger is kept out of the main scientific narrative and appears as an appendix-level traceability record.

Graphical Roadmap

The thesis is organized as a chain from electrical modeling to dynamic planning. The diagram below is the reader's map: each block introduces one object that must be carried forward without changing its meaning.

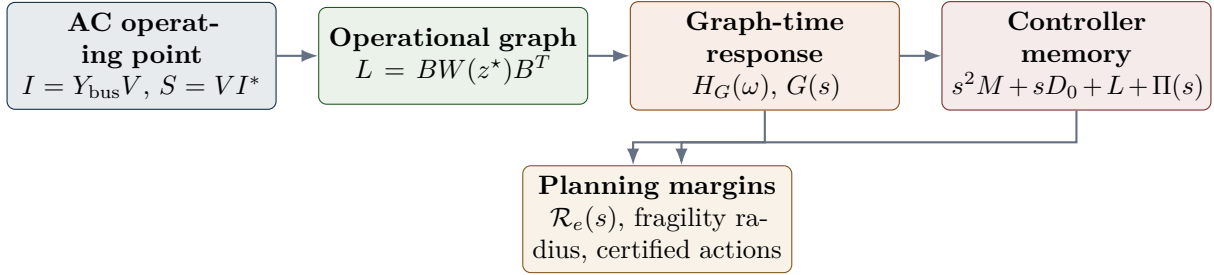


Figure 1. Visual roadmap of the dissertation. The thesis starts from the AC operating point and the physical network model, derives the operational graph $L = BWB^T$, builds graph-time response and the Green function of the fixed- D spectral dynamics, replaces fixed damping by rational self-energy when controllers are condensed, separates nonlinear memory from NEP theory, and ends with certified action and fragility planning.

How to read the dissertation

The main text follows the modular path: physical framework, mathematical foundations, operational graph dynamics, constant damping, dynamic Green functions, rational self-energy, nonlinear memory, and applications. A graph metric is accepted only after the dynamic operator, protected margin, action family, and certification gate attached to it are named.

Acknowledgments

This draft consolidates theory, reproducible experiments, and editorial material from the project into a thesis-oriented theoretical structure. I am grateful to the academic and engineering communities whose work on power-system stability, spectral graph methods, quadratic eigenvalue problems, nonlinear eigenvalue problems, and converter-dominated dynamics makes this synthesis possible.

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Part I

Foundations and State of the Art

Introduction

Power-system small-signal stability is a theory of operators before it is a theory of plots. A linearized power system is stable or unstable because the spectrum of its differential-algebraic representation lies in particular regions of the complex plane, because the corresponding eigenvectors project onto physical states in particular ways, and because perturbations of parameters move those eigenvalues through mechanisms that can be traced mathematically. The familiar language of damping ratio, participation factors, critical modes, and modal frequencies is a compact engineering vocabulary for that operator theory [1, 2, 3].

The same vocabulary becomes harder to use in converter-rich systems. A synchronous-machine dominated model permits a useful separation: inertia is stored in rotating masses, synchronizing stiffness comes from the network and power-angle relation, and damping can often be represented by a fixed local coefficient. Converter-interfaced resources alter this separation. A grid-following converter may contribute phase-locked-loop states, current-loop states, plant controllers, filters, and protection-related dynamics. A grid-forming converter may contribute droop, virtual oscillator, virtual synchronous-machine, voltage-loop, current-loop, and current-limiting dynamics. The extended IEEE/CIGRE stability taxonomy explicitly recognizes that power-electronic interfaces introduce faster and device-dependent dynamic phenomena that do not fit cleanly into the classical electromechanical classification [4].

This dissertation asks a sharper theoretical question:

What is the reduced dynamic operator that preserves grid strength, fragility, and actionable pole motion in a converter-rich power system?

That question has a planning consequence. If a utility asks how much converter-based generation can be hosted, where virtual inertia should be placed, or which lines should be reinforced before installing batteries, the answer cannot be read from voltage limits or graph connectivity alone. A hosting decision is dynamic only if it states the damping, resolvent, or protection margin that must survive the conversion. A line decision is robust only if it separates useful stiffness increase from damping-ratio loss or controller-resonant reinforcement risk.

The answer developed here is hierarchical, but the hierarchy is not the final claim. Fixed scalar damping is the uniform-decay member of a broader sequence of increasingly expressive operators. The exact decoupling condition for fixed damping is not $D = \gamma M$ alone. It is commutation of the inertia-normalized stiffness and damping operators. A fixed but non-commuting D keeps the problem quadratic, but it couples graph modes through the modal damping matrix. A controller-condensed converter-rich model changes the problem more fundamentally: the reduced correction depends on the spectral parameter s . The effective operator is then a rational self-energy $\Pi(s)$, obtained by a

Schur complement of the eliminated controller states. The object that turns this hierarchy into an action theory is the dynamic Green function $G_T(s) = T(s)^{-1}$, whose port projections measure how finite physical actions move poles.

1.1 Thesis Statement

The thesis statement is:

Dynamic grid strength in converter-rich power systems is an operator-valued property. Its static limit is encoded by graph stiffness, its oscillatory behavior by the Green function of the QEP, its controller-mediated behavior by rational self-energy $\Pi(s)$, its finite-action behavior by low-rank determinant identities, its fragility by structured pseudospectra, and its nonlinear extension by memory kernels. The damping-channel symbol $\Sigma(s)$ is reserved for the velocity channel after $\Pi(s) = \Pi_K(s) + s\Pi_D(s)$ has been declared. Spectral graph methods remain valuable, but their role changes from certificate to screen whenever damping and stiffness cease to share an eigenbasis, whenever eliminated controllers contribute frequency-dependent dynamics, or whenever an action claim requires a certified finite perturbation rather than a local derivative.

This statement does not claim novelty for Laplacian linearizations, generalized graph modes, effective resistance, QEP or NEP sensitivity formulas, Schur complements, or the existence of Braess-like phenomena in oscillator and power networks. Those are established mathematical and engineering ingredients. The novelty sought by this theoretical thesis is the synthesis: a single operator hierarchy that explains when classical damping intuition is mathematically justified, when it fails, and what reduced object must replace it.

Originality boundary

Known mathematics is used as the foundation; it is not claimed as the dissertation's invention. The doctoral contribution is the operator-valued planning synthesis. The strongest original claims are the graph roughness of D_i/M_i as a modal-coupling obstruction, the graph-time response atlas, the spectral effective resistance law for finite topology updates, and the controller-induced reinforcement-reversal certificate. Claims about universal weak nodes, completed sheaf-grid theory, or empirical benchmark closure remain validation targets rather than defended theorems.

Doctoral reading contract

Every planning statement in this dissertation must answer four questions: which operator is being used, which port or action family is being tested, which margin is being protected, and which validation gate turns the calculation from a screen into evidence.

1.2 The Operator Hierarchy

The hierarchy begins with the second-order swing equation

$$M\ddot{\theta} + D\dot{\theta} + L\theta = p, \tag{1.1}$$

where M is inertia, D is damping, L is synchronizing stiffness, and θ is the vector of bus or machine angles after removing the uniform reference. In inertia-normalized coordinates, the stiffness becomes $\tilde{L} = M^{-1/2}LM^{-1/2}$ and the damping becomes $\tilde{D} = M^{-1/2}DM^{-1/2}$.

The first theoretical floor is fixed damping. In inertia-normalized coordinates,

$$\tilde{L} = M^{-1/2}LM^{-1/2}, \quad \tilde{D} = M^{-1/2}DM^{-1/2}. \quad (1.2)$$

Exact modal decoupling occurs when

$$[\tilde{L}, \tilde{D}] = \tilde{L}\tilde{D} - \tilde{D}\tilde{L} = 0. \quad (1.3)$$

For symmetric matrices, this is the condition for simultaneous orthogonal diagonalization. The familiar $D = \gamma M$ case gives $\tilde{D} = \gamma I$, so it certainly satisfies (1.3); it is the special case in which every underdamped mode has the same real decay rate $-\gamma/2$. Rayleigh damping $D = \alpha M + \beta L$ also commutes in normalized coordinates and therefore decouples graph modes, but it does not produce a uniform decay rate.

The second floor keeps D fixed but removes proportionality. Now $Q^T \tilde{D} Q$ is not diagonal in the graph eigenbasis Q . Off-diagonal modal damping couples the modes. The eigenvalue problem remains a quadratic eigenvalue problem (QEP), but graph-frequency movement is no longer a damping certificate [5].

The third floor appears when auxiliary states are eliminated. If the full linearization is partitioned into retained electromechanical states x_e and condensed controller states x_c ,

$$A = \begin{bmatrix} A_{ee} & A_{ec} \\ A_{ce} & A_{cc} \end{bmatrix}, \quad (1.4)$$

then Schur condensation gives a nonlinear retained-state operator,

$$S(s) = sI - A_{ee} - A_{ec}(sI - A_{cc})^{-1}A_{ce}. \quad (1.5)$$

In second-order retained coordinates the general condensed operator is

$$T(s) = s^2M + sD_0 + \tilde{L} + \Pi(s), \quad \Pi(s) = \Pi_K(s) + s\Pi_D(s). \quad (1.6)$$

The rational correction may contain stiffness-like and damping-like channels. Only when the stiffness-like part is absent or explicitly absorbed does this reduce to $T(s) = s^2M + s\Sigma(s) + \tilde{L}$, with $\Sigma(s) = D_0 + \Pi_D(s)$. The reduced damping object is then rational in s , with poles inherited from the eliminated controller block.

The same operator also defines the thesis' action calculus. On the resolvent set of T , let

$$G_T(s) = T(s)^{-1}. \quad (1.7)$$

For a line $e = (i, j)$, with incidence vector $a_e = e_i - e_j$, define

$$\mathcal{R}_e(s) = a_e^T G_T(s) a_e. \quad (1.8)$$

At $s = 0$, when the reduced operator has the Laplacian limit and the reference angle is removed, this recovers classical effective resistance $a_e^T L^\dagger a_e$. Away from zero it is a dynamic effective resistance: it records stiffness, inertia, damping, non-normality, controller poles, residues, and modal coupling through the same port projection. A finite line change Δb_e then satisfies

$$\det(T(s) + \Delta b_e a_e a_e^T) = \det T(s) (1 + \Delta b_e \mathcal{R}_e(s)). \quad (1.9)$$

Thus a local line sensitivity is only the small- Δb_e limit of an exact scalar finite-action law.

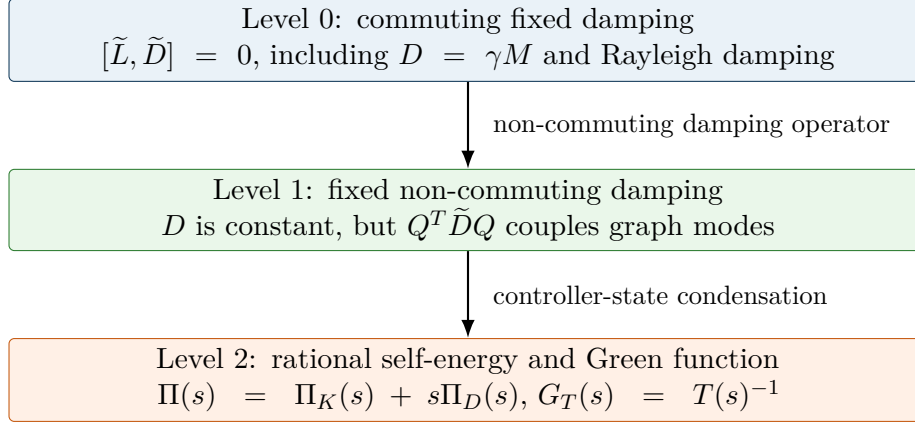


Figure 1.1. Damping-operator hierarchy developed in this thesis. Each transition keeps part of the previous intuition but removes one mathematical guarantee.

1.3 Core Calibrated Messages

The hierarchy leads to five messages that are strong enough to guide the thesis, but only if their assumptions are stated precisely.

1. **Pure inertia is not damping.** In the fixed-mode, fixed- D screen,

$$\frac{\partial \zeta_k}{\partial M_i} \approx -\frac{1}{2} \zeta_k \phi_k(i)^2. \quad (1.10)$$

Thus adding inertia at a modal antinode can reduce the scalar damping-ratio screen if damping is not co-designed. This does not say inertia is bad. It says inertia, damping ratio, RoCoF, and nadir are distinct objectives.

2. **Graph stiffness is not a damping certificate.** For a line $e = (i, j)$,

$$\frac{\partial \zeta_k}{\partial b_e} = -\frac{\zeta_k}{2\nu_k} (\phi_k(i) - \phi_k(j))^2 + \frac{\zeta_k}{\delta_k} \frac{\partial \delta_k}{\partial b_e}. \quad (1.11)$$

If the line action raises stiffness but does not raise modal damping, the scalar damping-ratio screen can decrease. Therefore algebraic connectivity and effective-resistance improvements are graph screens, not automatic small-signal damping certificates.

3. **Modal reduction and controller condensation share a resolvent structure.** Eliminating coupled graph modes from a fixed- D QEP produces a rational correction such as (7.8). Eliminating controller states produces the Schur self-energy $\Pi(s) = G(sI - A_{cc})^{-1}(B_\theta + sB_\omega)$. The common mechanism is not the physical origin of the eliminated states; it is the resolvent injected into the retained operator. The damping-channel notation $\Sigma(s)$ is used only after the rational correction has been projected onto the velocity channel.
4. **Static damping bridges have a spectral validity region.** In the damping-channel specialization, the NEP sensitivity denominator contains

$$T_s(s) = 2sM + \Sigma(s) + s\Sigma'(s). \quad (1.12)$$

The term $s\Sigma'(s)$ is the spectral memory of the condensed controller block. Setting it to zero is acceptable only when the rational self-energy is locally flat enough over the critical region. Near a controller resonance, a static bridge can predict the wrong pole-motion direction.

5. **Finite actions are not only derivatives.** For line, damping, inertia, and controller actions that enter $T(s)$ through low-rank port updates, the pole condition can be reduced to a determinant whose dimension is the number of action ports. This does not replace full-model validation, but it does give the right intermediate object between a scalar graph score and a large NEP recomputation.

None of these messages is presented as an industry-wide theorem without qualification. They are calibrated statements: exact in the commuting constant- D floor, first-order screens in fixed non-commuting damping, and validation gates after controller condensation.

1.4 Research Questions

The thesis is organized around eight research questions.

1. What exact assumptions make graph-mode damping analysis valid in the classical swing model?
2. How does the operating point induce the operational graph $L = BW(z^*)B^T$, and how do power-flow geometry, line loading, cut structure, and cycle structure shape dynamic modes?
3. How does fixed non-commuting damping create modal coupling, and how can that coupling be expressed as a rational correction?
4. How does Schur condensation of controller states transform a fixed damping matrix into a rational self-energy $\Pi(s)$, and when is its damping-channel projection $\Sigma(s)$ legitimate?
5. How does the dynamic Green function $G_T(s)$ turn effective resistance, line reinforcement, damping placement, inertia placement, and controller retuning into one low-rank action calculus?
6. What theoretical requirements must a planning metric satisfy before it can claim to identify weak buses, weak lines, or useful converter actions in a converter-rich grid?
7. How can dynamic hosting capacity, virtual-inertia placement, and line-reinforcement screening be expressed as spectral margin problems rather than as static graph or voltage heuristics?
8. Which certificates decide when a reduced QEP, a Schur NEP, a data-identified self-energy, or a full DAE is adequate for a protected decision?

The first five questions are mathematical. The last three are epistemic, planning-oriented, and computational: they define what future empirical validation must show before a screen becomes an engineering certificate.

1.5 Contributions

The dissertation makes one central contribution and five supporting contributions.

1. **Central contribution: operator-valued dynamic grid strength.** The dissertation treats graph stiffness, damping-to-inertia roughness, spectral effective resistance, rational controller self-energy, and nonlinear memory as successive mechanisms in one retained operator $T(s) = s^2M + sD + L + \Pi(s)$.

2. **Damping-to-inertia graph roughness.** It shows how non-commutation between inertia-normalized stiffness and damping becomes the graph-gradient energy of D_i/M_i , and how this roughness, together with small modal gaps, predicts modal mixing.
3. **Graph-time resonance atlas.** It defines $H_G(\omega) = [Q^T \tilde{T}(j\omega)Q]^{-1}$ as the joint spatial-mode and temporal-frequency response used to locate resonant modal conversion.
4. **Spectral effective resistance.** It defines $\mathcal{R}_e(s) = a_e^T G_T(s) a_e$ and derives the finite line-update law $1 + \Delta w_e \mathcal{R}_e(s) = 0$, recovering static effective resistance at the Laplacian limit.
5. **Controller-induced reinforcement reversal.** It formalizes the case in which a line or parameter action passes a static graph or fixed- D screen but reverses once the controller Schur resolvent is included.
6. **Certificate/screen/gate planning methodology.** It formulates weak elements, dynamic hosting, virtual-inertia placement, and line reinforcement as operator-, margin-, action-, and validation-gate-relative claims rather than graph-intrinsic labels.

1.6 Dissertation Organization

The dissertation is now organized as a modular monograph. Part I builds the problem statement, the physical grid framework, the spectral preliminaries, and the state of the art. The new physical framework chapter develops the AC network, phasor, power-flow, swing-equation, DAE, and operator chain before the more compact mathematical chapters use those objects. Part II develops the operational graph theory: power-flow geometry, total derivatives of $L = BW(z^*)B^T$, graph roughness of damping-to-inertia fields, graph-time response, dynamic effective resistance, and structured fragility radii. Part III closes the constant- D theory: QEP identities, commuting damping, uniform decay, Rayleigh damping, non-commuting coupling, and reduced-QEP gates. Part IV develops rational self-energy $\Pi(s)$, its damping-channel specialization $\Sigma(s)$, Schur condensation, finite-action laws, NEP sensitivities, port equivalence, passivity screens, structured fragility, and controller-resolvent reversals. Part V separates genuine nonlinear DAE stability from spectral NEP theory by treating equilibria, nonlinear memory, bifurcation, energy, and regions of attraction. Part VI turns the operators into applications: weak elements, planning risk vectors, dynamic hosting regions, reinforcement envelopes, and reproducible benchmark protocols.

The integrated source-to-claim ledger is not part of the main scientific progression. It is kept as an appendix-level traceability record that maps claims to sources, scripts, figures, and validation gates.

Physical and Theoretical Framework

This chapter explains the physical model behind the operator hierarchy. The later chapters are intentionally abstract, but the abstraction is not detached from the grid. Every matrix in the thesis comes from a physical layer: phasor voltages, line admittances, active-power balance, rotor or converter dynamics, controller states, and the operating point around which the system is linearized. The purpose of this chapter is to make that chain explicit before the thesis uses L , M , D , $\Pi(s)$, and $T(s)^{-1}$ as technical objects.

Chapter message

The problem is not only to compute eigenvalues. The problem is to identify which physical mechanism creates, moves, amplifies, or protects those eigenvalues. That is why the thesis needs a hierarchy from network stiffness to fixed damping, rational self-energy, Green functions, structured fragility, and nonlinear memory.

2.1 How to Read This Chapter Step by Step

The rest of the thesis uses compact notation, so this chapter starts from the physical chain and builds the operator one piece at a time. The intended reading order is:

1. Start with the electrical graph: buses are nodes, lines are edges, and line admittances create the network equation $I = Y_{\text{bus}}V$.
2. Fix an operating point: the same topology can have different small-signal stiffness under a different dispatch, voltage profile, or angle stress.
3. Linearize active-power balance: the derivative of power with respect to angles becomes the synchronizing stiffness matrix L .
4. Add dynamic storage and damping: inertia gives $M\ddot{\theta}$, damping or damping-like control gives $D\dot{\theta}$, and stiffness gives $L\theta$.
5. Introduce the spectral parameter s : an oscillatory response $\theta(t) = ve^{st}$ converts the differential equation into an operator $s^2M + sD + L$.
6. Escalate only when needed: if hidden controller states are eliminated, the missing frequency-dependent physics appears as $\Pi(s)$, and the object becomes $T(s) = s^2M + sD_0 + L + \Pi(s)$.

The key point is that each symbol is a physical commitment. A graph eigenvalue is not automatically a stability certificate; it is a certificate only when the dynamic terms that can change the protected

margin have been accounted for.

2.2 The Grid as a Networked Dynamic System

A power grid is a networked dynamical system, not a single aggregate machine. The buses form the electrical graph, the lines and transformers define the admittance matrix, the generators and converters inject dynamic behavior, and the loads close the power balance. A useful small-signal model must preserve all four roles: network geometry, operating-point physics, dynamic storage, and control memory.

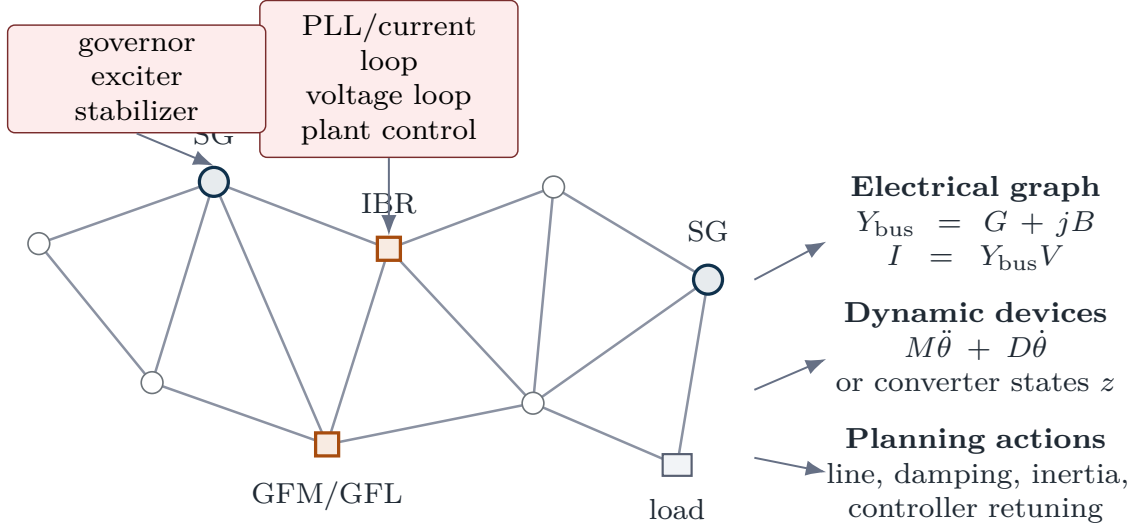


Figure 2.1. A converter-rich grid contains a physical graph, dynamic devices, and hidden controller states. The retained variables may be bus or machine angles, frequencies, voltages, and selected ports; the eliminated variables reappear as self-energy when they are condensed.

Figure 2.1 shows the modeling issue. The electrical graph tells us which buses are connected, but not which dynamic margin is protected. A synchronous generator contributes inertia, damping, exciter, governor, and stabilizer dynamics. A grid-following or grid-forming converter contributes synchronization, current-loop, voltage-loop, filter, limiter, and plant-level dynamics. If those internal states are removed from a reduced model, their effect must go somewhere. In this thesis, that place is the rational self-energy $\Pi(s)$.

2.3 AC Phasors, Power Flow, and Network Stiffness

The first physical object is the AC phasor voltage. At bus i ,

$$V_i = |V_i|e^{j\theta_i}, \quad (2.1)$$

where $|V_i|$ is voltage magnitude and θ_i is phase angle. The network admittance matrix maps bus voltages to currents,

$$I = Y_{\text{bus}}V, \quad Y_{\text{bus}} = G + jB, \quad (2.2)$$

and the complex power injected at the bus is

$$S_i = P_i + jQ_i = V_i I_i^*. \quad (2.3)$$

The operating point is therefore not only a graph. It is a solution of the AC power-flow equations at a particular dispatch, load, voltage profile, topology, and controller setting.

The active-power stiffness matrix is the first derivative of active power with respect to phase angles at the operating point. With voltage magnitudes fixed inside the small-signal angle block,

$$\Delta P \approx J_{P\theta} \Delta \theta, \quad J_{P\theta} = \left. \frac{\partial P}{\partial \theta} \right|_{(V^*, \theta^*)}. \quad (2.4)$$

Depending on the sign convention for exported electrical power, the synchronizing matrix is $L = J_{P\theta}$ or $L = -J_{P\theta}$. The thesis always states the convention before using a derivative. The important fact is that L is an operating-point derivative, not a purely topological matrix.

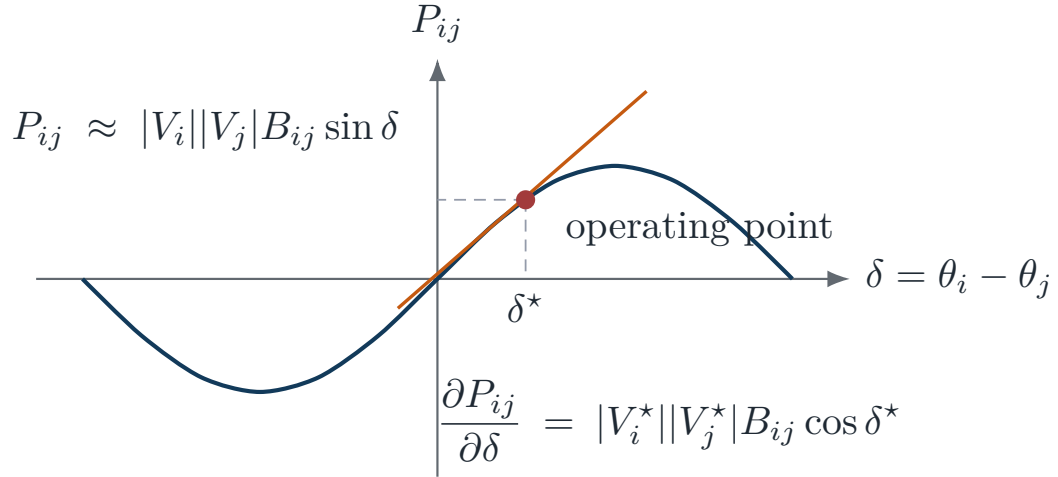


Figure 2.2. A line contributes synchronizing stiffness because active power is sensitive to angle separation. The slope at the operating point becomes the edge weight in the linearized stiffness matrix L .

Figure 2.2 gives the physical reason L acts like a stiffness matrix. For a lossless two-bus line,

$$P_{ij} \approx |V_i||V_j|B_{ij} \sin(\theta_i - \theta_j). \quad (2.5)$$

Linearization gives a slope proportional to $|V_i^*||V_j^*|B_{ij} \cos(\theta_i^* - \theta_j^*)$. This slope is large when the line is electrically strong and the operating point is not close to the angular stability limit. It decreases as the operating point approaches a stressed angle difference. Thus the same physical line can have different small-signal stiffness under different operating conditions.

2.4 From Physical Balance to the Swing Equation

The second physical object is rotational or emulated energy storage. For a synchronous machine, the rotor accelerates when mechanical input and electrical output are imbalanced:

$$M_i \dot{\omega}_i = P_{m,i} - P_{e,i} - D_i \omega_i, \quad \omega_i = \dot{\theta}_i. \quad (2.6)$$

After linearization around a balanced operating point,

$$M_i \Delta \ddot{\theta}_i + D_i \Delta \dot{\theta}_i + \Delta P_{e,i} = p_i. \quad (2.7)$$

The network supplies $\Delta P_e \approx L\Delta\theta$. Stacking retained angle variables gives

$$M\ddot{\theta} + D\dot{\theta} + L\theta = p. \quad (2.8)$$

The three matrices are not interchangeable. M multiplies acceleration, D multiplies velocity, and L multiplies angle displacement. A planning action that changes one term should not be interpreted as if it changed the others.

The uniform angle direction has no restoring force. Mathematically,

$$L\mathbf{1} = 0. \quad (2.9)$$

Physically, adding the same constant angle to every bus does not change any angle difference or power flow. All spectral statements therefore occur after fixing an angle reference, using center-of-inertia coordinates, or projecting onto the nonzero-angle subspace.

2.5 Converter States and the DAE Layer

Converter-rich grids add a hidden dynamic layer between the electrical graph and the retained swing-like variables. A generic local model has differential states x , algebraic variables y , and parameters or actions ρ :

$$\dot{x} = f(x, y, \rho), \quad (2.10)$$

$$0 = g(x, y, \rho). \quad (2.11)$$

The algebraic variables include network constraints such as nodal current balance and voltage equations. The differential states include machine rotor states, exciter states, governor states, PLL states, current controllers, voltage controllers, filters, and plant controls.

At a regular equilibrium, g_y is nonsingular and the algebraic variables can be eliminated locally. The first-order small-signal matrix is

$$A_{\text{red}} = f_x - f_y g_y^{-1} g_x. \quad (2.12)$$

This formula explains why the operating point matters. The Jacobians f_x , f_y , g_x , and g_y are all evaluated at (x^*, y^*, ρ^*) . If the dispatch, IBR penetration, line status, voltage setpoints, or controller gains change, the linearized operator changes.

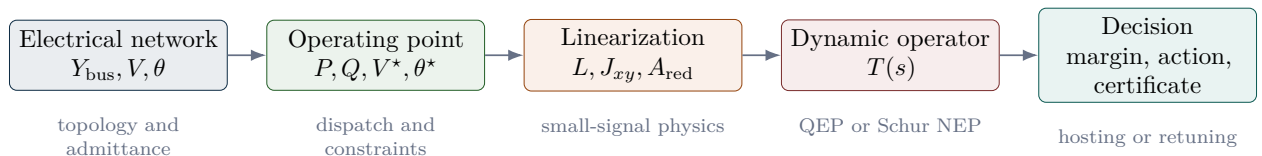


Figure 2.3. The thesis follows the physical modeling chain from network data to an operator-valued decision. A planning result is not reproducible unless the operating point, retained variables, action family, margin, and certification gate are all declared.

2.6 Why the Spectral Parameter Appears

Small-signal stability studies the free response after an infinitesimal disturbance. For the fixed- D second-order model, set $p = 0$ and try $\theta(t) = ve^{st}$. Since

$$\dot{\theta} = sve^{st}, \quad \ddot{\theta} = s^2 ve^{st},$$

substitution into (2.8) gives

$$(s^2 M + sD + L)v = 0. \quad (2.13)$$

This quadratic eigenvalue problem is the first spectral object of the thesis. The term $s^2 M$ is inertia, the term sD is velocity damping, and the term L is synchronizing stiffness.

When controller states are eliminated rather than retained, the operator is no longer a fixed-coefficient QEP. Partition the linearized dynamics into retained states x_r and condensed states x_c . Eliminating x_c gives a Schur complement with the resolvent $(sI - A_{cc})^{-1}$. In second-order retained coordinates this appears as

$$T(s) = s^2 M + sD_0 + L + \Pi(s), \quad \Pi(s) = \Pi_K(s) + s\Pi_D(s). \quad (2.14)$$

The rational term $\Pi(s)$ is the spectral memory of the eliminated controller block. It contains poles and residues. A static damping matrix cannot exactly represent those poles.

2.7 What a Pole Means Physically

A root $s = \sigma + j\omega$ of the retained operator produces a modal response of the form

$$e^{st} = e^{\sigma t} e^{j\omega t}.$$

The real part σ controls growth or decay, and the imaginary part ω controls oscillation frequency. The damping ratio is

$$\zeta = -\frac{\sigma}{\sqrt{\sigma^2 + \omega^2}}. \quad (2.15)$$

The same pole can be interpreted in several physical ways: a slow electromechanical oscillation, a local plant mode, a controller interaction mode, or a mixed network-control mode. This is why the thesis does not name a mode only by frequency. It asks which operator term and which port direction make the mode visible.

Figure 2.4 explains why a nominal eigenvalue plot is not enough. A system may be stable at the base point but fragile under small line uncertainty, controller retuning, model-identification error, or operating-point movement. The thesis therefore uses structured fragility rather than only nominal roots.

2.8 Why Scalar Grid Strength Is Not Enough

Classical strength metrics are useful because they compress a complicated network into a small number. Short-circuit ratio describes local electrical stiffness. Algebraic connectivity describes a global graph bottleneck. A damping ratio describes one pole. Effective resistance describes a static graph compliance. None of these scalars is wrong. The problem is that each one protects only one mechanism.

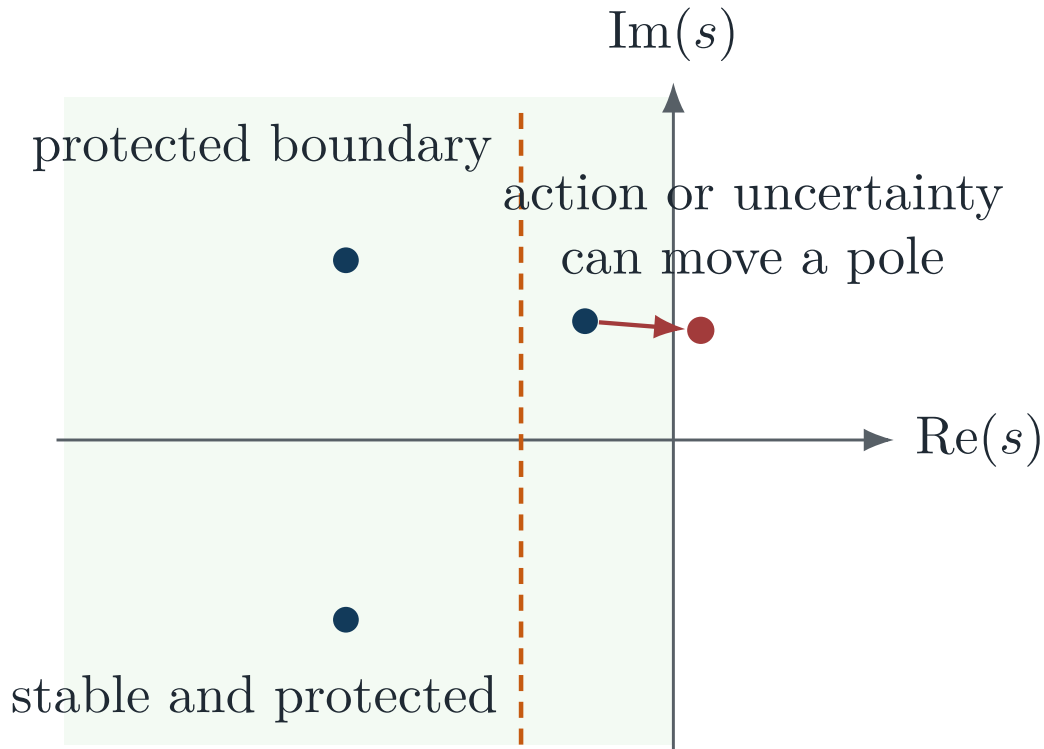


Figure 2.4. The spectral plane makes the margin explicit. A nominally stable pole can still be fragile if a physically admissible perturbation moves it across a protected damping, abscissa, resonance, or passivity boundary.

The dynamic replacement is not another scalar. It is the retained operator

$$T(s) = s^2 M + s D_0 + L + \Pi(s) \quad (2.16)$$

and its Green function

$$G_T(s) = T(s)^{-1}. \quad (2.17)$$

The scalar quantities used for planning are projections of this operator onto a declared port, action family, frequency band, or protected boundary.

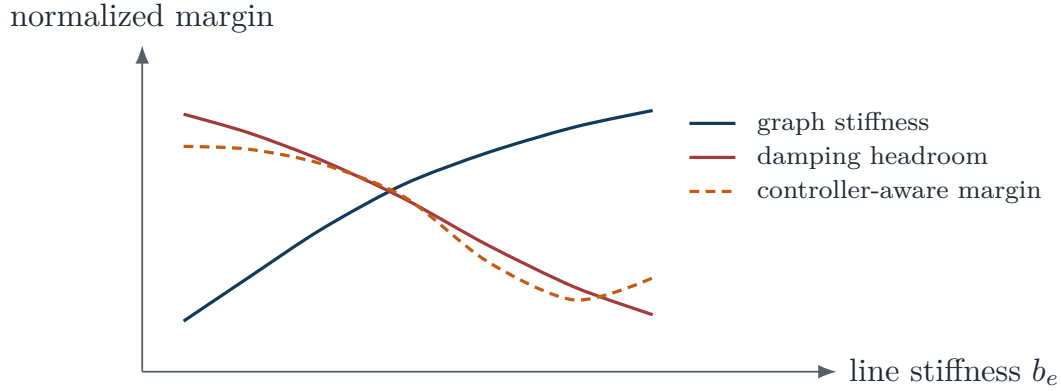
Figure 2.5 is the central planning motivation. If a line upgrade increases graph stiffness but reduces damping headroom, a scalar graph score gives the wrong engineering message. If the static screen accepts the line but the controller-aware model rejects it, the missing mechanism is not topology. It is controller memory.

2.9 A Three-Bus Example Where Classical Screens Fail

This section gives the smallest possible example that contains the dissertation problem. The example is not meant to represent a full test system. It is a teaching model that shows why a line action that is excellent for a classical graph metric can be bad for a dynamic margin.

Table 2.1. Why a scalar metric is not a complete dynamic-strength theory.

Scalar	What it sees	What it can miss
SCR	local fault-level or Thevenin stiffness	damping, controller poles, non-normal amplification, multi-port interactions
$\lambda_2(L)$	global graph connectivity	heterogeneous damping, inertia placement, operating-point residues
ζ_k	decay of one pole	finite-action fragility, nearby controller poles, hidden input-output gain
$a_e^T L^\dagger a_e$	static effective resistance	frequency-dependent compliance, resonant Green-function peaks

**Figure 2.5.** A stiffness-improving line action need not improve a dynamic margin. The graph metric, fixed- D damping screen, and controller-aware margin can move in different directions.

2.9.1 Step 1: Build the Three-Bus Graph

Consider the three-bus chain in Figure 2.6. Bus 1 is a synchronous source with appreciable damping, bus 2 is a passive transfer bus, and bus 3 is a converter-dominated bus with weak damping. The base system has two lines, 1–2 and 2–3, both with unit synchronizing weight. The candidate planning action is a new tie line 1–3 with incremental stiffness $p \geq 0$.

2.9.2 Step 2: Derive the Stiffness Matrix

With only the chain lines present, the Laplacian stiffness is

$$L_0 = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}. \quad (2.18)$$

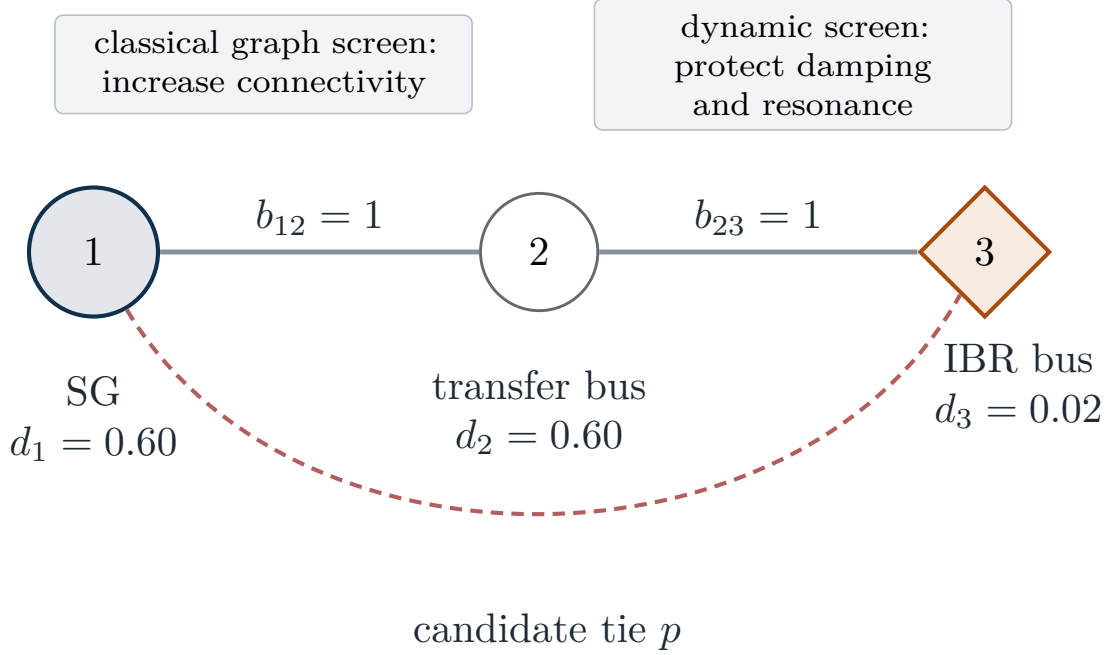


Figure 2.6. Three-bus teaching example. The candidate tie 1–3 improves the static graph, but it also changes the frequency of the endpoint mode that uses the weakly damped converter bus.

The nonzero graph modes are

$$\nu_1 = 1, \quad \phi_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \quad \nu_2 = 3, \quad \phi_2 = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}. \quad (2.19)$$

The candidate tie has incidence vector

$$a_{13} = e_1 - e_3 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \quad L(p) = L_0 + p a_{13} a_{13}^T. \quad (2.20)$$

Because a_{13} is aligned with ϕ_1 , the tie line changes the first nonzero stiffness exactly:

$$\nu_1(p) = 1 + 2p, \quad \nu_2(p) = 3. \quad (2.21)$$

For $0 \leq p \leq 1$, the algebraic connectivity therefore increases from 1 to 3. The static effective resistance between buses 1 and 3 also decreases:

$$R_{13}^{\text{stat}}(p) = \frac{2}{1 + 2p}. \quad (2.22)$$

A classical graph-only decision would say that increasing p is unambiguously good: the network becomes more connected and the endpoint resistance becomes smaller.

2.9.3 Step 3: Add the First Dynamic Screen

Now add unit inertia and heterogeneous damping,

$$M = I, \quad D = \text{diag}(0.60, 0.60, 0.02). \quad (2.23)$$

The endpoint mode ϕ_1 moves bus 1 against bus 3 while bus 2 stays nearly still. Its frozen graph-mode damping estimate is

$$\delta_1 = \phi_1^T D \phi_1 = \frac{d_1 + d_3}{2} = 0.31. \quad (2.24)$$

The same line that increases stiffness also raises the modal frequency. A first dynamic damping-ratio screen gives

$$\zeta_1(p) \approx \frac{\delta_1}{2\sqrt{\nu_1(p)}} = \frac{0.31}{2\sqrt{1+2p}}. \quad (2.25)$$

The sign of the conclusion changes. The graph metric improves with p , but the damping ratio falls with p . If the planning rule requires $\zeta \geq 0.10$, the base case passes and the strong tie $p = 1$ fails.

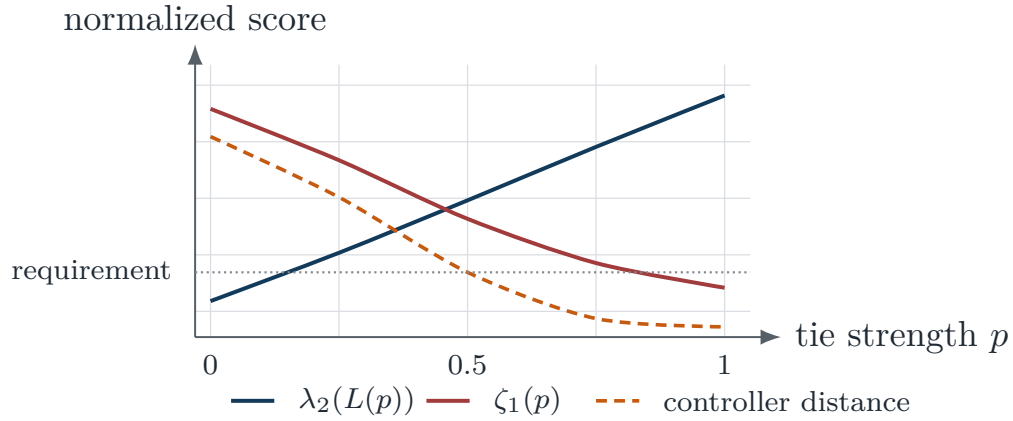


Figure 2.7. In the three-bus example, the classical graph score and the dynamic margin move in opposite directions. This is the simplest visible failure of a connectivity-only planning rule.

2.9.4 Step 4: Show the Classical Failure

Table 2.2 summarizes the contradiction. The line action is attractive under two classical static metrics: $\lambda_2(L(p))$ rises and $R_{13}^{\text{stat}}(p)$ falls. The same action is unattractive under the dynamic damping screen because it raises the frequency of a mode that uses the weakly damped converter bus.

Table 2.2. Three-bus example: static improvement but dynamic degradation.

p	λ_2	R_{13}^{stat}	$\omega_1 \approx \sqrt{1+2p}$	ζ_1	Decision if $\zeta_{\text{req}} = 0.10$
0	1.00	2.00	1.00	0.155	passes the damping screen
0.5	2.00	1.00	1.41	0.110	barely passes
1.0	3.00	0.67	1.73	0.089	fails even though graph strength improved

This is the basic reason the thesis cannot stop at graph strength. A graph method answers, “Did the network become stiffer?” The stability question is different: “Did the protected dynamic margin improve?” In this example the answer to the first question is yes, while the answer to the second is no.

2.9.5 Step 5: Add the Hidden Controller Mechanism

The previous step used only M , D , and $L(p)$. Converter-rich systems can fail one level higher. Suppose the eliminated converter/controller block at bus 3 has a lightly damped internal pole near

$$\mu_c = -0.05 + j1.70. \quad (2.26)$$

A static graph screen cannot see this pole. A fixed- D screen cannot represent it exactly either, because the reduced effect is frequency dependent. After Schur condensation, it appears inside $\Pi(s)$, and the mode must be tested through

$$T(s; p) = s^2 I + sD + L(p) + \Pi(s). \quad (2.27)$$

A simple resonance-distance gate is

$$d_c(p) = \frac{|j\sqrt{1+2p} - \mu_c|}{|j\sqrt{1+2p}| + |\mu_c|}. \quad (2.28)$$

For $p = 0$, $d_c \approx 0.260$. For $p = 0.5$, $d_c \approx 0.093$. For $p = 1$, $d_c \approx 0.017$. Thus the same tie that the classical graph method ranks best also moves the network mode toward the hidden controller pole. The correct escalation is not cosmetic; it changes the engineering decision. The controller-aware analysis asks whether $\Pi(s)$, residues, port direction, and dynamic effective resistance create a fragile interaction near the protected frequency band.

2.9.6 What the Example Teaches

The three-bus example separates three statements that are often conflated:

1. **Topology statement:** adding the tie 1–3 increases λ_2 and lowers static resistance.
2. **Fixed-dynamics statement:** the same tie can lower damping ratio because it raises frequency without adding proportional damping.
3. **Controller-aware statement:** if a hidden controller pole sits in the new frequency band, the reduced operator must include $\Pi(s)$, and the planning question becomes a port-resolvent question.

This is why the dissertation develops a hierarchy instead of a single graph index. Classical methods are valuable first screens, but they fail when the protected margin depends on heterogeneous damping, hidden controller memory, or finite line actions. The rest of the thesis formalizes exactly when each level is enough and when the analysis must escalate.

2.10 Dynamic Effective Resistance as the Bridge

The classical effective resistance of a line $e = (i, j)$ is

$$R_e^{\text{stat}} = a_e^T L^\dagger a_e, \quad a_e = e_i - e_j. \quad (2.29)$$

This quantity measures static graph compliance between the endpoints of the line. The thesis generalizes it by replacing the static pseudoinverse with the dynamic Green function:

$$\mathcal{R}_e(s) = a_e^T T(s)^{-1} a_e. \quad (2.30)$$

At $s = 0$ and $\Pi(0) = 0$, this reduces to $a_e^T L^\dagger a_e$. Away from zero, it is a frequency-dependent, controller-aware port compliance.

This definition matters because a finite line action has an exact scalar pole condition:

$$\det(T(s) + \Delta b_e a_e a_e^T) = \det T(s) (1 + \Delta b_e \mathcal{R}_e(s)). \quad (2.31)$$

Thus the new or shifted poles created by the line action satisfy

$$1 + \Delta b_e \mathcal{R}_e(s) = 0. \quad (2.32)$$

The usual line sensitivity is the local residue limit of this finite-action law. That is the reason dynamic effective resistance is the bridge between spectral graph theory and controller-aware planning.

2.11 Operator Hierarchy and Escalation Logic

The thesis uses a hierarchy because no single model level is always necessary or always sufficient. The cheapest valid model should be used, but it must pass the gate attached to the claim.

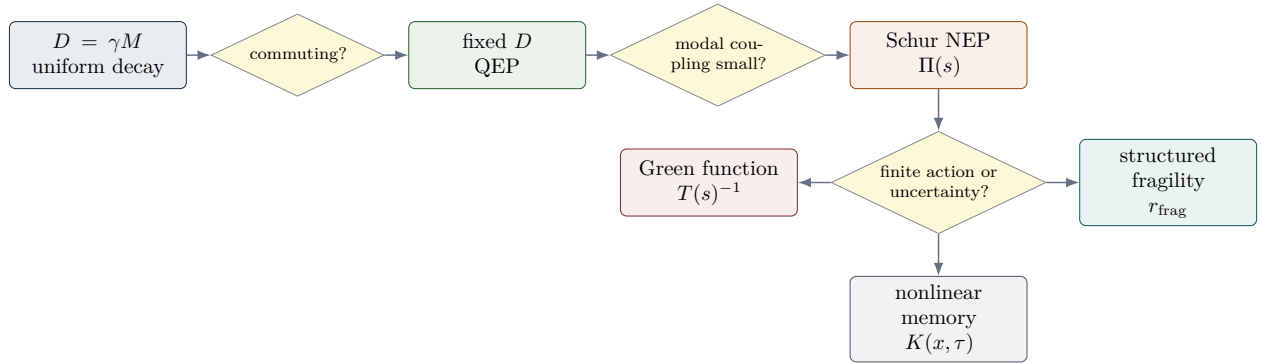


Figure 2.8. Escalation logic of the thesis. The analysis becomes richer only when the protected claim requires the richer operator: modal coupling, controller memory, finite actions, structured uncertainty, or nonlinear amplitude effects.

Figure 2.8 is the practical answer to why the full analysis is necessary. If damping commutes with stiffness, graph modes can certify more. If damping does not commute, the fixed- D QEP is needed. If controller states are condensed, $\Pi(s)$ is needed. If the question concerns a finite line, damping, inertia, or controller action, $T(s)^{-1}$ is needed. If the question concerns minimum physical perturbation to failure, structured fragility is needed. If the question concerns finite-amplitude behavior, nonlinear memory is needed.

2.12 Decision Contract for the Rest of the Thesis

Every claim in the rest of the dissertation should be readable through the same four-part contract:

1. **Operator:** Which model is being used: L , QEP, Schur NEP, Green function, structured fragility, or nonlinear memory?
2. **Port or action:** Which line, bus, converter, controller channel, input-output pair, or uncertainty family is being tested?

3. **Margin:** Which quantity is protected: damping ratio, spectral abscissa, protected gain, passivity, resonance distance, or hosting feasibility?
4. **Certificate:** Which theorem, bound, contour test, finite-action law, or full-model validation turns the calculation into evidence?

Without these four items, a weak-bus list, line-ranking plot, or hosting number is not reproducible. With these four items, the thesis can distinguish a graph screen from a dynamic certificate.

From Power Flow to Spectral Dynamics

This chapter derives the mathematical objects used throughout the thesis from the electrical model upward. The purpose is pedagogical as much as technical: the reader should see where the matrix L comes from, why the swing equation has the form $M\ddot{\theta} + D\dot{\theta} + L\theta = p$, and why substituting e^{st} produces the quadratic operator $s^2M + sD + L$. Later chapters use these equations aggressively, so this chapter builds them slowly.

3.1 The Physical Picture

A power grid is a network of buses connected by lines and transformers. A bus is a node where generators, loads, converters, transformers, or transmission lines meet. At each AC bus, the voltage is not only a magnitude. It is a sinusoid with a phase angle. In phasor notation,

$$V_i = |V_i|e^{j\theta_i}, \quad (3.1)$$

where $|V_i|$ is the voltage magnitude and θ_i is the phase angle. Power transfer in an AC network depends strongly on differences of angles, not on absolute angles. If every angle is shifted by the same constant, the physics is unchanged. This is the reason a Laplacian matrix will later have one zero eigenvalue.

The network itself is represented by the bus-admittance matrix

$$Y_{\text{bus}} = G + jB. \quad (3.2)$$

This matrix is built from the impedances of lines and transformers. It maps bus voltages to injected currents:

$$I = Y_{\text{bus}}V. \quad (3.3)$$

Equation (3.3) is Kirchhoff's current law in matrix form. It is the electrical network before any dynamic-machine model is added.

3.2 Complex Power and Power Flow

The complex power injected at bus i is

$$S_i = P_i + jQ_i = V_i I_i^*, \quad (3.4)$$

where P_i is active power and Q_i is reactive power. Substituting $I = Y_{\text{bus}}V$ gives the AC power-flow equations:

$$P_i(V, \theta) = |V_i| \sum_{j=1}^n |V_j| (G_{ij} \cos(\theta_i - \theta_j) + B_{ij} \sin(\theta_i - \theta_j)), \quad (3.5)$$

$$Q_i(V, \theta) = |V_i| \sum_{j=1}^n |V_j| (G_{ij} \sin(\theta_i - \theta_j) - B_{ij} \cos(\theta_i - \theta_j)). \quad (3.6)$$

These equations determine the operating point. A power-flow solution provides the steady voltages V^* , angles θ^* , and injections that balance generation and load. The dynamic model in this thesis is always a small-signal model around such an operating point.

Remark 3.1 (Where power flow enters the thesis). *The matrix L is not chosen by hand. It is the linearized active-power response around a solved operating point. If the loading, generation dispatch, network topology, line parameters, or converter injections change, the power flow changes, and therefore the stiffness matrix L changes.*

3.3 From Active-Power Jacobian to the Laplacian L

To obtain a dynamic model, perturb the angles around the operating point:

$$\theta = \theta^* + \Delta\theta. \quad (3.7)$$

The first-order change in active power is

$$\Delta P \approx J_{P\theta} \Delta\theta, \quad J_{P\theta} = \left. \frac{\partial P}{\partial \theta} \right|_{(V^*, \theta^*)}. \quad (3.8)$$

In this dissertation P denotes exported electrical power in the swing equation. With that convention, the synchronizing stiffness is this incremental power response:

$$L = J_{P\theta} \quad \text{on the retained angle coordinates.} \quad (3.9)$$

If instead P is defined as net injected disturbance with the opposite sign, the formula becomes $L = -J_{P\theta}$. The sign convention must be fixed before using any derivative formula.

Under the common lossless approximation $G_{ij} \approx 0$, fixed voltage magnitudes, and sufficiently small angle differences, the off-diagonal entries become

$$L_{ij} = -|V_i^*| |V_j^*| B_{ij} \cos(\theta_i^* - \theta_j^*), \quad i \neq j, \quad (3.10)$$

and the diagonal is chosen so that each row sums to zero:

$$L_{ii} = -\sum_{j \neq i} L_{ij}, \quad L\mathbf{1} = 0. \quad (3.11)$$

This is the graph Laplacian structure. The line weight is roughly $|V_i^*| |V_j^*| B_{ij} \cos(\theta_i^* - \theta_j^*)$. A line with high susceptance, high voltage magnitudes, and small angle separation contributes stronger synchronizing stiffness.

Example 3.1 (Two-bus stiffness). *For two buses connected by one lossless line, active power from bus 1 to bus 2 is approximately*

$$P_{12} = |V_1||V_2|B_{12}\sin(\theta_1 - \theta_2).$$

Linearizing around θ^ gives*

$$\Delta P_{12} \approx |V_1^*||V_2^*|B_{12}\cos(\theta_1^* - \theta_2^*)(\Delta\theta_1 - \Delta\theta_2).$$

Thus the line resists angle separation like a spring. That spring constant is the line weight that appears inside L .

3.4 The Single-Machine Swing Equation

The swing equation is Newton's second law for rotor angle. A synchronous machine stores kinetic energy in a rotating mass. Let θ_i be the electrical angle of machine i relative to a synchronous reference, and let $\omega_i = \dot{\theta}_i$ be the frequency deviation. The power imbalance accelerates the rotor:

$$M_i\dot{\omega}_i = P_{m,i} - P_{e,i} - D_i\omega_i. \quad (3.12)$$

Here $P_{m,i}$ is mechanical or commanded input power, $P_{e,i}$ is electrical air-gap power exported to the network, $D_i\omega_i$ is a linearized damping or primary-control term, and M_i is the inertia coefficient in the chosen per-unit convention. Since $\omega_i = \dot{\theta}_i$, (3.12) becomes

$$M_i\ddot{\theta}_i + D_i\dot{\theta}_i = P_{m,i} - P_{e,i}. \quad (3.13)$$

At the operating point, $P_{m,i}^* = P_{e,i}^*$. The dynamic equation therefore concerns deviations from that balance. Let p_i be a small input disturbance and let $\Delta P_{e,i}$ be the incremental electrical power. Then

$$M_i\Delta\ddot{\theta}_i + D_i\Delta\dot{\theta}_i + \Delta P_{e,i} = p_i. \quad (3.14)$$

The previous section showed that, to first order, $\Delta P_e \approx L\Delta\theta$. Therefore the electrical network supplies the restoring term in the swing equation.

3.5 The Networked Swing Model

Stack all retained angles into θ , all inertia coefficients into $M = \text{diag}(M_i)$, and all damping coefficients into $D = \text{diag}(D_i)$. Dropping the Δ notation for readability gives

$$M\ddot{\theta} + D\dot{\theta} + L\theta = p. \quad (3.15)$$

This is the model used in the fixed- D chapters. Each term has a physical meaning:

- $M\ddot{\theta}$: inertia resists acceleration;
- $D\dot{\theta}$: damping or primary control resists speed deviation;
- $L\theta$: network synchronizing stiffness resists angle separation;
- p : disturbance or control input.

Because $L\mathbf{1} = 0$, the uniform angle direction must be removed by selecting a reference bus, center-of-inertia coordinates, or a projection onto $\mathbf{1}^\perp$. After that reduction, L is positive definite on the retained nonzero-angle subspace if the network is connected and the operating point is regular.

3.6 Why the Exponential Ansatz Produces $s^2M + sD + L$

Small-signal stability asks what happens to a tiny perturbation if no further input is applied. Set $p = 0$ and look for modal solutions of the form

$$\theta(t) = ve^{st}, \quad (3.16)$$

where v is the mode shape and s is the complex growth rate. Then

$$\dot{\theta}(t) = sve^{st}, \quad \ddot{\theta}(t) = s^2ve^{st}. \quad (3.17)$$

Substituting (3.16) into (3.15) gives

$$Ms^2ve^{st} + Dsve^{st} + Lve^{st} = 0.$$

Since $e^{st} \neq 0$, it can be divided out:

$$(s^2M + sD + L)v = 0. \quad (3.18)$$

This is the quadratic eigenvalue problem (QEP). It is not an arbitrary mathematical object. The factor s^2 comes from acceleration, s comes from velocity damping, and the constant term comes from the network restoring force.

Table 3.1. Physical origin of the QEP terms.

Term	Physical source	Engineering interpretation
s^2M	rotor or virtual mass	acceleration and inertia
sD	damping, droop, primary response	decay of speed deviations
L	linearized active-power network	synchronizing stiffness
$(sI - A_{cc})^{-1}$	eliminated controller states	dynamic memory of controls

3.7 How to Read a Complex Pole

If a root is $s = \sigma + j\omega$, the perturbation contains

$$e^{st} = e^{\sigma t} e^{j\omega t}.$$

The real part σ controls decay or growth. The imaginary part ω controls oscillation frequency. Stability requires $\sigma < 0$. The damping ratio is

$$\zeta = -\frac{\sigma}{\sqrt{\sigma^2 + \omega^2}}. \quad (3.19)$$

This number measures how quickly the oscillation decays relative to its frequency. A line upgrade can increase ω by stiffening the network, but if σ does not become more negative, ζ may decrease. This is the basic reason graph stiffness and damping margin must be kept separate.

3.8 Where Inverters Enter the Same Chain

An inverter-based resource can enter this chain at several levels. In a simple reduced model, a grid-forming inverter may be represented by effective inertia, damping, or droop terms added to M and D . A grid-following inverter may alter the operating point and the Jacobian through its injection and controls. In a detailed small-signal model, inverter controllers add states: PLL states, current-loop states, voltage-loop states, filters, limits, and plant controls. If those states are retained, they appear in the full state matrix. If they are mathematically eliminated, they appear through a Schur complement and produce the rational self-energy $\Pi(s)$ developed in the self-energy part of the thesis. The dynamic damping object $\Sigma(s)$ is the special case where the rational correction enters through the velocity channel.

Thus the thesis path is:

$$\text{power flow} \rightarrow L \rightarrow M\ddot{\theta} + D\dot{\theta} + L\theta = p \rightarrow s^2M + sD + L \rightarrow s^2M + sD_0 + L + \Pi(s). \quad (3.20)$$

Every later result is a refinement of one arrow in this chain.

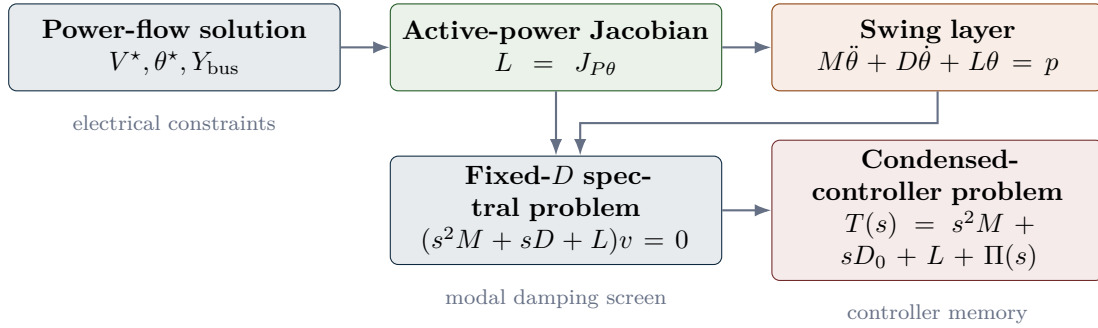


Figure 3.1. Modeling chain from AC power flow to the rational NEP. The same network operating point that sets voltages and flows also defines the stiffness operator L . Controller condensation changes the operator class, because the damping contribution becomes frequency dependent.

3.9 Linearization and Spectral Parameter

Let a power-system dynamic model be written locally as a differential-algebraic system,

$$\dot{x} = f(x, y, \rho), \quad (3.21)$$

$$0 = g(x, y, \rho), \quad (3.22)$$

where x are dynamic states, y are algebraic variables, and ρ is a parameter or action. At a regular equilibrium (x^*, y^*) , algebraic variables can be eliminated locally when g_y is nonsingular, giving a reduced state matrix

$$A_{\text{red}} = f_x - f_y g_y^{-1} g_x. \quad (3.23)$$

Small-signal stability is determined by the eigenvalues of A_{red} . The spectral parameter is denoted by $s \in \mathbb{C}$, and a pole is stable when $\text{Re}(s) < 0$.

For an oscillatory pole $s = \sigma + j\omega$, the frequency and damping ratio are

$$f = \frac{|\omega|}{2\pi}, \quad \zeta = -\frac{\sigma}{\sqrt{\sigma^2 + \omega^2}}. \quad (3.24)$$

The damping-ratio derivative satisfies

$$d\zeta = -\frac{\omega^2}{(\sigma^2 + \omega^2)^{3/2}} d\sigma + \frac{\sigma\omega}{(\sigma^2 + \omega^2)^{3/2}} d\omega, \quad (3.25)$$

so a frequency shift alone is not a damping certificate.

3.10 Graph Laplacians and Synchronizing Stiffness

Under a lossless active-power approximation, the synchronizing stiffness matrix has Laplacian structure. For $i \neq j$,

$$L_{ij} = -V_i V_j b_{ij} \cos(\theta_i^* - \theta_j^*), \quad (3.26)$$

with diagonal entries chosen so that

$$L\mathbf{1} = 0. \quad (3.27)$$

If the network is reciprocal and the operating point is sufficiently regular, $L = L^T \succeq 0$. The zero eigenvalue corresponds to uniform angle shift and is removed by fixing a reference or projecting onto $\mathbf{1}^\perp$.

Let $M = \text{diag}(M_i) \succ 0$. The inertia-normalized stiffness is

$$\tilde{L} = M^{-1/2} L M^{-1/2}. \quad (3.28)$$

On the nonzero subspace, write

$$\tilde{L}Q = Q\Lambda, \quad Q^T Q = I, \quad \Lambda = \text{diag}(\nu_1, \dots, \nu_n), \quad \nu_k > 0. \quad (3.29)$$

The columns of Q are graph-inertia modes. They diagonalize stiffness. They do not automatically diagonalize damping.

3.11 Second-Order Systems and QEPs

The retained second-order model is

$$M\ddot{\theta} + D\dot{\theta} + L\theta = p. \quad (3.30)$$

Substituting $\theta(t) = ve^{st}$ gives

$$P(s)v = (s^2 M + sD + L)v = 0. \quad (3.31)$$

This is a QEP. The corresponding first-order linearization is

$$\mathcal{A} = \begin{bmatrix} 0 & I \\ -M^{-1}L & -M^{-1}D \end{bmatrix}, \quad (3.32)$$

but the second-order form should be retained conceptually because the matrices M , D , and L have distinct physical meanings.

In inertia-normalized variables $\vartheta = M^{1/2}\theta$,

$$\ddot{\vartheta} + \tilde{D}\dot{\vartheta} + \tilde{L}\vartheta = M^{-1/2}p, \quad \tilde{D} = M^{-1/2}DM^{-1/2}. \quad (3.33)$$

The modal damping matrix is

$$\Gamma = Q^T \tilde{D}Q. \quad (3.34)$$

The equality $Q^T \tilde{L}Q = \Lambda$ is automatic by construction; the equality $Q^T \tilde{D}Q = \text{diag}(\cdot)$ is an additional damping assumption.

3.12 Simple Eigenvalue Sensitivity

For a polynomial or nonlinear eigenvalue problem

$$T(s, \rho)v = 0, \quad u^H T(s, \rho) = 0, \quad (3.35)$$

with a simple eigenvalue s_* , the first derivative is

$$\frac{ds_*}{d\rho} = -\frac{u^H T_\rho(s_*, \rho)v}{u^H T_s(s_*, \rho)v}. \quad (3.36)$$

For a QEP $T(s) = s^2 M + sD + L$, this becomes

$$\frac{ds_*}{d\rho} = -\frac{u^H (s_* D_\rho + L_\rho)v}{u^H (2s_* M + D)v} \quad (3.37)$$

when M is fixed. Equation (3.37) is central because it shows that line actions and damping actions enter the numerator differently, while the denominator depends on the mode and the damping operator.

3.13 Schur Complements

Partition a full first-order linearization into retained states x_e and condensed states x_c :

$$A = \begin{bmatrix} A_{ee} & A_{ec} \\ A_{ce} & A_{cc} \end{bmatrix}. \quad (3.38)$$

The eigenvalue equation $(sI - A)x = 0$ is equivalent, when $sI - A_{cc}$ is invertible, to

$$\left[sI - A_{ee} - A_{ec}(sI - A_{cc})^{-1}A_{ce} \right] x_e = 0. \quad (3.39)$$

This is an exact nonlinear eigenvalue problem for the retained states. It does not approximate the eliminated states; it represents them through their resolvent.

3.14 Rational Matrix Functions and Self-Energy

A rational matrix function has the form

$$R(s) = R_0 + \sum_{\ell=1}^m \frac{R_\ell}{s - \alpha_\ell} \quad (3.40)$$

after partial-fraction expansion when the poles are simple. The residues R_ℓ determine how strongly each eliminated pole α_ℓ couples to the retained coordinates. In the Schur operator, those residues are generated by A_{ec} , A_{ce} , and the eigenvectors of A_{cc} .

In a second-order retained representation, the rational contribution appears as a self-energy,

$$T(s) = s^2 M + sD_0 + L + \Pi(s), \quad \Pi(s) = G(sI - A_{cc})^{-1}(B_\theta + sB_\omega). \quad (3.41)$$

The term self-energy is useful because it emphasizes that the retained model is not isolated. It carries the dynamic influence of states that have been mathematically eliminated. When $\Pi(s) = s\Pi_D(s)$, the dynamic damping matrix is $\Sigma(s) = D_0 + \Pi_D(s)$.

3.15 Certificates, Screens, and Gates

The thesis uses three words with specific meanings.

Definition 3.1 (Certificate). *A certificate is a mathematical or numerical condition that directly implies a claimed margin property under stated assumptions.*

Definition 3.2 (Screen). *A screen is a lower-cost indicator that may identify candidates for action or inspection but does not imply the claimed margin property by itself.*

Definition 3.3 (Gate). *A gate is an accept/reject test that a proposed certificate or screen must pass before it is used for a stronger claim.*

The distinction matters throughout the dissertation. A graph eigenvalue can be a certificate in a proportional or commuting fixed- D theory, a screen in a non-commuting fixed- D theory, and an incomplete descriptor after controller condensation. The same number changes epistemic status when the operator class changes.

Literature Review and Theoretical Gap

The theory developed in this dissertation sits at the intersection of five literatures: classical power-system stability, spectral and network interpretations of synchronization, second-order and quadratic eigenvalue analysis, nonlinear eigenvalue problems generated by state condensation, and converter-dominated stability. The purpose of this chapter is not to catalogue every contribution in those areas. It is to identify the exact conceptual gap that motivates a damping-operator hierarchy.

4.1 Systematic Review Contract

In the expanded dissertation this chapter should become a systematic review, not only a narrative survey. The review protocol should report search strings, databases, dates, inclusion and exclusion criteria, preprint versus peer-reviewed status, extraction fields, and a method-assumption-data-certificate matrix. The central taxonomy is the operator hierarchy

$$\begin{aligned} D = \gamma M \rightarrow [\tilde{L}, \tilde{D}] = 0 \rightarrow D \text{ fixed and non-commuting} \rightarrow \Pi(s) \text{ rational} \\ \rightarrow G_T(s) = T(s)^{-1} \rightarrow \Lambda_\varepsilon^A(T) \rightarrow \text{nonlinear memory.} \end{aligned} \quad (4.1)$$

The first arrow marks the move from uniform damping-to-inertia ratio to the larger decouplable class. The second marks modal coupling within a constant QEP. The third marks controller or auxiliary-state condensation. The fourth marks the Green function that carries finite actions. The fifth marks structured fragility under physical perturbation families. The last arrow marks the move from linearized spectral memory to genuinely nonlinear memory. This taxonomy is the review's organizing principle.

4.2 Classical Small-Signal Stability

Classical power-system stability theory starts from an operating point and studies the linearized differential-algebraic model around that point. The standard engineering workflow is well established: solve the power flow, initialize dynamic models, linearize, compute eigenvalues, inspect damping ratios, and use eigenvectors or participation factors to interpret the critical modes [1, 2]. The IEEE/CIGRE stability definition paper formalized the classification of rotor-angle, frequency, and voltage stability and placed small-signal stability within a broader taxonomy of disturbances, time scales, and mechanisms [3].

For the present thesis, the central fact is that classical modal analysis is already an operator theory. A mode is not merely an oscillation observed in a time-domain trace. It is a spectral object of a linearized system. Its damping ratio is computed from the real and imaginary parts of a pole.

Its engineering interpretation depends on how the corresponding eigenvectors project onto angles, speeds, exciters, governors, loads, and controls. Thus, any reduced model that claims to preserve damping must preserve the relevant spectral mechanism, not just the visual shape of a response.

4.3 Graph and Synchronization Perspectives

Power networks are also graphs. Under lossless or nearly lossless assumptions, the active-power Jacobian has a Laplacian structure: off-diagonal terms encode coupling between connected buses and diagonal terms enforce row-sum balance. This structure links power-system dynamics to synchronization theory. The work of Doerfler, Chertkov, and Bullo gives a concise synchronization condition for complex oscillator networks and smart grids, clarifying how coupling strength, phase cohesiveness, and network structure interact [6].

The graph viewpoint is indispensable because it turns a high-dimensional network into interpretable objects: algebraic connectivity, effective resistance, cut structure, mode shapes, Kron-reduced Laplacians, and weighted network strength. Algebraic connectivity traces back to Fiedler's second Laplacian eigenvalue and its relation to graph connectivity [7]. Effective-resistance design provides another view of robustness by penalizing large pairwise electrical distances and by linking network reinforcement to convex edge-weight allocation [8]. Spectral sparsification shows why effective resistance is also a structural sampling quantity: preserving selected quadratic forms preserves the Laplacian action that matters to many slow-mode metrics [9]. Kron reduction gives the matching electrical-network operation for eliminating passive buses through a Schur complement while preserving a reduced Laplacian [10].

These tools motivate planning intuitions: reinforce a weak line, improve a weak cut, place a grid-forming converter at a strategic bus, or raise a critical graph eigenvalue. Those intuitions are valuable. The problem is that damping is not always a graph eigenvalue. Damping may be local, heterogeneous, non-proportional, controller-mediated, or frequency dependent. The graph literature gives the stiffness skeleton; the present thesis asks which damping operator must be attached to that skeleton.

This boundary is important for the originality claim. For example, Guo, Zhao, and Low already connect the scaled graph Laplacian spectrum with primary frequency regulation under uniform damping-to-inertia ratios [11]. The present thesis uses that kind of result as a known floor. Its contribution is not the existence of Laplacian modes, but the operator hierarchy that decides when those modes remain damping coordinates, when rough damping-to-inertia fields mix them, and when controller self-energy must replace a static graph screen.

4.4 Beyond Scalar Laplacians: Cellular Sheaves

An important structural extension is to look beyond scalar graph Laplacians. The closest mature mathematical object is the cellular sheaf Laplacian. Hansen and Ghrist develop spectral sheaf theory by lifting the ordinary graph Laplacian to a Hodge Laplacian acting on a sheaf of vector spaces over a cell complex [12]. In that setting, vertices and edges may carry vector spaces, and restriction maps encode how local data are compared across cells. The theory connects spectral data to sheaf cohomology and includes analogues of interlacing, sparsification, effective resistance, and synchronization.

This dissertation does not claim a completed sheaf theory of converter-rich power grids. The point is more precise: a scalar Laplacian assigns one number-valued state to each bus, whereas a converter-rich bus carries internal controller coordinates. A sheaf model is a natural candidate for a future operator in which the bus stalk contains retained electrical variables, converter-control states, or reduced controller coordinates, and each line restriction map describes which combinations are electrically compatible. Such an operator could turn the present scalar mode-family labels and self-energy-dominance diagnostics into a structured vector-bundle-like object.

The risk is equally clear. A sheaf Laplacian is useful only if it detects a weak node, weak link, or hidden margin that the scalar L , the fixed- D QEP, and the Schur self-energy screen do not already detect. In this thesis it is therefore placed as a speculative theoretical extension, not as evidence for any current planning claim.

4.5 Quadratic Eigenvalue Problems and Second-Order Models

The swing-equation model is naturally second order. In compact form it leads to a quadratic eigenvalue problem,

$$P(s)v = (s^2M + sD + L)v = 0. \quad (4.2)$$

Tisseur and Meerbergen’s survey establishes the QEP as a mature mathematical object, with well-developed linearizations, spectral properties, perturbation formulas, and numerical methods [5]. This literature matters because the power-system damping problem should not be reduced too quickly to a first-order matrix. The second-order structure carries physical information: mass or inertia, damping, and stiffness are distinct operators.

The distinction between proportional and non-proportional damping is also well known in structural dynamics. If damping and stiffness share a modal basis, modal decoupling is exact. If they do not, modal analysis becomes coupled, and non-proportionality indices are needed to measure the error of decoupled approximations [13]. This thesis uses the same mathematical distinction in a power-system setting. The first departure from graph-frequency intuition is not necessarily converter dynamics; it can already occur when D is fixed but heterogeneous relative to M .

4.6 Nonlinear Eigenvalue Problems and Schur Condensation

The next mathematical layer is the nonlinear eigenvalue problem (NEP). Guettel and Tisseur survey matrix-valued functions $T(s)$ whose dependence on the spectral parameter is nonlinear and show that such problems arise naturally when delayed, damped, or condensed dynamics are represented without expanding all states [14]. A Schur complement is a basic route from a larger linear eigenvalue problem to a smaller nonlinear one. If the eliminated block is A_{cc} , the retained operator contains the resolvent $(sI - A_{cc})^{-1}$.

For computation, contour integral methods and rational-interpolation viewpoints are particularly relevant. Brennan, Embree, and Gugercin recast contour integral methods for NEPs in a systems-theoretic framework using Loewner matrix pencils, allowing interpolation away from infinity and linking eigenvalue extraction to system realization and modal truncation [15]. This is important for a Schur self-energy thesis because $\Pi(s)$ and its damping-channel projection $\Sigma(s)$ are transfer-like rational objects. If the controller block is not fully known, a Loewner or data-driven contour formulation is a plausible route for identifying the visible part of $\Pi(s)$ from frequency-domain samples rather than from explicit state-space matrices.

Recent power-system work also strengthens the numerical side of the proposed planning workflow. Bouterakos, McKeon, and Tzounas develop continuation-based eigenvalue tracking for large-scale power-system models, supporting sparse representations and both explicit and semi-implicit DAE forms [16]. That result is directly aligned with the local pole-motion map used later in this dissertation: a planning screen that changes penetration, line weights, inertia, damping, or controller gains must track selected eigenvalue trajectories rather than repeatedly rediscovering the entire spectrum from scratch.

This observation is the mathematical hinge of the dissertation. Converter and auxiliary controller states can be retained explicitly, producing a large first-order model. They can also be condensed, producing a smaller nonlinear operator. The condensation is exact away from poles of the eliminated block, but the price of exactness is frequency dependence. A constant damping matrix cannot store the poles, zeros, and residues introduced by the eliminated controller states. The reduced controller object is therefore a rational self-energy; its damping-channel projection is the rational object denoted $\Sigma(s)$.

4.7 Pseudospectra, Fragility, and Model Adequacy

Nonlinear eigenvalue theory also raises a robustness question. A nominal root set may look safe while nearby structured perturbations can move a pole to a protected boundary, especially when the retained operator is non-normal. The classical pseudospectral viewpoint already warns that eigenvalues alone can miss transient amplification and perturbation sensitivity [17]. Recent work on nonlinear spectral problems extends the computational discussion of spectra and pseudospectra to broad NEP settings and emphasizes spectral pollution and invisibility risks in approximations [18].

For this dissertation, the distinction is that power-system perturbations must be structured. A generic matrix perturbation is not the same as uncertainty in line susceptance, PLL gains, converter delays, virtual inertia, damping, or an identified self-energy. This motivates a physical-action pseudospectrum $\Lambda_\epsilon^A(T)$ and a fragility radius: the smallest admissible perturbation that drives a retained root to a protected boundary. The related model-adequacy problem is contour based. If a reduced operator $\hat{T}$ is used for a decision, it should preserve the protected eigenvalue count on a contour before its derivative signs are trusted.

4.8 Sign Characteristics and Krein-Type Mechanisms

The mathematical language closest to the proposed controller-resolvent collision is not a new, free-floating “Krein signature for NEPs.” The citable foundation is the sign characteristic of Hermitian matrix polynomials and self-adjoint meromorphic matrix functions. Gohberg, Lancaster, and Rodman give the classical matrix-polynomial framework, including sign characteristics of spectral data [19]. Their related work on self-adjoint meromorphic matrix functions is closer to the rational structure created by Schur condensation [20]. Mehrmann, Noferini, Tisseur, and Xu develop sign characteristics for Hermitian matrix polynomials and clarify the role of structure-preserving transformations [21]. Kollár and Miller’s Evans–Krein survey provides a systems-oriented view in which rational Krein functions track spectral collisions [22].

Those references do not already solve the converter problem. They mostly live in Hamiltonian, self-adjoint, or Hermitian-polynomial settings. Converter self-energies are rational, dissipative, and generally non-normal. The thesis therefore treats sign characteristic as a foundation to be adapted,

not as a ready-made theorem. The open bridge is to show when a rational controller-condensed operator has an operational sign collision that coincides with weak-grid PLL or controller-resonance instability.

4.9 Converter-Dominated Stability

The increased penetration of converter-interfaced generation motivates a reconsideration of stability definitions and modeling practice. The revisited IEEE/CIGRE classification explicitly adds converter-driven and electric resonance phenomena to the stability vocabulary [4]. IEEE Std 2800-2022 reflects the same industrial shift by specifying performance and interconnection requirements for IBRs connected to transmission systems [23].

Within the research literature, grid-following and grid-forming controls are not interchangeable dynamic objects. Grid-following converters commonly depend on a PLL for synchronization, so PLL dynamics can contribute to small-signal instability under weak-grid conditions. Yang, Huang, Xin, and Ju analyze how placing grid-forming converters can improve the small-signal stability of PLL-integrated systems by modifying grid strength and Laplacian-related quantities [24]. Comparative studies of grid-forming and grid-following inverters further show that control architecture changes the state-space model and its eigenvalues [25].

Inertia and primary-control placement studies sharpen the same point. Optimal virtual inertia allocation can be formulated as an H_2 -type network performance problem [26]; disturbance propagation depends strongly on where inertia is located relative to slow network modes [27]; and perturbation theory shows that inertia and primary-control placement are not interchangeable under heterogeneous parameters [28]. These results support the thesis distinction between adding inertia, adding damping, and modifying the graph.

Passivity and port-Hamiltonian viewpoints provide a second, complementary control language. Passivity-based grid-forming work formulates converter controllers so that local subsystems have dissipativity properties that can be interconnected modularly. He and Dörfler derive decentralized stability conditions for grid-forming converters using passivity indices [29]. Huang et al. combine small-gain and small-phase arguments to obtain decentralized stability checks for heterogeneous power-electronics-dominated systems [30], while Gui, Subedi, and Xue use a port-controlled Hamiltonian form for passivity-based grid-forming control of distributed energy resources [31]. Recent matrix-valued passivity indices make the same issue sharper for MIMO ports: scalar indices can discard the direction and coupling structure of a passivity deficit [32]. Dynamic passivity multipliers further show why frequency-dependent certificates can be less conservative than strict static passivity tests for converter-dominated grids [33]. In the notation of this thesis, this literature says that the Hermitian, weighted directional, or multiplier-based behavior of the condensed self-energy $\Sigma(s)$ should not be a decorative diagnostic. It is a candidate certificate for controller retuning, provided that the passivity screen is tied back to the retained pole motion and to full-model validation.

These studies justify the engineering relevance of the present theory, but they do not remove the need for a general operator language. A placement rule derived for a particular control model or grid-strength metric may not transfer to a different controller, parameter regime, or retained-state partition. The common mathematical issue is not the label GFL or GFM by itself; it is whether the reduced model preserves the spectral contribution of the eliminated control states.

4.10 Hosting Capacity, Network Reinforcement, and Braess Effects

Hosting capacity is usually introduced as a planning quantity: how much distributed generation, load, or converter-interfaced equipment can be added before a registered performance limit is violated. Bollen and Roennberg frame hosting capacity as a distribution-planning tool under uncertainty [34]. Later dynamic hosting-capacity studies emphasize that snapshot limits can miss the time duration, severity, and controller response of violations [35, 36]. Those works motivate the same methodological principle used in this thesis: capacity is not a single number until the limiting metric has been named.

The present dissertation uses the phrase dynamic hosting capacity in a different but compatible sense. Instead of asking only how long a voltage or thermal violation lasts, it asks how much IBR conversion can be admitted before a small-signal spectral margin is lost. Static hosting constraints remain necessary, but they are insufficient in low-inertia systems because a voltage-safe operating point can still have an unacceptable damping ratio, resolvent gain, or controller-resonance exposure.

Recent stability-manifold work for converter-dominated systems reinforces the need to map controller-parameter regions under multiple operating scenarios [37]. The distinction here is mechanism: the present thesis seeks not only a sampled stability boundary, but an analytic normal, a physical self-energy label, an error certificate, and a minimum admissible action associated with that boundary.

Network reinforcement has a similar ambiguity. In a purely stiffness-oriented graph view, increasing a line weight or adding a line seems beneficial. Braess’ original traffic result already showed that adding capacity can worsen system performance under decentralized routing [38]. Power-system analogues have been studied for oscillator synchronization and electric grids. Witthaut and Timme showed that adding links in oscillator networks can destroy synchronization [39], while Coletta and Jacquod analyzed how added couplings can enhance or reduce linear stability in coupled-oscillator and power-grid models [40]. Schaefer and coauthors further demonstrated Braess’ paradox in AC power-grid experiments and large-grid studies, emphasizing that line upgrades can increase loading elsewhere in the network [41].

For this thesis, the Braess literature gives a warning rather than a license to overclaim. A line reinforcement that increases modal frequency but reduces damping ratio is a Braess-like damping-margin reversal, not automatically a classical flow Braess paradox. A controller-condensed case in which the line move pulls a network-family pole toward a nearby control pole is a stronger mechanistic claim, but it still requires a registered NEP or full-model validation gate. This calibrated vocabulary is used in the planning chapters.

4.11 Computational Modeling and Reproducible Dynamics

Detailed simulation environments are essential for testing the theory. ANDES, for example, implements a hybrid symbolic-numeric framework for power-system DAE modeling and analysis, including eigenvalue analysis and time-domain simulation [42]. Such tools make it possible to build converter-rich cases, extract Jacobians, compare full and reduced spectra, and test whether a proposed self-energy approximation is valid.

The next modeling frontier is to identify the port self-energy when the internal converter model is unavailable. Recent gray-box converter identification work estimates voltage-source converter parameters from terminal time-series data and validates the resulting model with frequency-

domain metrics [43]. Structure-preserving identification of port-Hamiltonian DAEs points in a complementary direction: learn from noisy input-output data while preserving interconnection and passivity properties [44]. These lines support the thesis requirement that an identified $\hat{\Pi}(s)$ must carry an uncertainty bound, not just a fitted curve.

This dissertation, however, intentionally separates theory from validation. The present manuscript develops the operator framework. A later empirical chapter or companion article may use ANDES or another DAE platform to test the framework on specific systems. That separation is methodological: without a clear theoretical object, simulation can produce plausible plots without explaining which damping mechanism has been preserved.

4.12 Theoretical Gap

The reviewed literatures leave a precise gap. Classical small-signal stability provides modal damping ratios and participation factors. Spectral graph theory provides an interpretable stiffness skeleton. QEP theory explains second-order fixed-coefficient dynamics. NEP theory explains frequency-dependent operators created by condensation. Converter-dominated stability shows why controller dynamics matter. What is missing is a unified damping-operator hierarchy that tells the engineer which level is mathematically sufficient for a given claim.

The gap can be stated as four questions.

1. When does a graph mode carry a true damping certificate rather than only a frequency screen?
2. When does fixed non-commuting damping create enough modal coupling to invalidate decoupled modal reasoning?
3. When do controller states require a rational self-energy $\Pi(s)$, and when can its damping-channel projection $\Sigma(s)$ replace a constant reduced damping matrix?
4. When does a dynamic Green function or structured fragility radius change the decision that would be made from a scalar graph or damping metric?

The remainder of the dissertation answers these questions by developing the hierarchy from uniform damping $D = \gamma M$, to the broader commuting constant- D class, to fixed non-commuting damping, to rational self-energy $\Pi(s)$, to Green-function action calculus, to structured fragility, and finally to nonlinear memory.

Part II

Operational Graph Dynamics

Operational Graph Dynamics and Graph-Time Resonance

The graph should not appear only as a static stiffness screen. In a converter-rich power system, the graph is the spatial support on which power flow, inertia, damping, controller memory, resonance, fragility, and design act together. This chapter builds the missing dynamic graph layer. Its central operator is

$$T(s; z^*) = s^2 M + sD + BW(z^*)B^T + \Pi(s; z^*), \quad (5.1)$$

where B is an oriented incidence matrix, $W(z^*)$ contains incremental line stiffnesses at the AC equilibrium z^* , M is a nodal inertia measure, D is a nodal or device-level dissipative field, and $\Pi(s; z^*)$ is the condensed controller self-energy. This turns the thesis into a theory of damped, controlled waves on electrical graphs.

Graph-dynamic thesis

Dynamic stability is a joint property of topology, operating point, spatial inertia and damping fields, and controller memory. A resonance occurs when temporal proximity, spatial alignment, port visibility, low damping, and resolvent amplification coincide.

5.1 Power-Flow Geometry of the Operational Graph

Consider a connected lossless graph with n buses, m lines, and an arbitrarily oriented incidence matrix $B \in \mathbb{R}^{n \times m}$. Let

$$\eta = B^T \theta \quad (5.2)$$

be the vector of line angle differences. For line $e = (i, j)$, define

$$\kappa_e = b_e V_i V_j > 0. \quad (5.3)$$

The oriented active line flow is

$$f_e(\theta, V) = \kappa_e \sin \eta_e, \quad (5.4)$$

and the nodal active-power injection map is

$$p(\theta, V) = B \operatorname{diag}(\kappa) \sin(B^T \theta). \quad (5.5)$$

At an equilibrium $z^* = (\theta^*, V^*, \dots)$,

$$f_e^* = \kappa_e^* \sin \eta_e^*, \quad w_e^* = \kappa_e^* \cos \eta_e^*. \quad (5.6)$$

Thus the operating-point stiffness is

$$L^* = BW^*B^T, \quad W^* = \text{diag}(w_e^*). \quad (5.7)$$

If $|\eta_e^*| < \pi/2$ for all lines, then $W^* \succ 0$. For a connected graph, $L^* \succeq 0$ and $\ker L^* = \text{span}\{\mathbf{1}\}$. The uniform angle direction is removed by projection onto $\mathbf{1}^\perp$.

The flow-stiffness pair in (5.6) is the first dynamic graph fact:

$$\frac{\partial w_e}{\partial \eta_e} = -\kappa_e \sin \eta_e = -f_e. \quad (5.8)$$

A heavily loaded line can lose synchronizing stiffness rapidly under an additional angular perturbation. This is the direct bridge from power flow to modal fragility.

5.2 Total Derivative of the Operational Laplacian

Let ρ be a planning parameter that may change injections, susceptances, voltages, topology, or the operating point. The reduced equilibrium equation is

$$F(\theta, \rho) = B \text{diag}(\kappa(\rho)) \sin(B^T \theta) - p(\rho) = 0. \quad (5.9)$$

Assume the reduced Laplacian L is invertible on $\mathbf{1}^\perp$. Then

$$\theta_\rho = L^\dagger [p_\rho - B \text{diag}(\sin \eta) \kappa_\rho], \quad \eta_\rho = B^T \theta_\rho. \quad (5.10)$$

Since $w = \kappa \odot \cos \eta$,

$$w_\rho = \cos \eta \odot \kappa_\rho - f \odot \eta_\rho, \quad (5.11)$$

and

$$L_\rho = B \text{diag}(w_\rho) B^T. \quad (5.12)$$

Equation (5.11) separates the direct line-parameter channel from the operating-point channel:

$$L_\rho = B \text{diag}(\cos \eta \odot \kappa_\rho) B^T - B \text{diag}(f \odot \eta_\rho) B^T. \quad (5.13)$$

The second term is often the one missed by frozen-graph screens. A physical line change modifies flows and angles; those changes modify many incremental line stiffnesses, not only the line being touched.

5.3 Mode Sensitivity to Power-Flow Geometry

For generalized graph modes

$$L\phi_k = \nu_k M\phi_k, \quad \phi_k^T M\phi_\ell = \delta_{k\ell}, \quad (5.14)$$

a simple eigenvalue satisfies

$$(\nu_k)_\rho = \phi_k^T (L_\rho - \nu_k M_\rho) \phi_k. \quad (5.15)$$

Let a_e be the incidence vector of line e , and define the modal line deformation

$$g_{e,k} = a_e^T \phi_k. \quad (5.16)$$

Then

$$(\nu_k)_\rho = \sum_{e=1}^m (w_e)_\rho g_{e,k}^2 - \nu_k \sum_{i=1}^n (M_i)_\rho \phi_k(i)^2. \quad (5.17)$$

If only injections change while κ and M are fixed, then $\theta_\rho = L^\dagger p_\rho$, and

$$(\nu_k)_\rho = - \sum_e f_e g_{e,k}^2 a_e^T L^\dagger p_\rho. \quad (5.18)$$

Equivalently,

$$\nabla_p \nu_k = -L^\dagger B \operatorname{diag}(f) c_k, \quad c_k = (g_{1,k}^2, \dots, g_{m,k}^2)^T. \quad (5.19)$$

The flow-weighted modal centrality

$$\chi_{e,k}^{\text{flow}} = |f_e| |g_{e,k}|^2 \quad (5.20)$$

flags lines that simultaneously carry high flow and sit across a large jump of the critical mode. This is more physical than degree, betweenness, or $\partial\lambda_2/\partial b_e$ alone.

5.4 Cycle and Cut Spaces

The DC flow induced by $p \in \mathbf{1}^\perp$ is

$$f^\star = WB^T L^\dagger p. \quad (5.21)$$

It is the unique solution of

$$\min_f \frac{1}{2} f^T W^{-1} f \quad \text{subject to} \quad Bf = p. \quad (5.22)$$

The Lagrange multiplier proof is immediate: stationarity gives $f = WB^T \lambda$, and the constraint gives $L\lambda = p$.

If C spans the cycle space, $BC = 0$, every feasible flow can be written as $f = f_0 + Cq$. Changing a line weight changes the metric W^{-1} , modifies cycle redistribution, changes the operating point, and then changes L . This is the formal bridge between topology control, Braess-like effects, and modal dynamics.

For a cut $S \subset \mathcal{V}$, define

$$m_S = \sum_{i \in S} M_i, \quad m_{S^c} = \sum_{i \notin S} M_i, \quad w(\partial S) = \sum_{e \in \partial S} w_e. \quad (5.23)$$

The first nonzero generalized eigenvalue satisfies

$$\nu_2 \leq w(\partial S) \left(\frac{1}{m_S} + \frac{1}{m_{S^c}} \right) \quad (5.24)$$

for every nontrivial cut. The proof is the Rayleigh quotient evaluated on the two-area vector that is constant on S and S^c and has zero center-of-inertia mean.

5.5 Graph Roughness of Damping-to-Inertia Fields

Assume $M = \text{diag}(M_i)$ and $D = \text{diag}(D_i)$, and define the nodal field

$$\gamma_i = \frac{D_i}{M_i}. \quad (5.25)$$

Then

$$\tilde{D} = M^{-1/2} D M^{-1/2} = \text{diag}(\gamma_i). \quad (5.26)$$

For $i \neq j$,

$$[\tilde{L}, \tilde{D}]_{ij} = \tilde{L}_{ij}(\gamma_j - \gamma_i). \quad (5.27)$$

Therefore

$$\|[\tilde{L}, \tilde{D}]\|_F^2 = 2 \sum_{e=(i,j)} \frac{w_e^2}{M_i M_j} (\gamma_i - \gamma_j)^2. \quad (5.28)$$

This is a Dirichlet energy over an auxiliary graph with weights $\hat{w}_e = w_e^2/(M_i M_j)$. Within the diagonal nodal damping class, and for a connected graph,

$$[\tilde{L}, \tilde{D}] = 0 \iff \gamma_i = \gamma \text{ for all } i \iff D = \gamma M. \quad (5.29)$$

Thus non-proportionality is spatial roughness of the damping-to-inertia field.

Let $\tilde{L}Q = Q\Lambda$ and $\Gamma = Q^T \tilde{D}Q$. Since

$$Q^T [\tilde{L}, \tilde{D}] Q = [\Lambda, \Gamma], \quad (5.30)$$

for $k \neq \ell$,

$$(\nu_k - \nu_\ell) \Gamma_{k\ell} = q_k^T [\tilde{L}, \tilde{D}] q_\ell. \quad (5.31)$$

Therefore

$$|\Gamma_{k\ell}| \leq \frac{\|[\tilde{L}, \tilde{D}]\|_2}{|\nu_k - \nu_\ell|} \leq \frac{\left(2 \sum_{e=(i,j)} \frac{w_e^2}{M_i M_j} (\gamma_i - \gamma_j)^2\right)^{1/2}}{|\nu_k - \nu_\ell|}. \quad (5.32)$$

Large graph roughness and small spectral separation are the two ingredients of strong modal mixing. The design reading is direct: smooth the field D_i/M_i , avoid abrupt local damping-to-inertia jumps across high-stiffness edges, or escalate from a graph-mode damping screen to the full QEP whenever (5.32) is not small in the protected mode band.

5.6 Dynamic Roughness of Local Control Fields

If the damping-channel self-energy is approximately local,

$$\Sigma(s) = \text{diag}(\sigma_i(s)), \quad \gamma_i(s) = \frac{\sigma_i(s)}{M_i}, \quad (5.33)$$

then, frequency by frequency,

$$\|[\tilde{L}, \tilde{\Sigma}(s)]\|_F^2 = 2 \sum_{e=(i,j)} \frac{w_e^2}{M_i M_j} |\gamma_i(s) - \gamma_j(s)|^2. \quad (5.34)$$

The band function

$$\mathcal{E}_\gamma(\omega) = \sum_{e=(i,j)} \frac{w_e^2}{M_i M_j} |\gamma_i(j\omega) - \gamma_j(j\omega)|^2 \quad (5.35)$$

identifies frequencies where the spatial distribution of control is incompatible with the electrical topology. If $\Pi(s)$ is dense, it should be decomposed into edge-like, shunt-like, and irreducibly nonlocal parts:

$$\Pi(j\omega) = BK_\Pi(j\omega)B^T + \text{diag}(k_\Pi(j\omega)) + \Pi_{\text{nonlocal}}(j\omega). \quad (5.36)$$

This is the dynamic extension of the line-shunt decomposition used later for static perturbations.

5.7 Joint Graph-Time Frequency Response

Normalize the retained operator:

$$\tilde{T}(s) = M^{-1/2}T(s)M^{-1/2}. \quad (5.37)$$

Let $\tilde{L} = Q\Lambda Q^T$. In graph-frequency coordinates,

$$T_G(s) = Q^T \tilde{T}(s)Q = s^2I + s\Gamma + \Lambda + \Pi_G(s), \quad (5.38)$$

where $\Pi_G(s) = Q^T M^{-1/2}\Pi(s)M^{-1/2}Q$. The graph-time transfer function is

$$H_G(\omega) = T_G(j\omega)^{-1}. \quad (5.39)$$

The index k is a spatial graph frequency, while ω is temporal frequency. $H_{G,kk}(\omega)$ measures amplification of spatial mode k , and $H_{G,k\ell}(\omega)$ measures conversion from spatial mode ℓ to spatial mode k .

Let

$$d_k(\omega) = \nu_k - \omega^2 + j\omega\Gamma_{kk} + [\Pi_G(j\omega)]_{kk}, \quad (5.40)$$

and write

$$T_G(j\omega) = D_0(\omega) + E(\omega), \quad D_0(\omega) = \text{diag}(d_1, \dots, d_n). \quad (5.41)$$

If $\alpha(\omega) = \|D_0(\omega)^{-1}E(\omega)\|_2 < 1$, then

$$\|H_G(\omega) - D_0(\omega)^{-1}\|_2 \leq \frac{\|D_0^{-1}\|_2^2 \|E\|_2}{1 - \alpha(\omega)}. \quad (5.42)$$

To first order,

$$[H_G(\omega)]_{k\ell} \approx -\frac{E_{k\ell}(\omega)}{d_k(\omega)d_\ell(\omega)}, \quad k \neq \ell. \quad (5.43)$$

A cross-modal resonance therefore requires temporal amplification in both modes and a strong off-diagonal channel from damping, self-energy, nonnormal AC stiffness, or nonlocal control.

5.8 Resonance Taxonomy

The graph-time response separates six mechanisms:

1. **Natural resonance:** $\det T(s) = 0$.
2. **Forced resonance:** $j\Omega$ lies near a visible pole and the input shape has strong projection on the corresponding left eigenvector.
3. **Cross-modal graph resonance:** $|H_{G,k\ell}(\omega)|$ is large for $k \neq \ell$.
4. **Network-control resonance:** a pole of $\Pi(s)$ is close to a network-family pole and has a visible residue.
5. **Parametric topology resonance:** a time-varying line weight modulates $L(t) = L_0 + \varepsilon L_1 \cos \Omega t$.
6. **Internal nonlinear resonance:** finite-amplitude dynamics satisfy $m\omega_k \approx n\omega_\ell$ and require normal forms, continuation, or harmonic balance.

For parametric line modulation,

$$L_1 = B\widehat{W}B^T, \quad c_{k\ell} = \phi_k^T L_1 \phi_\ell = \sum_e \widehat{w}_e (a_e^T \phi_k)(a_e^T \phi_\ell). \quad (5.44)$$

The coefficient $c_{k\ell}$ is the exact spatial bridge from a modulated edge to the two interacting graph modes.

5.9 Dynamic Effective Resistance and Low-Rank Updates

Let

$$G(s) = T(s)^{-1} \quad (5.45)$$

on the projected nonzero-angle subspace. For line e , define the dynamic effective resistance

$$\mathcal{R}_e(s) = a_e^T G(s) a_e. \quad (5.46)$$

If $s = 0$, $\Pi(0) = 0$, and $T(0) = L$, this reduces to the static effective resistance $a_e^T L^\dagger a_e$. For $s \neq 0$, it contains topology, operating point, inertia, damping, controller poles, memory, and resolvent amplification.

For a frozen operating point and a rank-one stiffness update,

$$T_\Delta(s) = T(s) + \Delta w_e a_e a_e^T, \quad (5.47)$$

the determinant lemma gives

$$\det T_\Delta(s) = \det T(s) [1 + \Delta w_e \mathcal{R}_e(s)]. \quad (5.48)$$

New or shifted poles caused by this finite frozen update satisfy

$$1 + \Delta w_e \mathcal{R}_e(s) = 0. \quad (5.49)$$

The corresponding Green function update is

$$G_\Delta(s) = G(s) - \frac{\Delta w_e G(s) a_e a_e^T G(s)}{1 + \Delta w_e \mathcal{R}_e(s)}. \quad (5.50)$$

For multiple candidate lines with incidence matrix A_C ,

$$K_C(s) = A_C^T G(s) A_C, \quad (5.51)$$

and

$$\det T_\Delta(s) = \det T(s) \det[I + \Delta W K_C(s)]. \quad (5.52)$$

This reduces a large spectral update to a matrix whose dimension is the number of candidate actions. It is exact for frozen low-rank stiffness updates and a predictor for physical line changes that also move the operating point.

5.10 Dynamic Edge Centrality

For an input shape b , an output shape c , and

$$h(s) = c^T G(s) b, \quad (5.53)$$

the frozen line-weight derivative is

$$\frac{\partial h(s)}{\partial w_e} = - \left[c^T G(s) a_e \right] \left[a_e^T G(s) b \right]. \quad (5.54)$$

Thus

$$\mathcal{C}_e(\omega; b, c) = |c^T G(j\omega) a_e| |a_e^T G(j\omega) b| \quad (5.55)$$

is a frequency-specific dynamic centrality. It identifies a line that is reachable from the disturbance, transmits energy toward the protected output, and matters in the resonant band. It need not have high static degree, betweenness, or Fiedler sensitivity.

5.11 Structured Fragility Radii

There is no single maximum perturbation. At least four different radii are needed.

For topology or parameter changes, define a protected spectral region $\mathcal{S}$. With candidate line matrix A_C , the structured topological radius is

$$r_{\text{top}} = \inf \left\{ \|\Delta W\| : \exists s \in \partial \mathcal{S}, \det[T(s) + A_C \Delta W A_C^T] = 0 \right\}. \quad (5.56)$$

Using (5.52), the reduced condition is

$$\det[I + \Delta W K_C(s)] = 0. \quad (5.57)$$

With a declared physical uncertainty structure, this becomes the structured radius

$$r_{\text{top}} = \left[\sup_{s \in \partial \mathcal{S}} \mu_\Delta(K_C(s)) \right]^{-1}. \quad (5.58)$$

For a single frozen line update, the complex unstructured line radius along a protected boundary is especially transparent:

$$r_e^{\text{C}} = \left[\sup_{s \in \partial \mathcal{S}} |\mathcal{R}_e(s)| \right]^{-1}. \quad (5.59)$$

This is the smallest complex scalar Δw_e that can satisfy $1 + \Delta w_e \mathcal{R}_e(s) = 0$ for some boundary point. A physical line weight is real, so the real frozen-line radius is

$$r_e^{\mathbb{R}} = \min_{\substack{s \in \partial \mathcal{S} \\ \text{Im } \mathcal{R}_e(s) = 0}} \frac{1}{|\text{Re } \mathcal{R}_e(s)|}, \quad (5.60)$$

with the convention $r_e^{\mathbb{R}} = \infty$ if the feasibility set is empty. Thus dynamic effective resistance is not only a ranking score; it gives an exact single-line fragility radius for the declared spectral boundary. For a forced input in a stable linear model, increasing amplitude does not move poles. It can still violate an operational limit:

$$|\hat{u}|_{\max}^{\Omega} = \frac{y_{\max}}{\sup_{\omega \in \Omega} \sigma_{\max}(C_y G(j\omega) B_u)}. \quad (5.61)$$

For nonlinear loss of stability, the relevant object is a region-of-attraction radius,

$$r_{\text{ROA}} = \inf_{x \in \partial \mathcal{B}(x^*)} \|W_x(x - x^*)\|. \quad (5.62)$$

The first is a parametric fragility radius, the second is an input-output exposure bound, and the third is a nonlinear stability quantity.

5.12 Stability Recovery Design

The graph-dynamic design problem combines line, damping, inertia, and controller actions:

$$u = [\Delta w \quad \Delta D \quad \Delta M \quad \Delta \kappa]^T. \quad (5.63)$$

A unified recovery problem is

$$\begin{aligned} \min_u \quad & c_w^T |\Delta w| + c_D^T |\Delta D| + c_M^T |\Delta M| + c_{\kappa}^T |\Delta \kappa| \\ \text{subject to} \quad & F(z^*, u) = 0, \\ & |f_e(z^*, u)| \leq \bar{f}_e, \\ & |B^T \theta^*| \leq \eta_{\max}, \\ & \text{Re } s_k(u) \leq -\alpha_{\star}, \quad \zeta_k(u) \geq \zeta_{\star}, \\ & \sup_{\omega \in \Omega} \sigma_{\max}(C_y T(j\omega, u)^{-1} B_u) \leq g_{\star}, \\ & r_{\text{top}}(u) \geq r_{\star}. \end{aligned} \quad (5.64)$$

Sequential versions should use pole sensitivities and resolvent-gradient steps only inside a trust region, followed by continuation of the operating point and the protected poles.

5.13 AC, Directed, and Matrix-Weighted Extensions

The scalar lossless Laplacian is the symmetric floor, not the universal model. For the AC Jacobian,

$$\begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix} = \begin{bmatrix} J_{P\theta} & J_{PV} \\ J_{Q\theta} & J_{QV} \end{bmatrix} \begin{bmatrix} \Delta \theta \\ \Delta V \end{bmatrix}, \quad (5.65)$$

eliminating voltage increments under $\Delta Q = 0$ gives

$$L_{AC} = J_{P\theta} - J_{PV} J_{QV}^{-1} J_{Q\theta}. \quad (5.66)$$

In general $L_{AC} \neq L_{AC}^T$. Its symmetric part is stiffness-like; its skew part is circulatory and nonconservative. This nonnormal layer requires left-right eigenvectors and resolvent gains.

For multivariable ports, assign each bus an r -dimensional vector and set $\mathcal{B} = B \otimes I_r$. With line admittances $Y_e(s) \in \mathbb{C}^{r \times r}$,

$$\mathcal{L}(s) = \mathcal{B} \text{blkdiag}(Y_1(s), \dots, Y_m(s)) \mathcal{B}^T. \quad (5.67)$$

For a matrix update on line e , with $\mathcal{A}_e = a_e \otimes I_r$,

$$\mathcal{R}_e^{(r)}(s) = \mathcal{A}_e^T \mathcal{T}(s)^{-1} \mathcal{A}_e, \quad (5.68)$$

and

$$\det \mathcal{T}_\Delta(s) = \det \mathcal{T}(s) \det \left[I_r + \Delta Y_e(s) \mathcal{R}_e^{(r)}(s) \right]. \quad (5.69)$$

This is the natural extension to dq coupling, voltage-frequency channels, frequency-dependent line models, directed effects, and multivariable converter ports.

5.14 Paper Extraction

This chapter supports a focused Transactions-style paper:

Graph-Time Resonance, Spectral Effective Resistance, and Structured Topology Fragility in Converter-Rich Power Systems.

The broad dissertation contribution is operator-valued dynamic grid strength. The narrower article should be sold as controller-resolvent-aware spectral graph planning. Its minimum publishable package is:

1. graph roughness of D_i/M_i and the modal-mixing bound (5.32);
2. graph-time transfer $H_G(\omega)$ and cross-resonance atlas;
3. finite dynamic-effective-resistance updates $1 + \Delta w_e \mathcal{R}_e(s) = 0$;
4. structured topology-fragility radii (5.59)–(5.60);
5. a controller-induced reinforcement-reversal gate, using the Schur self-energy machinery developed later in the thesis.

The novelty claim should not be that Laplacians or effective resistance are new, that Schur complements are new, or that adding a line can never hurt in existing literature. The defensible claim is narrower and stronger: power-flow-induced stiffness, damping-to-inertia roughness, graph-time modal conversion, finite low-rank topology updates, rational controller self-energy, and structured stability radii are projections of one operator $T(s; z^*)$, and that operator determines when a topological improvement really improves a protected dynamic margin.

Part III

Constant Damping Theory

Fixed-Damping Spectral Theory

The first level of the hierarchy assumes that damping is a fixed matrix. This is the last level at which one can still try to extract a large amount of engineering information from the graph Laplacian L , the inertia matrix M , and the damping matrix D without retaining converter-control states. The chapter develops that analysis aggressively. It derives the spectral screens, the exact first-order eigenvalue sensitivities, the decoupled modal formulas for d/dL , d/dD , and d/dM , and the limits of graph-only reinforcement.

The conclusion is deliberately sharp. With fixed D , adding inertia is not automatically stabilizing for oscillatory damping margins. Adding a line is not automatically stabilizing either. Damping ratio is a quotient: it improves when modal damping rises faster than modal frequency. If an action raises stiffness or inertia but does not raise effective modal damping, the damping ratio can fall even while a graph metric improves.

6.1 The Classical Swing Operator

Consider

$$M\ddot{\theta} + D\dot{\theta} + L\theta = p, \quad (6.1)$$

with $M \succ 0$, $D \succeq 0$, and $L = L^T \succeq 0$ on the projected subspace orthogonal to uniform angle shift. The homogeneous modal ansatz $\theta(t) = ve^{st}$ gives the QEP

$$P(s)v = (s^2M + sD + L)v = 0. \quad (6.2)$$

This is the basic fixed- D operator. All planning screens in this chapter are approximations or consequences of (6.2).

In inertia-normalized coordinates,

$$\ddot{\vartheta} + \tilde{D}\dot{\vartheta} + \tilde{L}\vartheta = 0, \quad (6.3)$$

where

$$\tilde{L} = M^{-1/2}LM^{-1/2}, \quad \tilde{D} = M^{-1/2}DM^{-1/2}. \quad (6.4)$$

Let $\tilde{L}Q = Q\Lambda$. In modal coordinates $z = Q^T\vartheta$,

$$\ddot{z} + \Gamma\dot{z} + \Lambda z = 0, \quad \Gamma = Q^T\tilde{D}Q. \quad (6.5)$$

The graph diagonalizes stiffness. It diagonalizes damping only when

$$\tilde{L}\tilde{D} = \tilde{D}\tilde{L} \quad (6.6)$$

on the nonzero modal subspace. Equation (6.6) is the first hidden assumption behind graph-only damping arguments.

A basis-dependent off-diagonal ratio is useful only when the eigenvalues of $\tilde{L}$ are well separated. For clustered eigenvalues, the basis inside the cluster can rotate without changing the physics. A coordinate-free global gate is

$$\eta_{\text{comm}} = \frac{\|\tilde{L}\tilde{D} - \tilde{D}\tilde{L}\|_F}{2\|\tilde{L}\|_F\|\tilde{D}\|_F + \varepsilon_{\text{reg}}}. \quad (6.7)$$

For two spectral clusters I and J , with orthogonal projectors P_I and P_J , the cross-cluster damping gate is

$$\eta_{IJ} = \frac{\|P_I\tilde{D}P_J\|_2}{\text{sep}(\Lambda_I, \Lambda_J) + \varepsilon_{\text{reg}}}. \quad (6.8)$$

This form is invariant to rotations of the modal basis inside each cluster and is the safer trigger for adaptive reduced-QEP construction.

6.2 Exact QEP Energy Identities

The scalar decoupled formulas used later as screens have an exact ancestor in the full QEP. Let $P(s)x = 0$ for

$$P(s) = s^2M + sD + L, \quad M \succ 0, \quad D \succeq 0, \quad L \succ 0$$

on the nonzero-angle subspace. Assume $s = \sigma + j\omega$ with $\omega \neq 0$. Multiplying by x^H gives

$$s^2m + sd + \ell = 0, \quad m = x^H Mx, \quad d = x^H Dx, \quad \ell = x^H Lx. \quad (6.9)$$

Since M, D, L are Hermitian in this fixed- D layer, m, d, ℓ are real and $m > 0$. Separating real and imaginary parts yields

$$\sigma = -\frac{x^H Dx}{2x^H Mx}, \quad (6.10)$$

$$|s|^2 = \frac{x^H Lx}{x^H Mx}, \quad (6.11)$$

$$\zeta = \frac{x^H Dx}{2\sqrt{(x^H Mx)(x^H Lx)}}. \quad (6.12)$$

These identities do not require $D = \gamma M$. They require the fixed Hermitian QEP structure and an oscillatory root.

Proposition 6.1 (Absence of undamped oscillatory modes). *For the fixed Hermitian QEP, an oscillatory eigenvalue on the imaginary axis $s = j\omega$, $\omega \neq 0$, exists only if there is a nonzero vector x such that*

$$Dx = 0, \quad Lx = \omega^2 Mx. \quad (6.13)$$

Proof. If $s = j\omega$, then (6.10) gives $x^H Dx = 0$. Since $D \succeq 0$, this implies $Dx = 0$. The QEP then reduces to $(-\omega^2 M + L)x = 0$. The converse follows by substitution. $\square$

6.3 Generalized Graph Modes

For placement calculations it is often cleaner to use the generalized eigenproblem

$$L\phi_k = \nu_k M\phi_k, \quad \phi_k^T M\phi_\ell = \delta_{k\ell}. \quad (6.14)$$

The zero mode is removed. The scalar modal stiffness is ν_k , and the frozen-mode scalar damping is

$$\delta_k = \phi_k^T D\phi_k. \quad (6.15)$$

If the modes are decoupled, mode k is governed by

$$s^2 + \delta_k s + \nu_k = 0. \quad (6.16)$$

The damping ratio is

$$\zeta_k = \frac{\delta_k}{2\sqrt{\nu_k}}. \quad (6.17)$$

This formula is a screen outside the decoupled case. It is still extremely useful because it exposes the competing effects:

$$\frac{d\zeta_k}{\zeta_k} = \frac{d\delta_k}{\delta_k} - \frac{1}{2} \frac{d\nu_k}{\nu_k}. \quad (6.18)$$

An action improves damping ratio only if it raises modal damping faster than half the relative increase in modal stiffness.

6.4 Uniform-Decay Case $D = \gamma M$

The most restrictive decoupled case is

$$D = \gamma M, \quad \gamma > 0. \quad (6.19)$$

Then $\delta_k = \gamma$, $\Gamma = \gamma I$, and every nonzero graph mode satisfies

$$s^2 + \gamma s + \nu_k = 0. \quad (6.20)$$

The roots are

$$s_{k,\pm} = \frac{-\gamma \pm \sqrt{\gamma^2 - 4\nu_k}}{2}. \quad (6.21)$$

When $\nu_k > \gamma^2/4$, the roots are underdamped and

$$\text{Re}(s_{k,\pm}) = -\gamma/2. \quad (6.22)$$

Proposition 6.2 (Proportional-damping floor). *Under $D = \gamma M$, every underdamped nonzero mode has the same decay rate $-\gamma/2$. Network topology changes ν_k and therefore modal frequency, but it does not change the real decay rate of an underdamped mode inside the proportional class.*

Proof. Equation (6.20) gives the two roots directly. If $\nu_k > \gamma^2/4$, the square root is imaginary and the real part is $-\gamma/2$. The expression contains no dependence on ν_k in its real part. $\square$

The proposition is sometimes counterintuitive because it does not say topology is irrelevant. Topology can move a mode between overdamped and underdamped regions, and it changes frequency. It states only that, after a mode is underdamped, its decay rate is fixed by γ in this uniform damping-to-inertia model.

6.5 Commuting and Generalized Proportional Damping

Uniform damping is not the whole decouplable class. The exact condition is

$$[\tilde{L}, \tilde{D}] = 0. \quad (6.23)$$

For real symmetric matrices, (6.23) is equivalent to simultaneous orthogonal diagonalization. Therefore there exists an orthogonal Q such that

$$Q^T \tilde{L} Q = \text{diag}(\nu_k), \quad Q^T \tilde{D} Q = \text{diag}(\delta_k). \quad (6.24)$$

Each mode then satisfies

$$s_k^2 + \delta_k s_k + \nu_k = 0, \quad \zeta_k = \frac{\delta_k}{2\sqrt{\nu_k}}. \quad (6.25)$$

Rayleigh damping is the most important generalized proportional example:

$$D = \alpha M + \beta L. \quad (6.26)$$

In normalized coordinates,

$$\tilde{D} = \alpha I + \beta \tilde{L}, \quad (6.27)$$

so $[\tilde{L}, \tilde{D}] = 0$. Mode k satisfies

$$s_k^2 + (\alpha + \beta \nu_k) s_k + \nu_k = 0, \quad (6.28)$$

and its underdamped real part and damping ratio are

$$\text{Re}(s_k) = -\frac{\alpha + \beta \nu_k}{2}, \quad (6.29)$$

$$\zeta_k = \frac{\alpha}{2\sqrt{\nu_k}} + \frac{\beta \sqrt{\nu_k}}{2}. \quad (6.30)$$

Unlike the $D = \gamma M$ floor, Rayleigh damping lets network stiffness change both frequency and decay. Depending on the modal range and on the signs and magnitudes of α and β , reinforcement can improve one set of modes while degrading another damping-ratio target. This is why the thesis uses the commuting condition as the decoupling test and reserves $D = \gamma M$ for the uniform-decay case.

6.6 Exact QEP Sensitivities

Let a scalar action ρ perturb all three operators:

$$P(s, \rho) = s^2 M(\rho) + s D(\rho) + L(\rho). \quad (6.31)$$

For a simple eigenvalue s_* , right vector v , and left vector w , the QEP sensitivity is

$$\frac{ds_*}{d\rho} = -\frac{w^H (s_*^2 M_\rho + s_* D_\rho + L_\rho) v}{w^H (2s_* M + D) v}. \quad (6.32)$$

This formula, standard in QEP perturbation theory [5], is the most compact answer to how d/dL , d/dD , and d/dM mix.

The three numerator terms have different phase factors:

$$L_\rho, \quad s_* D_\rho, \quad s_*^2 M_\rho. \quad (6.33)$$

For an oscillatory pole, those terms are not collinear in the complex plane. Thus, line reinforcement, damping addition, and inertia addition do not point in the same eigenvalue direction. A scalar ranking that ignores these phase weights is not a damping certificate.

For $s = \sigma + j\omega$, the damping-ratio derivative is

$$\frac{d\zeta}{d\rho} = -\frac{\omega^2}{(\sigma^2 + \omega^2)^{3/2}} \operatorname{Re} \left(\frac{ds}{d\rho} \right) + \frac{\sigma\omega}{(\sigma^2 + \omega^2)^{3/2}} \operatorname{Im} \left(\frac{ds}{d\rho} \right). \quad (6.34)$$

The real-part movement alone is insufficient when the frequency movement is large.

6.7 Derivative with Respect to the Graph L

Let a candidate line $e = (i, j)$ have incidence vector $a_e = e_i - e_j$ and susceptance-like weight b_e . In the Laplacian approximation,

$$L = \sum_e b_e a_e a_e^T, \quad \frac{\partial L}{\partial b_e} = a_e a_e^T. \quad (6.35)$$

For the generalized graph mode (6.14),

$$\frac{\partial \nu_k}{\partial b_e} = (\phi_k(i) - \phi_k(j))^2. \quad (6.36)$$

Equation (6.36) is the core weak-link screen: edges across large modal angle differences have the strongest effect on modal frequency.

For the scalar decoupled damping-ratio screen with fixed D ,

$$\frac{\partial \zeta_k}{\partial b_e} = -\frac{\zeta_k}{2\nu_k} (\phi_k(i) - \phi_k(j))^2 + \frac{\zeta_k}{\delta_k} \frac{\partial \delta_k}{\partial b_e}. \quad (6.37)$$

If modal damping is frozen under the line action, the second term is zero and line reinforcement reduces ζ_k for that mode. It may still improve angle coherence, synchronizing torque, or protected resolvent gain, but it does not improve scalar damping ratio by itself.

Example 6.1 (A reinforced cut can hide a damping loss). *Suppose a critical mode has δ_k fixed and a line across its modal cut has $(\phi_k(i) - \phi_k(j))^2$ large. Reinforcing that line raises ν_k , which is good for graph stiffness. Yet (6.37) gives $\partial \zeta_k / \partial b_e < 0$ if $\partial \delta_k / \partial b_e = 0$. The graph has become stiffer, but the damping ratio has decreased. This is a hidden margin failure.*

6.8 Derivative with Respect to Damping D

For a local damping addition at bus i ,

$$D_\rho = e_i e_i^T. \quad (6.38)$$

The frozen-mode damping derivative is

$$\frac{\partial \delta_k}{\partial D_i} = \phi_k(i)^2. \quad (6.39)$$

Therefore

$$\frac{\partial \zeta_k}{\partial D_i} = \frac{\phi_k(i)^2}{2\sqrt{\nu_k}} \quad (6.40)$$

in the decoupled scalar screen. The best damping location for a single mode is an antinode of that mode, not necessarily the largest generator, the largest load, or the highest-degree bus.

For multiple modes, a risk-weighted damping placement score is

$$\mathcal{D}_i = \sum_{k \in \mathcal{K}} \omega_k \frac{\phi_k(i)^2}{2\sqrt{\nu_k}}, \quad (6.41)$$

where $\mathcal{K}$ is the set of monitored modes and $\omega_k \geq 0$ weights criticality. This is a screen. The exact acceptance test remains (6.32) or full eigenvalue recomputation.

Example 6.2 (A low-degree bus can be the best damping site). *A bus with modest graph degree can have large $\phi_k(i)^2$ for the critical mode. Equation (6.40) ranks it highly for damping placement even if graph centrality ranks it poorly. This is a weak-node mechanism: the bus is weak because the critical mode lives there.*

6.9 Derivative with Respect to Inertia M

For a local inertia addition at bus i ,

$$M_\rho = e_i e_i^T. \quad (6.42)$$

The generalized eigenvalue derivative is

$$\frac{\partial \nu_k}{\partial M_i} = -\nu_k \phi_k(i)^2. \quad (6.43)$$

Local inertia lowers the modal frequency most strongly where the mode has large amplitude. That fact supports inertia placement for frequency nadir or RoCoF objectives, and it is consistent with studies showing that virtual-inertia placement can be posed as a network performance problem [26], that inertia location and slow modes shape disturbance propagation [27], and that inertia and primary-control placement separate under heterogeneous parameters [28].

For oscillatory damping ratio, however, the conclusion is more delicate. The exact answer is still the QEP derivative (6.32). As a fixed-mode planning screen, if physical damping D is not co-added, the frozen-shape normalization gives

$$\frac{\partial \delta_k}{\partial M_i} \approx -\delta_k \phi_k(i)^2. \quad (6.44)$$

Combining (6.18), (6.43), and (6.44) yields

$$\frac{\partial \zeta_k}{\partial M_i} \approx -\frac{1}{2} \zeta_k \phi_k(i)^2. \quad (6.45)$$

Thus, in a fixed- D model, inertia-only placement at a modal antinode can decrease the scalar damping-ratio screen. It may still improve RoCoF or reduce frequency peaks, but it is not a free damping improvement. Any candidate that matters operationally should therefore be rechecked with (6.32) or with a full eigenvalue recomputation.

If inertia is co-designed with proportional damping,

$$D_i = \gamma M_i, \quad (6.46)$$

then the damping-to-inertia ratio is preserved. The real decay floor remains $-\gamma/2$, and the damping ratio increases as ν_k decreases. This is the engineering difference between adding inertia alone and adding a properly-damped inertial device.

Example 6.3 (Inertia placement has two different objectives). *Placing inertia where $\phi_k(i)^2$ is large is attractive if the objective is to slow a disturbance. Under fixed D , the same placement can reduce ζ_k through damping dilution. A correct planning statement must say whether the target is RoCoF, frequency nadir, decay rate, damping ratio, or input-output amplification.*

6.10 Graph Robustness Without Adding Inertia

If inertia is unavailable or undesirable, the graph can be strengthened through line weights, switching, topology control, or series compensation. The oldest spectral screen is algebraic connectivity: maximize ν_2 , the first nonzero Laplacian eigenvalue introduced by Fiedler's graph connectivity theory [7]. The local derivative is

$$\frac{\partial \nu_2}{\partial b_e} = (\phi_2(i) - \phi_2(j))^2. \quad (6.47)$$

This favors lines across the Fiedler cut.

A different robustness screen is total effective resistance, also called the Kirchhoff index:

$$K_f = n \operatorname{trace}(L^+) = n \sum_{k=2}^n \frac{1}{\lambda_k}. \quad (6.48)$$

Minimizing effective resistance is a convex edge-weight allocation problem in standard graph settings [8]. Its derivative is

$$\frac{\partial \operatorname{trace}(L^+)}{\partial b_e} = -a_e^T (L^+)^2 a_e = - \sum_{k=2}^n \frac{(q_k(i) - q_k(j))^2}{\lambda_k^2}. \quad (6.49)$$

This favors lines that reduce many slow modes, not only the Fiedler mode. Effective resistance also underlies spectral sparsification, where edges are sampled or weighted according to their resistance contribution while preserving quadratic forms [9].

The hidden damping question remains:

$$\text{Does the graph action also improve } \zeta_k, \operatorname{Re}(s_k), \text{ or } g_{\text{prot}}? \quad (6.50)$$

If not, the action is graph-robust but damping-ambiguous. It must be paired with damping, controller retuning, or a certificate that the chosen margin is not damping ratio.

6.11 Weak Nodes, Weak Links, and Hidden Margins

The fixed- D theory gives precise first-order definitions.

Definition 6.1 (Weak node for damping). *For a monitored mode set $\mathcal{K}$, bus i is weak for damping if its modal damping placement score $\mathcal{D}_i$ in (6.41) is large while the current local damping contribution $D_i\phi_k(i)^2$ is small for critical modes.*

Definition 6.2 (Weak link for graph stiffness). *Line e is weak for a graph metric if the derivative of that metric with respect to b_e is large. Examples are (6.47) for algebraic connectivity and (6.49) for total effective resistance.*

Definition 6.3 (Hidden damping margin). *A graph action has hidden damping risk for mode k if*

$$\frac{\partial \nu_k}{\partial \rho} > 0 \quad \text{but} \quad \frac{\partial \zeta_k}{\partial \rho} < 0 \quad (6.51)$$

under the selected damping model. It improves a stiffness screen while degrading a damping-ratio screen.

This definition prevents a common overclaim: a stronger graph is not the same as a better damping margin.

6.12 A Fixed- D Spectral Planning Screen

The following screen extracts the maximum information available before moving to the full QEP or Schur NEP.

1. Compute generalized modes $L\phi_k = \nu_k M\phi_k$.
2. Compute modal damping $\delta_k = \phi_k^T D\phi_k$ and $\zeta_k = \delta_k/(2\sqrt{\nu_k})$.
3. Compute node damping scores $\mathcal{D}_i$.
4. Compute inertia-dilution scores

$$\mathcal{M}_i = \sum_{k \in \mathcal{K}} \omega_k \zeta_k \phi_k(i)^2. \quad (6.52)$$

5. Compute line graph scores using $\partial \nu_2 / \partial b_e$ and $\partial \text{trace}(L^+) / \partial b_e$.
6. Flag hidden-margin conflicts where graph scores are positive but $\partial \zeta_k / \partial b_e < 0$.
7. Verify accepted candidates with the exact QEP derivative (6.32).

This screen is intentionally not the final thesis method. It is the strongest fixed- D layer before controller self-energy is introduced.

Fixed Non-Commuting Damping and Modal Coupling

The second level of the hierarchy keeps the damping matrix fixed but drops the commutation condition $[\tilde{L}, \tilde{D}] = 0$. The eigenvalue problem remains a QEP, but graph modes are no longer independent. This chapter develops the resulting modal coupling and shows that it already has the algebraic form of a rational correction. The term non-proportional is used only in this precise sense: damping is non-proportional for the present modal analysis when it is not simultaneously diagonalizable with stiffness in inertia-normalized coordinates.

7.1 Modal Damping Matrix

In graph modal coordinates,

$$\ddot{z} + \Gamma \dot{z} + \Lambda z = 0, \quad \Gamma = Q^T \tilde{D} Q. \quad (7.1)$$

Write

$$\Gamma = \Gamma_0 + E, \quad \Gamma_0 = \text{diag}(\Gamma_{11}, \dots, \Gamma_{nn}), \quad (7.2)$$

where E contains the off-diagonal entries. If $E = 0$, modes are decoupled. If $E \neq 0$, the velocity of one graph mode enters the equation of another. Equivalently, for a simple spectrum of $\tilde{L}$, nonzero off-diagonal entries of Γ are the modal expression of the commutator $[\tilde{L}, \tilde{D}]$.

The off-diagonal entries are not numerical noise. They are the representation of damping heterogeneity in the stiffness eigenbasis. A bus with large local damping can still be ineffective for a critical mode if that mode has small shape at the bus. Conversely, a small local damping change can matter if it projects strongly onto the left-right structure of a sensitive eigenpair.

7.2 Two-Mode Illustration

For two graph modes, the QEP is

$$\begin{bmatrix} s^2 + s\gamma_1 + \nu_1 & s\epsilon \\ s\epsilon & s^2 + s\gamma_2 + \nu_2 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = 0. \quad (7.3)$$

The characteristic equation is

$$(s^2 + s\gamma_1 + \nu_1)(s^2 + s\gamma_2 + \nu_2) - s^2\epsilon^2 = 0. \quad (7.4)$$

Eliminating z_2 yields an exact scalar nonlinear equation for z_1 :

$$s^2 + s\gamma_1 + \nu_1 - \frac{s^2 \epsilon^2}{s^2 + s\gamma_2 + \nu_2} = 0. \quad (7.5)$$

The correction is rational in s . Even before converter controls are introduced, eliminating coupled modal coordinates converts a fixed-coefficient QEP into a frequency-dependent scalar equation for the retained mode.

7.3 General Modal Correction

Partition the modal coordinates into one target mode k and the remaining modes $\bar{k}$. The modal QEP can be written as

$$\begin{bmatrix} p_k(s) & sE_{k\bar{k}} \\ sE_{\bar{k}k} & s^2I + s\Gamma_{\bar{k}\bar{k}} + \Lambda_{\bar{k}\bar{k}} \end{bmatrix} \begin{bmatrix} z_k \\ z_{\bar{k}} \end{bmatrix} = 0, \quad (7.6)$$

where

$$p_k(s) = s^2 + s\Gamma_{kk} + \nu_k. \quad (7.7)$$

Eliminating $z_{\bar{k}}$ gives

$$p_k(s) - s^2 E_{k\bar{k}} \left(s^2 I + s\Gamma_{\bar{k}\bar{k}} + \Lambda_{\bar{k}\bar{k}} \right)^{-1} E_{\bar{k}k} = 0. \quad (7.8)$$

This expression shows three facts.

1. Off-diagonal damping affects a mode through products of coupling terms, not through diagonal damping alone.
2. Nearby modal denominators matter more than distant denominators.
3. The correction has the same resolvent structure that later appears in controller condensation.

7.4 Perturbation View

For small E , one may evaluate the correction near an unperturbed root $s_{k,0}$ of $p_k(s)$. The leading eigenvalue shift is governed by

$$\Delta s_k \approx \frac{s_{k,0}^2 E_{k\bar{k}} \left(s_{k,0}^2 I + s_{k,0} \Gamma_{\bar{k}\bar{k}} + \Lambda_{\bar{k}\bar{k}} \right)^{-1} E_{\bar{k}k}}{2s_{k,0} + \Gamma_{kk}}. \quad (7.9)$$

The denominator $2s_{k,0} + \Gamma_{kk}$ is the derivative of the decoupled scalar polynomial. This is the scalar version of the general QEP sensitivity formula. The modal correction can move both real and imaginary parts, so its effect on damping ratio must be computed rather than inferred from graph frequency alone.

7.5 Non-Proportionality as a Gate

The previous equations suggest a theoretical gate before using graph metrics as damping screens. A fixed- D graph screen is plausible when

$$\|E\| \ll \|\Gamma_0\| \quad (7.10)$$

and when the resolvent denominators in (7.8) are not near resonance over the frequency band of interest. This does not prove that the screen is accurate; it states the conditions under which the modal coupling neglected by the screen is small.

Structural-dynamics non-proportionality indices formalize similar ideas: measure how far the damping operator is from the modal subspace in which the undamped system is diagonal [13]. The power-grid interpretation is more specific. The off-diagonal damping matrix identifies which graph modes exchange damping through heterogeneous device locations and damping-to-inertia ratios.

7.6 Connection to the Next Level

Fixed non-commuting damping is not the same as converter self-energy. In (7.8), the eliminated coordinates are other graph modes of the same fixed-QEP system. In controller self-energy, the eliminated coordinates are physical control and auxiliary states. The algebraic similarity is nevertheless important. It shows that rational dependence is not an exotic artifact; it is the natural result of eliminating coupled dynamic coordinates. The next chapter applies that logic to controller-state condensation.

Part IV

Rational Self-Energy

Rational Self-Energy and Schur Nonlinear Eigenproblems

The third level of the hierarchy begins when converter and auxiliary controller states are condensed. The reduced problem is no longer a fixed-coefficient QEP. It is a nonlinear eigenvalue problem whose retained correction is rational in the spectral parameter. This chapter derives that object and explains why $\Pi(s)$, not a static damping matrix, is the general theoretical replacement after controller or algebraic states are eliminated.

8.1 Full and Retained State Partitions

Let the linearized state matrix be partitioned as

$$A = \begin{bmatrix} A_{ee} & A_{ec} \\ A_{ce} & A_{cc} \end{bmatrix}. \quad (8.1)$$

The retained states x_e may include angles, speeds, or selected interface variables. The condensed states x_c may include exciters, governors, stabilizers, PLLs, voltage and current controllers, plant controllers, filters, and other auxiliary dynamics. The eigenvalue equation is

$$(sI - A_{ee})x_e - A_{ec}x_c = 0, \quad (8.2)$$

$$-A_{ce}x_e + (sI - A_{cc})x_c = 0. \quad (8.3)$$

When $sI - A_{cc}$ is invertible,

$$x_c = (sI - A_{cc})^{-1}A_{ce}x_e. \quad (8.4)$$

Substitution gives

$$S(s)x_e = 0, \quad S(s) = sI - A_{ee} - A_{ec}(sI - A_{cc})^{-1}A_{ce}. \quad (8.5)$$

Proposition 8.1 (Exact Schur equivalence). *If $sI - A_{cc}$ is invertible, then s is an eigenvalue of the full matrix A with retained component $x_e \neq 0$ if and only if $S(s)x_e = 0$.*

Proof. The proof follows by the two substitutions above. Starting from a full eigenpair gives $S(s)x_e = 0$. Conversely, if $S(s)x_e = 0$, define $x_c = (sI - A_{cc})^{-1}A_{ce}x_e$. Then the block eigenvalue equations are satisfied. $\square$ $\square$

The result is exact, but the retained operator is nonlinear in s . Any constant reduced matrix that tries to replace $A_{ec}(sI - A_{cc})^{-1}A_{ce}$ must be an approximation.

8.2 Descriptor Schur Complement

Power-system linearizations are often descriptor systems rather than explicit ODEs. The more general eigenvalue equation is

$$(sE - A)x = 0. \quad (8.6)$$

Partition retained and condensed variables as

$$sE - A = \begin{bmatrix} sE_{rr} - A_{rr} & sE_{rc} - A_{rc} \\ sE_{cr} - A_{cr} & sE_{cc} - A_{cc} \end{bmatrix}. \quad (8.7)$$

When $sE_{cc} - A_{cc}$ is invertible, the exact retained pencil is

$$S(s) = sE_{rr} - A_{rr} - (sE_{rc} - A_{rc})(sE_{cc} - A_{cc})^{-1}(sE_{cr} - A_{cr}). \quad (8.8)$$

This form is the correct starting point for DAEs, semi-explicit models, algebraic variables that are not eliminated first, and identified port models. The explicit-state formula (8.5) is the special case $E = I$ with compatible zero off-diagonal descriptor blocks. The descriptor form also makes visible which eliminated poles are genuine dynamic poles and which are algebraic or infinite-eigenvalue artifacts of the chosen representation.

8.3 Port Equivalence of Self-Energy

The self-energy must be treated as a port map, not as a property of one hidden state realization. Let two condensed subsystems have hidden coordinates z_1 and z_2 , possibly of different dimensions, and suppose they produce the same retained correction on a spectral domain $\mathcal{D}$:

$$\Pi_1(s) = \Pi_2(s), \quad s \in \mathcal{D}. \quad (8.9)$$

Then every retained-port statement that uses only

$$T(s) = s^2M + sD_0 + L + \Pi(s) \quad (8.10)$$

is invariant under the change of hidden realization.

Theorem 8.1 (Spectral sufficiency of the port self-energy). *Assume that the retained matrices M, D_0, L are the same and that two hidden subsystems satisfy (8.9) on a domain $\mathcal{D}$ that contains the protected roots and the contours used for certification. Then the two reduced operators have the same retained roots in $\mathcal{D}$, the same retained Green function $T(s)^{-1}$ wherever the inverse exists, and the same low-rank action determinants for any action that is expressed only through the retained ports.*

Proof. Under (8.9), the two retained operators (8.10) are identical on $\mathcal{D}$. Equality of roots, inverses on the resolvent set, and any determinant formula built from that inverse follows directly. The full hidden-state eigenvectors may differ, but those coordinates are not observables of a retained-port decision. $\square$

This theorem is deliberately modest. It does not say that hidden participation factors, Euclidean controller-state norms, or internal energy coordinates are invariant. It says that a retained-port planning claim should depend on the map $\Pi(s)$, the port normalization, and the protected spectral region. Similarity transformations $z = S\hat{z}$, changes of hidden units, or a different order of Schur elimination are harmless only to the extent that they preserve the same port self-energy on the domain being used.

8.4 General Second-Order Self-Energy Form

The exact Schur operator (8.5) does not by itself imply that the reduced second-order model contains only a dynamic damping term. To see the required structure, write a retained mechanical model with eliminated controller states z :

$$M\ddot{\theta} + D_0\dot{\theta} + \tilde{L}\theta + Gz = 0, \quad (8.11)$$

$$\dot{z} = A_{cc}z + B_\theta\dot{\theta} + B_\omega\dot{\theta}. \quad (8.12)$$

After Laplace transformation and elimination of z ,

$$T(s) = s^2M + sD_0 + \tilde{L} + \Pi(s), \quad \Pi(s) = G(sI - A_{cc})^{-1}(B_\theta + sB_\omega). \quad (8.13)$$

The sign of G, B_θ, B_ω depends on the port convention; the important point is the channel separation. In general, $\Pi(s)$ contains a rational stiffness contribution and a rational damping contribution:

$$\Pi(s) = \Pi_K(s) + s\Pi_D(s). \quad (8.14)$$

Only under the additional assumption $\Pi_K(s) = 0$, or after a defensible absorption of Π_K into a stated stiffness correction, does the operator reduce to the dynamic-damping form

$$T(s) = s^2M + s\Sigma(s) + \tilde{L}, \quad (8.15)$$

where

$$\Sigma(s) = D_0 + \Pi_D(s) = D_0 + C(sI - A_{cc})^{-1}B \quad \text{in the damping-channel special case.} \quad (8.16)$$

This is the precise transition from a fixed D to a rational self-energy $\Pi(s)$. The broader self-energy is $\Pi(s)$; $\Sigma(s)$ names the dynamic damping component.

The notation $\Sigma(s)$ is therefore deliberately restricted. It is not simply damping in the classical viscous sense. It can include dissipative, anti-dissipative, phase-lag, resonant, and cross-channel effects, depending on the poles and residues of the condensed block. Its Hermitian part on $s = j\omega$ may suggest dissipative behavior over a frequency band, but the full eigenvalue effect also depends on eigenvectors and on $T_s(s)$.

8.5 Dynamic Green Function and Effective Resistance

The inverse of the retained operator is the dynamic Green function of the condensed network:

$$G_T(s) = T(s)^{-1}, \quad s \in \rho(T), \quad (8.17)$$

where $\rho(T)$ denotes the resolvent set. This object is the map from a retained port injection to the retained response at spectral parameter s . It is therefore the natural carrier of dynamic grid strength: it contains the graph stiffness, inertia, fixed damping, controller self-energy, and non-normal amplification in one operator.

Definition 8.1 (Dynamic effective resistance). *For a line or candidate line $e = (i, j)$, let $a_e = e_i - e_j$. The dynamic effective resistance of the line is*

$$\mathcal{R}_e(s) = a_e^T G_T(s) a_e = a_e^T \left[s^2M + sD_0 + L + \Pi(s) \right]^{-1} a_e. \quad (8.18)$$

When $s = 0$, $\Pi(0) = 0$, and the uniform angle reference is removed, this quantity reduces to the classical graph effective resistance

$$\mathcal{R}_e(0) = a_e^T L^\dagger a_e. \quad (8.19)$$

For $s \neq 0$, $\mathcal{R}_e(s)$ is not a physical resistor. It is a meromorphic port compliance. Its poles are the retained roots of $T(s)$, and its residues measure how strongly each mode is seen across the endpoint difference of line e .

Theorem 8.2 (Exact finite line-action law). *Let a finite line-weight change enter the retained operator as*

$$T_{\text{new}}(s) = T(s) + \Delta b_e a_e a_e^T. \quad (8.20)$$

For all s where $T(s)$ is invertible,

$$\det T_{\text{new}}(s) = \det T(s) [1 + \Delta b_e \mathcal{R}_e(s)]. \quad (8.21)$$

Thus any new or shifted root created by this rank-one line action and not already a root of T satisfies

$$1 + \Delta b_e \mathcal{R}_e(s) = 0. \quad (8.22)$$

Proof. Apply the matrix determinant lemma to $T(s) + \Delta b_e a_e a_e^T$. This gives

$$\det(T + \Delta b_e a_e a_e^T) = \det T \left(1 + \Delta b_e a_e^T T^{-1} a_e\right),$$

which is (8.21). □

This result is stronger than a local derivative. It gives an exact scalar equation for a finite rank-one line modification even when $T(s)$ is a large rational NEP. The same equation also explains why classical effective resistance is a static shadow of a richer dynamic object: the graph quantity is the zero-frequency limit of the port Green function.

8.6 Low-Rank Dynamic Action Calculus

Line reinforcement is only one member of a broader action family. Suppose a planning or retuning action enters through a low-rank retained-port update

$$T_u(s) = T(s) + U(s) K_u(s) V(s)^*. \quad (8.23)$$

Define the action interaction matrix

$$\mathcal{G}_u(s) = V(s)^* T(s)^{-1} U(s). \quad (8.24)$$

Then the determinant lemma gives

$$\det T_u(s) = \det T(s) \det[I + K_u(s) \mathcal{G}_u(s)]. \quad (8.25)$$

The large NEP has been reduced, for the purpose of this action, to a determinant whose dimension is the number of action ports.

Several planning levers become special cases of (8.25):

$$\text{line } e : \quad U = V = a_e, \quad K_u(s) = \Delta b_e, \quad (8.26)$$

$$\text{local damping at bus } i : \quad U = V = e_i, \quad K_u(s) = s\Delta d_i, \quad (8.27)$$

$$\text{local inertia at bus } i : \quad U = V = e_i, \quad K_u(s) = s^2\Delta m_i, \quad (8.28)$$

$$\text{controller retuning through ports : } \quad U = U_c, \quad V = V_c, \quad K_u(s) = \Delta K_c(s). \quad (8.29)$$

Thus line, damping, inertia, and controller actions are not four unrelated sensitivity formulas. They are four kernels projected through the same Green function $T(s)^{-1}$.

8.7 Modal Sensitivity as a Local Limit

The usual NEP sensitivity is recovered by expanding the Green function near a simple root. Let $T(s_k)x_k = 0$, $y_k^*T(s_k) = 0$, and $y_k^*T_s(s_k)x_k \neq 0$. Then

$$T(s)^{-1} \sim \frac{x_k y_k^*}{(s - s_k) y_k^* T_s(s_k) x_k}. \quad (8.30)$$

For a line e ,

$$\mathcal{R}_e(s) \sim \frac{(a_e^T x_k)(y_k^* a_e)}{(s - s_k) y_k^* T_s(s_k) x_k}. \quad (8.31)$$

Substituting this expansion into the finite-action equation (8.22) gives

$$s - s_k \approx -\Delta b_e \frac{y_k^* a_e a_e^T x_k}{y_k^* T_s(s_k) x_k}, \quad (8.32)$$

which is exactly the first-order line sensitivity

$$\frac{ds_k}{db_e} = -\frac{y_k^* a_e a_e^T x_k}{y_k^* T_s(s_k) x_k}. \quad (8.33)$$

The sensitivity is therefore not an isolated formula. It is the local residue limit of an exact finite-action law.

8.8 Pole-Residue Structure

Assume for exposition that A_{cc} is diagonalizable:

$$A_{cc} = R\Lambda_c R^{-1}, \quad \Lambda_c = \text{diag}(\alpha_1, \dots, \alpha_m). \quad (8.34)$$

Then

$$C(sI - A_{cc})^{-1}B = \sum_{\ell=1}^m \frac{Cr_\ell \ell_\ell^T B}{s - \alpha_\ell}, \quad (8.35)$$

where r_ℓ and ℓ_ℓ are right and left eigenvectors of A_{cc} , normalized so that $\ell_\ell^T r_\ell = 1$. The residue $Cr_\ell \ell_\ell^T B$ measures how strongly the eliminated controller pole α_ℓ is seen by retained coordinates.

This expression explains why a nearby controller pole can dominate a damping margin. If s is close to α_ℓ , the denominator is small. If the residue projects strongly onto the retained eigenvectors, the self-energy term can dominate the eigenvalue sensitivity.

8.9 NEP Sensitivity with Self-Energy

For $T(s, \rho)v = 0$, the simple-root sensitivity is

$$\frac{ds_\star}{d\rho} = -\frac{u^H T_\rho(s_\star, \rho)v}{u^H T_s(s_\star, \rho)v}. \quad (8.36)$$

For the general operator (8.13),

$$T_s(s) = 2sM + D_0 + \Pi_s(s). \quad (8.37)$$

In the damping-channel specialization (8.15),

$$T_s(s) = 2sM + \Sigma(s) + s\Sigma'(s), \quad (8.38)$$

with

$$\Sigma'(s) = -C(sI - A_{cc})^{-2}B. \quad (8.39)$$

Thus even the denominator of the sensitivity contains controller-resolvent terms. A fixed scalar damping bridge cannot reproduce this dependence unless the rational term is negligible over the critical spectral region.

The additional denominator term $s\Sigma'(s)$ is the precise mathematical place where the condensed controller block retains spectral memory. It is not another static damping coefficient; it is a slope of the self-energy with respect to the eigenvalue being solved for. Using (8.39), this slope contains the squared resolvent $(sI - A_{cc})^{-2}$. Hence it becomes large when the retained pole is close to a visible condensed-controller pole. In that regime, replacing $\Sigma(s)$ by a constant matrix may preserve one fitted value of the self-energy while losing the local pole-motion geometry.

8.10 Why Static Damping Bridges Fail

A static bridge assumes that the reduced damping mechanism can be represented by a constant matrix D_{red} . In the self-energy theory, that assumption requires

$$C(sI - A_{cc})^{-1}B \approx D_{\text{red}} - D_0 \quad (8.40)$$

over all poles and frequency bands relevant to the claim. This is a strong condition. It can hold locally if the controller poles are far away and the resolvent varies slowly. It fails when the critical pole is near a controller pole, when residues are large, or when the relevant mode is dominated by condensed states.

The failure is structural rather than cosmetic. A constant matrix has no poles. It cannot represent the denominator $s - \alpha_\ell$. It can approximate the rational term at a selected point, but it cannot preserve the full spectral mechanism over a region unless additional assumptions are imposed.

Remark 8.1 (Static models are local models). *The conclusion is not that every static reduced model is useless. The conclusion is that a static bridge has a validity region. It is defensible when controller poles are far from the protected region, when residues are small in the retained eigenvector directions, and when $\Sigma'(s)$ is small enough that the denominator of (8.36) is not materially changed. It is not defensible as a universal damping-margin surrogate near a controller resonance.*

8.11 Certified Adequacy of Reduced Operators

The hierarchy is useful only if it contains an escalation rule. Let $\hat{T}(s)$ be a reduced or approximate operator, such as a static bridge, a fitted self-energy, or a lower-order rational approximation of $\Pi(s)$. On a closed contour Γ that bounds a protected region, define

$$\eta_\Gamma = \sup_{s \in \Gamma} \left\| \hat{T}(s)^{-1} [T(s) - \hat{T}(s)] \right\|_2. \quad (8.41)$$

Theorem 8.3 (Contour adequacy certificate). *Assume that T and $\hat{T}$ are regular analytic matrix functions on and inside Γ , that $\hat{T}(s)$ is nonsingular on Γ , and that*

$$\eta_\Gamma < 1. \quad (8.42)$$

Then T and $\hat{T}$ have the same number of eigenvalues inside Γ , counted with algebraic multiplicity.

Proof. The statement is the matrix-function analogue of Rouché's theorem. Since $\|\hat{T}^{-1}(T - \hat{T})\|_2 < 1$ on Γ , the homotopy $\hat{T} + \tau(T - \hat{T})$ remains nonsingular on Γ for $0 \leq \tau \leq 1$. The argument principle for regular analytic matrix functions then preserves the eigenvalue count inside the contour [45, 14]. $\square$

This certificate turns the operator hierarchy into a computational policy:

$$\eta_\Gamma < 1 \implies \text{the reduced operator preserves the protected eigenvalue count,} \quad (8.43)$$

whereas

$$\eta_\Gamma \geq 1 \implies \text{escalate to a richer NEP, identified self-energy, or full DAE.} \quad (8.44)$$

A local decision also needs a sign certificate. If an approximate model predicts a margin derivative $\hat{g}$, and an error analysis gives

$$|g - \hat{g}| \leq \varepsilon_g, \quad (8.45)$$

then

$$|\hat{g}| > \varepsilon_g \implies \text{sign}(g) = \text{sign}(\hat{g}). \quad (8.46)$$

Without a bound of this type, a ranked action is a screen rather than a certified recommendation.

8.12 Data-Identified Self-Energy

The self-energy theory should not require access to proprietary hidden matrices. If a manufacturer, EMT model, or field experiment provides only terminal data, the retained-port object can be estimated as

$$\hat{\Pi}(s) \approx \Pi(s) \quad (8.47)$$

from frequency scans, injected perturbations, PMU records, or black-box simulation. The estimation method may be vector fitting, Loewner realization, gray-box calibration, or a structure-preserving DAE identification method. Recent converter-identification work is moving in this direction by combining terminal time-series data with physics-informed converter structure and frequency-domain validation [43]; recent port-Hamiltonian DAE identification similarly emphasizes preserving interconnection and passivity structure under noisy input-output data [44].

An identified self-energy is useful for planning only if it comes with an error model. On a protected frequency or contour set Ω , require

$$\|\Pi(s) - \hat{\Pi}(s)\|_2 \leq \varepsilon_\Pi(s), \quad s \in \Omega. \quad (8.48)$$

This bound propagates through the same adequacy certificate:

$$\hat{T}(s) = s^2 M + s D_0 + L + \hat{\Pi}(s), \quad \eta_\Gamma = \sup_{s \in \Gamma} \left\| \hat{T}(s)^{-1} [\Pi(s) - \hat{\Pi}(s)] \right\|_2. \quad (8.49)$$

The practical chain is therefore

$$\text{data} \rightarrow \hat{\Pi}(s) \rightarrow \varepsilon_\Pi \rightarrow \eta_\Gamma \rightarrow \text{certified or rejected action}. \quad (8.50)$$

The thesis contribution is not merely to identify a converter. It is to decide how much identification error can be tolerated before a planning recommendation changes sign or loses its protected eigenvalue count.

8.13 Mode Families

The Schur partition motivates a mode-family diagnostic, but a raw Euclidean ratio of retained and condensed state norms is not invariant under a change of units or controller realization. If physically meaningful weights are available, use the weighted index

$$\pi_c^{(W)} = \frac{x_c^H W_c x_c}{x_e^H W_e x_e + x_c^H W_c x_c}, \quad (8.51)$$

where $W_e, W_c \succeq 0$ represent compatible energy, power, or per-unit normalization choices. Small $\pi_c^{(W)}$ indicates a retained-state or network-family mode. Large $\pi_c^{(W)}$ indicates a control-family mode. Intermediate values indicate mixed modes.

When no defensible W_c exists, the more invariant diagnostic is not internal state size but port visibility:

$$\mathcal{V}_c(s) = \left\| C(sI - A_{cc})^{-1} B \right\|_2. \quad (8.52)$$

This quantity asks whether the condensed block is visible from retained coordinates at the frequency being tested. It changes under input-output scaling, so the ports must still be normalized, but it is not changed by a similarity transformation of the hidden controller state.

8.14 Structured Pseudospectrum and Fragility Radius

Nominal eigenvalues do not fully describe a non-normal converter-rich model. The unstructured NEP pseudospectrum is

$$\Lambda_\varepsilon(T) = \{s : \sigma_{\min}(T(s)) \leq \varepsilon\}. \quad (8.53)$$

This set is useful as a numerical warning, but it permits arbitrary matrix perturbations. Planning perturbations are not arbitrary: they come from line uncertainty, damping placement, inertia placement, controller gains, delays, operating-point errors, or identified-model errors. The relevant object is therefore the action-structured pseudospectrum

$$\Lambda_\varepsilon^{\mathcal{A}}(T) = \{s : \exists u \in \mathcal{A}, \|W_u u\| \leq \varepsilon, \det T(s, u) = 0\}, \quad (8.54)$$

where $\mathcal{A}$ is the declared physical action or uncertainty family and W_u normalizes units and costs. Recent nonlinear spectral algorithms make the broader computation of spectra and pseudospectra increasingly relevant for NEPs, but the power-system decision must still restrict perturbations to physically admissible directions [18].

For a protected spectral boundary $\partial\mathcal{S}$, define the dynamic fragility radius

$$r_{\text{frag}} = \inf \{ \|W_u u\| : \exists s \in \partial\mathcal{S}, \det T(s, u) = 0 \}. \quad (8.55)$$

This number answers the planning question directly: what is the smallest physical perturbation that can violate the protected margin? A damping-ratio margin, an abscissa margin, a distance-to-resonance margin, a passivity margin, or a protected-gain margin can each be used to define $\partial\mathcal{S}$. Local pole-motion cones and hidden-margin derivatives are then first-order approximations to (8.55), not substitutes for it.

8.15 Passivity and Resolvent Gates

A stable A_{cc} block does not guarantee that the condensed self-energy is benign. A passivity statement requires a declared input-output pairing and a supply rate in the sense of dissipativity theory [46]. For a MIMO retained port, reducing passivity to one scalar index can hide the direction that causes the deficiency. With input and output normalizers W_u and W_y , define the directional passivity matrix

$$\mathcal{P}_\Sigma(\omega) = \frac{1}{2} \left[W_y \Sigma(j\omega) W_u^{-1} + W_u^{-*} \Sigma(j\omega)^* W_y^* \right]. \quad (8.56)$$

This matrix records both intensity and direction of passivity excess or deficit. Recent matrix-valued passivity work is useful here precisely because scalar indices can lose MIMO channel structure [32]. The unweighted Hermitian screen used below is the special case

$$H_\Sigma(\omega) = \frac{\Sigma(j\omega) + \Sigma(j\omega)^H}{2}. \quad (8.57)$$

Negative eigenvalues of $H_\Sigma(\omega)$, or of the weighted matrix $\mathcal{P}_\Sigma(\omega)$, indicate frequencies where the condensed contribution acts like negative damping in a retained port direction. Large singular values of the resolvent identify frequencies where the eliminated block strongly amplifies retained-state motion. Dynamic passivity multipliers provide a broader certificate family when strict passivity in the trivial metric is too conservative [33].

Recent passivity-based grid-forming work suggests a useful interpretation of this gate [29, 31]. If a controller parameter κ changes the condensed block, write $A_{cc} = A_{cc}(\kappa)$ and define the band-limited self-energy passivity screen

$$\mu_\Sigma(\Omega; \kappa) = \inf_{\omega \in \Omega} \lambda_{\min}(H_\Sigma(\omega; \kappa)), \quad (8.58)$$

over the frequency band Ω that contains the protected electromechanical or controller-interaction modes. If $\mu_\Sigma(\Omega; \kappa) > 0$, the condensed controller contribution is dissipative in the retained coordinates over that band. If $\mu_\Sigma(\Omega; \kappa) < 0$, there exists a retained direction and a frequency where the condensed block supplies negative damping.

Theorem 8.4 (Local self-energy passivity-retuning certificate). *Let $\Omega \subset \mathbb{R}$ be compact. Assume $A_{cc}(\kappa)$, $B(\kappa)$, and $C(\kappa)$ are continuously differentiable on an interval $\mathcal{K}$, and that $j\omega I - A_{cc}(\kappa)$ is invertible for all $(\omega, \kappa) \in \Omega \times \mathcal{K}$. At a fixed κ , suppose the infimum in (8.58) is attained at a unique*

ω_* , and that the smallest eigenvalue of $H_\Sigma(\omega_*; \kappa)$ is simple with unit eigenvector q_* . Then μ_Σ is differentiable at κ , and

$$\frac{d\mu_\Sigma}{d\kappa} = q_*^H \frac{\partial H_\Sigma(\omega_*; \kappa)}{\partial \kappa} q_*. \quad (8.59)$$

If this quantity is nonnegative throughout an interval, then μ_Σ is nondecreasing on that interval. If it is positive before a single zero and negative afterward, the band-limited passivity margin is locally unimodal and the zero gives a candidate retuning optimum.

Proof. For fixed ω , $H_\Sigma(\omega; \kappa)$ is Hermitian and differentiable under the stated invertibility assumptions. The standard simple-eigenvalue perturbation formula gives $\partial_\kappa \lambda_{\min} = q_*^H (\partial_\kappa H_\Sigma) q_*$ at the active frequency. Since the margin is an infimum over the compact frequency set, Danskin's envelope theorem applies to the active minimizer [47, 48]. Under uniqueness and simplicity, the directional formula collapses to (8.59). Monotonicity follows by integrating the derivative over the interval. The unimodal statement is the same first-order condition with one sign change. $\square$

The derivative in (8.59) is computable from the same resolvent that defines the self-energy. With $R(s, \kappa) = (sI - A_{cc}(\kappa))^{-1}$,

$$\frac{\partial \Sigma}{\partial \kappa} = C_\kappa R B + C R_\kappa B + C R B_\kappa, \quad R_\kappa = R \frac{\partial A_{cc}}{\partial \kappa} R. \quad (8.60)$$

If only the controller matrix A_{cc} depends on κ , the first and third terms vanish. The sign of $d\mu_\Sigma/d\kappa$ is therefore not a generic algebraic fact; it is a physical property of the controller family, operating point, and frequency band.

For a grid-following PLL, this distinction matters. Increasing PLL bandwidth can improve phase tracking over one range and degrade weak-grid damping over another. The thesis therefore treats monotonicity as a local certificate, not as a global law. The actionable design problem is

$$\kappa_* \in \arg \max_{\kappa \in \mathcal{K}} \mu_\Sigma(\Omega; \kappa), \quad \text{subject to a protected-pole gate.} \quad (8.61)$$

This formulation also explains what must be validated after retuning: μ_Σ should increase in the relevant band, and the protected pole should move in the desired direction.

This is not a complete stability theorem for the retained NEP. It is a certificate candidate. The missing companion step is the pole sensitivity:

$$\frac{d}{d\rho} \mu_\Sigma(\Omega; \rho), \quad \frac{ds_*}{d\rho} = -\frac{u^H T_\rho(s_*, \rho) v}{u^H T_s(s_*, \rho) v}. \quad (8.62)$$

A controller-retuning action is credible only if it increases the passivity screen in the relevant band and moves the protected pole in the correct direction. This gives a principled version of “PLL retuning”: the action is not merely moving an eigenvalue numerically; it is attempting to restore a band-limited dissipativity margin of the Schur self-energy.

These diagnostics are gates, not standalone stability theorems. They must be connected to eigenvalue sensitivities and full-model validation before they can support planning decisions. Their theoretical value is that they point to the operator that must be inspected: $\Sigma(s)$, not a scalar damping constant.

8.16 Sign Characteristic and Controller-Resolvent Collisions

The language of Krein signature is mature for Hamiltonian and self-adjoint spectral problems, and the matrix-polynomial literature uses the more precise term *sign characteristic* [19, 20, 21]. Kollár and Miller give a useful graphical interpretation through Evans–Krein functions for polynomial operator pencils [22]. That literature should not be cited as if it already proved the converter result needed here. The thesis uses it as the correct mathematical foundation for a narrower question: can a rational, controller-condensed self-energy produce an operational sign collision that predicts weak-grid synchronization loss?

For a Hermitian matrix polynomial, the sign characteristic is invariant under structure-preserving transformations and is tied to the perturbation behavior of real or imaginary-axis eigenvalues. A converter-condensed operator

$$T(s) = s^2 M + s D_0 + \tilde{L} + \Pi(s)$$

is usually not Hermitian as a function of s , and its rational term may be dissipative and non-normal. Therefore the following object is only an operational signature gate, not a completed Krein theorem:

$$\chi_k^{\text{op}} = \text{sign Re}\left(v_k^H T_s(s_k) v_k\right), \quad T(s_k) v_k = 0. \quad (8.63)$$

If left and right eigenvectors differ strongly, the bilinear variant $u_k^H T_s(s_k) v_k$ should be tracked together with eigenvalue conditioning. If $s_k = j\omega_k$ lies on a symmetry line where a Hermitian reduction is available, one may instead inspect the inertia of $(T_s(j\omega_k) + T_s(j\omega_k)^H)/2$ on the active eigenspace.

The reason this is relevant to controller-resonant instability is the derivative term

$$T_s(s) = 2sM + D_0 + \Pi_s(s) \quad \text{or, in the damping-channel case,} \quad T_s(s) = 2sM + \Sigma(s) + s\Sigma'(s).$$

Near a pole of A_{cc} , the resolvent derivative contains squared-resolvent growth. A mode can then become dominated by the controller self-energy rather than by the mechanical part of the QEP. The proposed mechanism is a controller-resolvent sign collision: two tracked modes approach while their operational signatures differ, and the denominator $u^H T_s(s) v$ in the NEP sensitivity becomes small or ill-conditioned.

This bridge is a research claim, not a settled theorem. It is credible only if a benchmark shows the same event in three views: a sign change or singularity in (8.63), growth of the absolute self-energy dominance index, and a full-model eigenvalue or time-domain loss of damping. The general theory of dissipation-induced instabilities explains why such collisions are plausible [49]; the new power-system claim would be that the Schur self-energy identifies the controller channel that causes them.

Part V

Nonlinear Models and Memory

Nonlinear Models, Memory, and Stability Beyond the NEP

The rational nonlinear eigenvalue problem developed in the previous part is nonlinear in the spectral parameter s . It is still a small-signal object: it comes from a linearization in time and from exact or approximate elimination of linearized states. This chapter separates that NEP theory from the genuinely nonlinear power-system problem. The latter concerns equilibria, local stability, bifurcations, regions of attraction, amplitude-dependent damping, and memory generated by eliminating nonlinear controllers.

Separation of roles

The Schur NEP answers: what rational operator preserves the linearized controller memory around an operating point? The nonlinear chapter answers: which equilibria exist, how they lose stability, and whether a spectral margin survives finite perturbations.

9.1 Nonlinear DAE Layer

Let the full converter-rich model be written as a semi-explicit DAE,

$$\dot{x} = f(x, y, p), \quad (9.1)$$

$$0 = g(x, y, p), \quad (9.2)$$

where x collects dynamic states, y collects algebraic network variables, and p is an operating or planning parameter. A regular equilibrium (x^*, y^*) satisfies

$$f(x^*, y^*, p) = 0, \quad g(x^*, y^*, p) = 0, \quad g_y(x^*, y^*, p) \text{ nonsingular}. \quad (9.3)$$

When the algebraic Jacobian is nonsingular, the implicit-function theorem gives a local algebraic manifold $y = h(x, p)$. The reduced ODE is

$$\dot{x} = F(x, p) = f(x, h(x, p), p), \quad (9.4)$$

and the small-signal state matrix is

$$A_{\text{red}} = f_x - f_y g_y^{-1} g_x \quad (9.5)$$

at the equilibrium. The NEP chapters use this Jacobian, or a structured second-order projection of it. The nonlinear theory must also study what happens when the equilibrium moves, becomes singular, or ceases to attract finite disturbances.

9.2 Canonical Converter-Rich Family

A useful hierarchical model for later chapters is

$$M\ddot{\theta} + D_0\dot{\theta} + P_e(\theta, V) + Gz = p, \quad (9.6)$$

$$\dot{z} = f_c(z, \theta, \dot{\theta}, V, p), \quad (9.7)$$

$$0 = g(\theta, V, z, p). \quad (9.8)$$

The state z may represent a synchronous-machine exciter or governor, a grid-following PLL and current loop, a grid-forming droop or virtual synchronous-machine controller, a voltage loop, a filter, or a plant-level control state. Limiters and protections should be treated as smooth models only inside a declared operating region; outside it they create hybrid dynamics and require a different analysis.

Linearizing (9.6)–(9.8) recovers the QEP or Schur NEP used earlier. The nonlinear chapter keeps the prelinearized equations visible so that a small-signal margin is not mistaken for a large-signal certificate.

9.3 Linear Controller Elimination as Memory

For a linear controller block

$$\dot{z} = A_c z + B_\theta \theta + B_\omega \dot{\theta}, \quad (9.9)$$

the exact variation-of-constants formula is

$$z(t) = e^{A_c t} z(0) + \int_0^t e^{A_c(t-\tau)} [B_\theta \theta(\tau) + B_\omega \dot{\theta}(\tau)] d\tau. \quad (9.10)$$

Substitution into the retained equation produces an integro-differential model. Its temporal kernel is

$$K(t) = G e^{A_c t} B, \quad (9.11)$$

and its Laplace transform is the Schur-resolvent term

$$G(sI - A_c)^{-1} B. \quad (9.12)$$

Thus the rational self-energy $\Pi(s)$ is the spectral representation of linear memory. This is the conceptual bridge between the NEP part and the nonlinear-memory part of the dissertation.

Theorem 9.1 (Self-energy as the tangent of memory). *Consider a smooth nonlinear retained-hidden system near a regular equilibrium, and suppose the hidden variables can be eliminated locally as a causal memory functional acting on the retained trajectory. If this memory functional is Fréchet differentiable at the equilibrium trajectory, then the Laplace transform of its first variation is the Schur self-energy $\Pi(s)$ of the linearized hidden block.*

Proof. Linearize the full nonlinear DAE or ODE around the equilibrium and partition the linearization into retained and hidden coordinates. The hidden linear subsystem has the variation-of-constants representation (9.10). Its contribution to the retained equation is a convolution with kernel $K(t) = G e^{A_c t} B$. Taking the Laplace transform gives $G(sI - A_c)^{-1} B$, with the appropriate stiffness and velocity-channel blocks collected as $\Pi(s)$. This is exactly the first variation of the nonlinear memory functional. $\square$

9.4 Nonlinear Elimination

If the controller block is nonlinear,

$$\dot{z} = f_c(z, \theta, \dot{\theta}, V, p), \quad (9.13)$$

then eliminating z no longer gives a single matrix-valued rational function. The reduced dynamics may contain history-dependent nonlinear functionals, amplitude-dependent damping, and higher-order kernels. Locally, one can organize those effects through:

- Volterra-series kernels when the nonlinear response is analytic and small enough for a convergent expansion;
- center-manifold or invariant-manifold reductions near critical eigenvalues;
- Mori-Zwanzig-type memory terms when unresolved variables are projected out of the dynamics;
- amplitude-dependent damping and stiffness coefficients when only the dominant oscillatory envelope is retained.

The thesis will use this list as a research contract. A nonlinear memory claim must state which reduction is being used and which finite-amplitude validation gate supports it.

The same idea can be expressed as an amplitude-dependent self-energy when a dominant oscillatory envelope is meaningful:

$$\Pi(s; A) = \Pi_1(s) + A^2\Pi_3(s) + A^4\Pi_5(s) + \cdots. \quad (9.14)$$

Here $\Pi_1(s)$ is the small-signal self-energy from Chapter 6, while higher terms represent finite-amplitude stiffness and damping corrections. This expansion is only legitimate under a declared Volterra, harmonic-balance, or normal-form approximation. Its value is conceptual and testable: it asks whether a small-signal damping reversal survives finite amplitude, disappears through saturation, or becomes a Hopf mechanism. Recent non-Markovian closure work reinforces the same principle that hidden variables generally reappear as memory terms when only partial state information is retained [50].

9.5 Bifurcation and Large-Signal Stability Contracts

Small-signal stability asks whether the eigenvalues of the linearization have negative real part. Nonlinear stability asks additional questions:

1. Does a regular equilibrium exist for the planned operating point?
2. Does the equilibrium lose stability through a Hopf, saddle-node, or other bifurcation as p changes?
3. If a Hopf bifurcation occurs, what is the first Lyapunov coefficient and does the emerging oscillation remain bounded?
4. What region of attraction is certified by an energy, Lyapunov, contraction, or incremental-passivity argument?
5. Does a controller-induced damping reversal seen in the NEP persist under finite perturbations, or is it only a local pole-motion effect?

For a simple Hopf pair $s(p) = \sigma(p) \pm j\omega(p)$, the spectral crossing condition is

$$\sigma(p_*) = 0, \quad \omega(p_*) \neq 0, \quad \frac{d\sigma}{dp}(p_*) \neq 0. \quad (9.15)$$

This condition locates a candidate bifurcation. It does not state the amplitude, stability, or basin of the post-bifurcation motion. Those require the nonlinear terms of $F(x, p)$, not only the Schur NEP.

9.6 Chapter Deliverables

The nonlinear module is complete only when it contains the following deliverables:

1. a regular DAE equilibrium theorem or a clearly stated regularity assumption for each benchmark family;
2. a local-stability result tied to the QEP or Schur NEP;
3. at least one bifurcation calculation with finite-difference or complex step verification of the derivative;
4. an energy, passivity, contraction, or Lyapunov-region argument for a restricted model;
5. a finite-perturbation simulation showing whether the small-signal margin predicts the observed nonlinear response;
6. a paper extraction section that states which nonlinear-memory result is new, which result is known, and which result is only numerical evidence.

This chapter is therefore not a replacement for the Schur NEP. It is the next layer above it. The NEP preserves linearized memory; nonlinear analysis decides which parts of that memory survive beyond infinitesimal perturbations.

Part VI

Applications and Reproducible Results

Risk Vectors, Weak Elements, and Planning Metrics

The damping-operator hierarchy changes how weak elements and planning actions should be defined. A weak bus, weak line, or useful converter placement is not an intrinsic property of a graph alone. It is a property relative to a dynamic margin, an operator class, an action family, and a validation gate. This chapter turns the spectral derivatives into a planning language.

10.1 Margins Must Name the Mechanism

The most common small-signal margin is the critical damping ratio,

$$S_\zeta = \log \frac{\zeta_{\text{req}}}{\zeta_{\text{crit}}}. \quad (10.1)$$

The case satisfies the requirement when $S_\zeta \leq 0$. This margin is natural for oscillatory eigenvalues, but it does not measure non-normal amplification, input-output gain, or protection-weighted response. For a state matrix A , selected inputs B , selected outputs C , and weights Γ_u, Γ_y , a protected resolvent gain can be defined as

$$g_{\text{prot}} = \sup_{\omega \in \Omega} \sigma_1 \left(\Gamma_y C (j\omega I - A)^{-1} B \Gamma_u \right). \quad (10.2)$$

The corresponding log margin is

$$S_g = \log g_{\text{prot}}. \quad (10.3)$$

The two margins can rank actions differently. That disagreement is not an error; it means the mechanisms being measured are different.

10.2 Action Coordinates

Let a planning action be parameterized as

$$\rho = \begin{bmatrix} \rho_L \\ \rho_D \\ \rho_M \end{bmatrix}, \quad (10.4)$$

where ρ_L changes line weights or topology, ρ_D changes damping or primary-control gain, and ρ_M changes physical or virtual inertia. For a simple pole $s_\star$, the total first-order movement is

$$ds_\star = \nabla_L s_\star^T d\rho_L + \nabla_D s_\star^T d\rho_D + \nabla_M s_\star^T d\rho_M. \quad (10.5)$$

The three gradients are computed from the exact QEP sensitivity (6.32) or, after condensation, from the NEP sensitivity (8.36). The key point is that the gradients are not interchangeable. *L*-moves are stiffness moves, *D*-moves are dissipation moves, and *M*-moves are inertia and normalization moves.

10.3 Finite-Action Coordinates

The action coordinates above are local, but the rational self-energy chapter gives a finite-action object. For any low-rank action

$$T_u(s) = T(s) + U(s)K_u(s)V(s)^*,$$

the action interaction matrix is

$$\mathcal{G}_u(s) = V(s)^*T(s)^{-1}U(s), \quad (10.6)$$

and the protected roots created or shifted by the action satisfy

$$\det[I + K_u(s)\mathcal{G}_u(s)] = 0. \quad (10.7)$$

This formula changes how a planning action should be reported. A derivative ds_*/du is the tangent of (10.7); it is not the full action law. For a finite line increment, the scalar interaction is the dynamic effective resistance $\mathcal{R}_e(s) = a_e^T T(s)^{-1} a_e$. For finite damping, inertia, and controller-retuning actions, the relevant object is the corresponding block of $\mathcal{G}_u(s)$.

A local risk vector should therefore be interpreted as the first-order linearization of three finite objects:

$$\text{action determinant} \quad \text{protected margin boundary} \quad \text{structured fragility radius}. \quad (10.8)$$

This interpretation prevents a candidate from being accepted only because its infinitesimal derivative is favorable at the base point. The finite action must remain inside the protected margin and outside the structured pseudospectral boundary over the declared budget.

10.4 Risk Vector for a Candidate Action

For any candidate action a , define a local risk vector after normalizing the action coordinate and each margin. A gain component must be dimensionless, e.g. $S_g = \log(g_{\text{prot}}/g_{\text{req}})$, not the logarithm of a dimensional quantity. With a declared action scale $\bar{a}$, derivatives below are taken with respect to $\hat{a} = a/\bar{a}$:

$$r(a) = \begin{bmatrix} r_\sigma(a) \\ r_\zeta(a) \\ r_g(a) \\ r_h(a) \\ r_f(a) \end{bmatrix}. \quad (10.9)$$

The components are:

$$r_\sigma(a) = \left[\frac{d \operatorname{Re}(s_{\text{crit}})}{da} \right]_+, \quad (10.10)$$

$$r_\zeta(a) = \left[\frac{dS_\zeta}{da} \right]_+, \quad (10.11)$$

$$r_g(a) = \left[\frac{dS_g}{da} \right]_+, \quad (10.12)$$

$$r_h(a) = \mathbf{1} \left\{ \frac{d\nu_k}{da} > 0, \frac{d\zeta_k}{da} < 0 \right\}, \quad (10.13)$$

$$r_f(a) = \text{normalized feasibility penalty}. \quad (10.14)$$

Here $[x]_+ = \max(x, 0)$. The first entry detects rightward pole movement. The second detects damping-ratio margin loss. The third detects protected input-output amplification. The fourth detects the hidden-margin conflict from Chapter 4. The fifth records voltage, flow, current, protection, or dispatch constraints.

All entries in (10.9) are dimensionless local rates after the chosen normalization. Without that convention, the angle between two risk vectors can change simply because the same action is expressed in MW, per-unit, or percent penetration.

A scalar screening score may be formed as

$$\mathcal{R}(a) = w^T r(a), \quad w \geq 0, \quad (10.15)$$

but the vector should be reported before scalarization. Otherwise, a strong graph improvement can hide a damping loss.

10.5 Bus Risk Vector

For bus i , define a fixed- D spectral bus vector

$$r_i^{\text{bus}} = \begin{bmatrix} \mathcal{D}_i \\ \mathcal{M}_i \\ \Pi_i \\ H_i \end{bmatrix}, \quad (10.16)$$

where

$$\mathcal{D}_i = \sum_{k \in \mathcal{K}} \omega_k \frac{\phi_k(i)^2}{2\sqrt{\nu_k}} \quad (10.17)$$

is damping placement leverage,

$$\mathcal{M}_i = \sum_{k \in \mathcal{K}} \omega_k \zeta_k \phi_k(i)^2 \quad (10.18)$$

is inertia-dilution exposure under fixed D ,

$$\Pi_i = \sum_{k \in \mathcal{K}} \omega_k \phi_k(i)^2 \quad (10.19)$$

is modal participation exposure, and H_i records whether incident graph actions produce hidden damping conflicts. A high $\mathcal{D}_i$ says “put damping here.” A high $\mathcal{M}_i$ says “do not put inertia here without co-designed damping if the objective is damping ratio.” A high Π_i says the bus is dynamically visible.

10.6 Line Risk Vector

For a line $e = (i, j)$, define

$$r_e^{\text{line}} = \begin{bmatrix} G_e^{(2)} \\ G_e^{(K)} \\ Z_e \\ H_e \\ F_e \end{bmatrix}. \quad (10.20)$$

The algebraic-connectivity gain is

$$G_e^{(2)} = (\phi_2(i) - \phi_2(j))^2. \quad (10.21)$$

The effective-resistance gain is

$$G_e^{(K)} = a_e^T (L^+)^2 a_e. \quad (10.22)$$

The damping-ratio risk for a monitored mode k is

$$Z_e^{(k)} = -\frac{\partial \zeta_k}{\partial b_e}, \quad Z_e = \max_{k \in \mathcal{K}} Z_e^{(k)}. \quad (10.23)$$

The hidden-margin flag is

$$H_e = \mathbf{1} \left\{ G_e^{(2)} > 0, Z_e > 0 \right\}, \quad (10.24)$$

and F_e is a feasibility penalty for thermal, voltage, protection, or right-of-way constraints. A high $G_e^{(2)}$ or $G_e^{(K)}$ is not enough: the line is attractive only if Z_e and F_e are acceptable.

10.7 Weak Nodes Are Relative Objects

A bus is weak only after the following items are specified:

1. the base operating point and dynamic model;
2. the operator class being used: uniform $D = \gamma M$, commuting fixed D , non-commuting fixed D , rational $\Pi(s)$, or damping-channel $\Sigma(s)$;
3. the margin: damping ratio, decay rate, protected resolvent gain, or another registered metric;
4. the action family: line change, damping addition, inertia change, converter placement, controller retuning, or protection setting;
5. the accept/reject gate that turns a screen into evidence.

Without those items, a weak-node list is not reproducible. A bus may be weak for damping-ratio improvement under controller retuning and irrelevant for protected-gain reduction under line reinforcement.

10.8 Improving the Graph Without Adding Inertia

When the objective is to improve graph robustness without adding inertia, the candidate action should be selected from graph derivatives and then passed through damping gates. A conservative procedure is:

1. rank candidate lines by $G_e^{(2)}$ if the target is algebraic connectivity;
2. rank candidate lines by $G_e^{(K)}$ if the target is total effective resistance or broad slow-mode coherence;
3. remove candidates with unacceptable F_e ;
4. remove or pair candidates with damping support if $H_e = 1$;
5. verify the remaining candidates with QEP or NEP sensitivity.

This gives a mathematically defensible answer to the question “how do we improve L without adding inertia?” Improve L where it moves the slow graph modes most, but do not confuse that with damping unless the damping gate passes.

10.9 Substitutability of Actions

Two actions are substitutable only if they improve the same margin through compatible mechanisms. Adding physical damping, changing virtual damping, placing a grid-forming converter, retuning a PLL, reinforcing a line, or placing inertia can all move the critical eigenvalue, but they do not generally act through the same operator term.

Let a and b be candidate actions. A local substitutability test is

$$\cos \angle(r(a), r(b)) = \frac{r(a)^T r(b)}{\|r(a)\| \|r(b)\|}. \quad (10.25)$$

High alignment means the actions stress or relieve the same risks. Low alignment means they are not substitutes even if both improve one scalar metric. If one action improves S_ζ by adding damping and another improves S_g by reducing resolvent amplification, their risk vectors will not generally align.

10.10 Theory-to-Validation Protocol

The theoretical hierarchy implies a protocol for future empirical work.

1. Start with the proportional model only as a baseline and state which margin it certifies.
2. Measure non-commutation through $\Gamma = Q^T \tilde{D} Q$ before trusting graph-mode damping claims.
3. If controller states are eliminated, construct or approximate $\Pi(s)$, then inspect its poles, residues, channel decomposition, and any damping-channel $\Sigma(s)$ over the frequency band of interest.
4. Use eigenvalue sensitivity to decompose action effects into retained and condensed contributions.
5. Validate screens against full eigenvalue recomputation before using them as planning rules.

This protocol prevents a visually plausible graph or time-domain result from being mistaken for a damping certificate.

10.11 Implications for Converter-Rich Systems

Converter-rich systems make the risk vector more important because converter controls can supply both damping and destabilizing feedback. A grid-following PLL, a grid-forming droop loop, a current controller, and a plant controller do not merely add scalar damping. They add states. If those states are retained, they appear in the full matrix A . If they are condensed, they appear in $\Pi(s)$, or in its declared damping-channel specialization $\Sigma(s)$.

The planning implication is direct. A reduced model that removes converter states must show where their spectral effect went. If the answer is a constant coefficient, the model is making a static approximation. If the answer is a rational self-energy, the model has preserved the mechanism in the reduced operator. The practical value of the theory is to make that distinction visible before numerical ranking begins.

Dynamic Hosting Capacity and Spectral Planning Levers

The preceding chapters define the damping operator hierarchy. This chapter uses that hierarchy to formulate the planning questions that motivated the project: how much inverter-based resource (IBR) conversion can be hosted, where virtual inertia or virtual damping should be placed, and which network reinforcements can improve robustness before resorting to batteries or other large storage devices. The common theme is that every planning answer must be attached to a margin. Voltage feasibility, graph connectivity, damping ratio, protected resolvent gain, and controller-resonance distance are different objects.

11.1 From Static Hosting to Spectral Hosting

Let an operating point be described by complex bus voltages $V_i = |V_i|e^{j\theta_i}$ and bus admittance matrix $Y_{\text{bus}} = G + jB$. The exact AC active- and reactive-power injections are

$$P_i(V, \theta) = |V_i| \sum_{j=1}^n |V_j| (G_{ij} \cos(\theta_i - \theta_j) + B_{ij} \sin(\theta_i - \theta_j)), \quad (11.1)$$

$$Q_i(V, \theta) = |V_i| \sum_{j=1}^n |V_j| (G_{ij} \sin(\theta_i - \theta_j) - B_{ij} \cos(\theta_i - \theta_j)). \quad (11.2)$$

The linearized active-power stiffness is the angle block of the AC Jacobian. After removing the uniform angle reference, write the corresponding positive semidefinite stiffness as L_{op} . In the lossless small-angle approximation, L_{op} reduces to a weighted Laplacian.

A static hosting set for a vector of converter injections or conversions $\rho \in \mathcal{U}$ can be written abstractly as

$$\mathcal{H}_{\text{stat}} = \{\rho \in \mathcal{U} : c_V(\rho) \leq 0, c_{\text{th}}(\rho) \leq 0, c_q(\rho) \leq 0\}, \quad (11.3)$$

where the constraints may represent voltage, thermal, protection, power quality, or operational limits. This is the classical hosting-capacity language. It is indispensable, but it does not by itself certify small-signal damping.

Definition 11.1 (Dynamic spectral hosting set). *For a target damping ratio ζ_{req} , protected resolvent threshold g_{req} , and admissible static set $\mathcal{H}_{\text{stat}}$, define the dynamic spectral hosting set*

$$\mathcal{H}_{\text{dyn}} = \{\rho \in \mathcal{H}_{\text{stat}} : \zeta_{\min}(\rho) \geq \zeta_{\text{req}}, g_{\text{prot}}(\rho) \leq g_{\text{req}}, T(s; \rho) \text{ has no forbidden roots}\}. \quad (11.4)$$

For a penetration direction $\alpha \geq 0$, the directional dynamic hosting capacity is

$$h_{\text{dyn}}(\alpha) = \sup\{\eta \geq 0 : \eta\alpha \in \mathcal{H}_{\text{dyn}}\}. \quad (11.5)$$

This definition deliberately extends time-varying hosting-capacity ideas [35, 36]. In distribution planning, dynamic hosting often means that voltage or thermal violations are counted with their duration and severity. Here it means that the electromechanical and controller spectra also impose capacity limits. A system can be voltage safe and still dynamically unsafe if the critical pole violates ζ_{req} , if the protected resolvent gain is too large, or if a stable network-family pole lies close to a controller pole with large self-energy sensitivity.

11.2 First-Order Spectral Hosting Screen

In the fixed- D layer, let

$$L(\rho)\phi_k(\rho) = \nu_k(\rho)M(\rho)\phi_k(\rho), \quad \phi_k(\rho)^T M(\rho)\phi_\ell(\rho) = \delta_{k\ell}. \quad (11.6)$$

The diagonal damping-region screen from Chapter 4 is

$$h_k(\rho) = \delta_k(\rho) - 2\zeta_{\text{req}}\sqrt{\nu_k(\rho)} \geq 0, \quad \delta_k(\rho) = \phi_k(\rho)^T D(\rho)\phi_k(\rho). \quad (11.7)$$

The quantity h_k is a modal headroom, not an eigenvalue. It is exact only when the relevant modes decouple and becomes a screen otherwise.

For a simple generalized mode and small perturbation ρ_a ,

$$\frac{\partial \nu_k}{\partial \rho_a} = \phi_k^T (L_a - \nu_k M_a) \phi_k, \quad (11.8)$$

where $L_a = \partial L / \partial \rho_a$ and $M_a = \partial M / \partial \rho_a$. The direct damping derivative is $\phi_k^T D_a \phi_k$, with modal-rotation corrections when damping is non-proportional. The first-order hosting-gap derivative is therefore

$$g_{ka} = \frac{\partial h_k}{\partial \rho_a} = \frac{\partial \delta_k}{\partial \rho_a} - \frac{\zeta_{\text{req}}}{\sqrt{\nu_k}} \frac{\partial \nu_k}{\partial \rho_a}. \quad (11.9)$$

Proposition 11.1 (Linear spectral hosting screen). *Assume $h_k(0) > 0$ for every monitored mode $k \in \mathcal{K}$, and assume that the first-order expansion of (11.7) is used as a tangent screen. Define*

$$A_{ka} = -g_{ka}, \quad b_k = h_k(0). \quad (11.10)$$

Then the polytope

$$\mathcal{P}_{\text{host}}^{(1)} = \{\rho \in \mathcal{U} : A\rho \leq b\} \quad (11.11)$$

is the first-order spectral hosting screen associated with the damping-region margin. It is not, by itself, a certified inner approximation of $\mathcal{H}_{\text{dyn}}$.

Proof. The first-order expansion gives

$$h_k(\rho) = h_k(0) + \sum_a g_{ka}\rho_a + O(\|\rho\|^2).$$

Dropping the higher-order term and imposing $h_k(\rho) \geq 0$ gives $-\sum_a g_{ka}\rho_a \leq h_k(0)$, which is (11.11). $\square$

Proposition 11.2 (Curvature-certified hosting region). *Suppose that, on a trust region $\mathcal{B}_r = \{\rho : \|\rho\|_2 \leq r\}$, each monitored headroom satisfies*

$$\left\| \nabla^2 h_k(\rho) \right\|_2 \leq \beta_k.$$

Then every $\rho \in \mathcal{U} \cap \mathcal{B}_r$ satisfying

$$h_k(0) + g_k^T \rho - \frac{\beta_k}{2} \|\rho\|_2^2 \geq 0, \quad k \in \mathcal{K}, \quad (11.12)$$

also satisfies the second-order lower bound for the monitored modal headrooms.

Proof. Taylor's theorem with the stated Hessian bound gives

$$h_k(\rho) \geq h_k(0) + g_k^T \rho - \frac{\beta_k}{2} \|\rho\|_2^2$$

for all $\rho \in \mathcal{B}_r$. Imposing (11.12) therefore certifies nonnegative headroom for the monitored screen. $\square$

The certified set is generally quadratic or conic, not a polytope. This is the right object when the claim is an inner dynamic-hosting region rather than a first-order ranking.

11.3 Certified Hosting and Model Escalation

A dynamic-hosting screen becomes a certified hosting statement only after the model level is certified over the protected region. Let Γ enclose the set of eigenvalues whose damping ratio, abscissa, or controller-resonance distance limits hosting. For a reduced screen $\hat{T}(s; \rho)$, define

$$\eta_\Gamma(\rho) = \sup_{s \in \Gamma} \left\| \hat{T}(s; \rho)^{-1} \left[T(s; \rho) - \hat{T}(s; \rho) \right] \right\|_2. \quad (11.13)$$

If $\eta_\Gamma(\rho) < 1$, the reduced model preserves the protected eigenvalue count inside Γ . If $\eta_\Gamma(\rho) \geq 1$, the hosting decision must escalate from the diagonal screen to the QEP, from the QEP to the Schur NEP, or from the identified NEP to the full DAE, depending on where the uncertainty enters.

For finite penetration increments, the same principle should be stated as a fragility budget. Given a protected boundary $\partial\mathcal{S}$, define

$$r_{\text{host}}(\rho) = \inf \{ \|W_u u\| : \exists s \in \partial\mathcal{S}, \det T(s; \rho, u) = 0 \}. \quad (11.14)$$

A hosting point is not robust merely because the first-order headroom is positive. It is robust to the declared uncertainty family only if r_{host} exceeds the uncertainty or action budget assigned to that point. This is the dynamic analogue of a voltage or thermal security margin.

Remark 11.1 (SG-to-IBR conversion). *If action ρ_i replaces synchronous inertia and damping at bus i without changing the network stiffness, then locally*

$$M_i(\rho_i) = M_i^0 - \rho_i \Delta M_i, \quad D_i(\rho_i) = D_i^0 - \rho_i \Delta D_i.$$

The direct mode-frequency contribution is

$$\frac{\partial \nu_k}{\partial \rho_i} = \nu_k \Delta M_i \phi_k(i)^2.$$

The direct damping contribution is approximately

$$\frac{\partial \delta_k}{\partial \rho_i} \approx -\Delta D_i \phi_k(i)^2 + \text{modal-rotation terms.}$$

Therefore SG-to-IBR conversion is most restrictive where the monitored mode has large $\phi_k(i)^2$ and the removed damping-to-inertia package is large. This is the mathematical reason a hosting limit is not only a voltage limit.

11.4 Where to Place Virtual Inertia

Virtual inertia is not a single planning lever. It may mean inertia-only emulation, virtual synchronous-machine control with damping, grid-forming droop with filtering, or a controller whose reduced effect is frequency dependent. The fixed- D screen is still useful because it identifies what would be true if the virtual device were represented locally by added mass and added damping:

$$M_i(\mu_i) = M_i^0 + \mu_i, \quad D_i(\mu_i) = D_i^0 + \gamma_i^v \mu_i. \quad (11.15)$$

Using the fixed-mode approximations of Chapter 4,

$$\frac{\partial h_k}{\partial \mu_i} \approx (\gamma_i^v - \delta_k + \zeta_{\text{req}} \sqrt{\nu_k}) \phi_k(i)^2. \quad (11.16)$$

Thus inertia-only support ($\gamma_i^v = 0$) can reduce the damping-region headroom near the boundary $\delta_k \approx 2\zeta_{\text{req}} \sqrt{\nu_k}$. The same bus can still be excellent for RoCoF or frequency-nadir objectives because inertia reduces modal acceleration. It is not automatically excellent for damping ratio.

For a set of monitored modes, a placement score that respects this distinction is

$$J_i^{\text{VI}} = \sum_{k \in \mathcal{K}} \omega_k \left[\eta_f \nu_k \phi_k(i)^2 + \eta_h (\gamma_i^v - \delta_k + \zeta_{\text{req}} \sqrt{\nu_k}) \phi_k(i)^2 \right], \quad (11.17)$$

where the first term rewards frequency-security leverage and the second term rewards damping-headroom leverage. The weights η_f and η_h must be declared before ranking. Otherwise the ranking hides a tradeoff between slower frequency motion and better modal damping.

11.5 Line-Shunt Taxonomy and Substitutability

The project also asks whether the network can be improved before installing batteries or local inertial devices. A structural answer comes from separating line and shunt directions.

Let $b_{ij} = e_i - e_j$ and define

$$\mathcal{L}_{\text{all}} = \text{span}\{b_{ij} b_{ij}^T : 1 \leq i < j \leq n\}, \quad \mathcal{D}_{\text{sh}} = \{\text{diag}(g) : g \in \mathbb{R}^n\}. \quad (11.18)$$

Theorem 11.1 (Line-shunt decomposition). *Every real symmetric matrix $X \in \mathbb{R}^{n \times n}$ has a unique decomposition*

$$X = X_{\mathcal{L}} + X_{\mathcal{D}}, \quad X_{\mathcal{L}} \in \mathcal{L}_{\text{all}}, \quad X_{\mathcal{D}} \in \mathcal{D}_{\text{sh}}. \quad (11.19)$$

Equivalently,

$$\text{Sym}(n) = \mathcal{L}_{\text{all}} \oplus \mathcal{D}_{\text{sh}}.$$

Proof. Set the off-diagonal entries of $X_{\mathcal{L}}$ equal to those of X and choose the diagonal of $X_{\mathcal{L}}$ so that every row sums to zero. Then $X_{\mathcal{L}}$ is a Laplacian-type matrix and $X_{\mathcal{D}} = X - X_{\mathcal{L}}$ is diagonal. The intersection of the two spaces is zero because a diagonal zero-row-sum matrix must be zero. $\square$

For local inertia,

$$\frac{\partial \tilde{L}}{\partial M_i} = -\frac{1}{2M_i} (E_i \tilde{L} + \tilde{L} E_i), \quad E_i = e_i e_i^T. \quad (11.20)$$

This perturbation cannot create off-diagonal support on a non-existing edge. Therefore it lies structurally in existing-line directions plus a shunt residual:

$$\frac{\partial \tilde{L}}{\partial M_i} \in \mathcal{L}_{\text{ex}} \oplus \mathcal{D}_{\text{sh}}. \quad (11.21)$$

It has no new-line component. This statement does not mean inertia and line reinforcement are dynamically equivalent. It says where the inertia action lives in the perturbation space.

Let

$$-\frac{\partial \tilde{L}}{\partial M_i} = X_i^{\text{ex}} + \text{diag}(r_i), \quad X_i^{\text{ex}} \in \mathcal{L}_{\text{ex}}. \quad (11.22)$$

For a critical mode q_c , define the modal substitutability score

$$s_i = 1 - \frac{|r_i^T(q_c \odot q_c)|}{\|r_i\|_2 \|q_c \odot q_c\|_2 + \varepsilon_{\text{reg}}}. \quad (11.23)$$

High s_i means the shunt residual is nearly invisible to the critical mode; existing-line actions may be tried before local inertia. Low s_i means the shunt residual aligns with the critical mode; topology-only actions are less likely to substitute for a local inertial or synchronous-condenser lever. Internal random-graph checks in `temp/poster/` verified the algebraic residual of this decomposition to approximately 10^{-14} , which is a structural verification rather than a power-system performance proof. The score is still not a feasibility certificate for construction. A physical line-only substitute must solve a constrained projection with nonnegative line increments, corridor limits, and costs; membership in a linear span is only the algebraic first screen.

$$\min_{\Delta b \in \mathcal{C}_{\text{line}}} \left\| -\frac{\partial \tilde{L}}{\partial M_i} - \sum_{e \in \mathcal{E}_{\text{cand}}} \Delta b_e a_e a_e^T \right\|_W, \quad \mathcal{C}_{\text{line}} = \{\Delta b \geq 0 : c^T \Delta b \leq B, \Delta b_e \leq \bar{b}_e\}. \quad (11.24)$$

11.6 Reinforcing Lines Without Batteries

For a line $e = (i, j)$, the modal-frequency derivative is

$$\frac{\partial \nu_k}{\partial b_e} = (\phi_k(i) - \phi_k(j))^2. \quad (11.25)$$

The damping-headroom derivative is

$$\frac{\partial h_k}{\partial b_e} = \frac{\partial \delta_k}{\partial b_e} - \frac{\zeta_{\text{req}}}{\sqrt{\nu_k}} (\phi_k(i) - \phi_k(j))^2. \quad (11.26)$$

This equation is the cleanest answer to the line-reinforcement question. A line that strongly raises modal stiffness can still reduce damping headroom unless it also raises modal damping through eigenvector rotation, damping placement, or controller action.

The network-only benefits should still be used. If the objective is to improve algebraic connectivity, use $G_e^{(2)} = (\phi_2(i) - \phi_2(j))^2$. If the objective is to reduce total effective resistance, use

$$G_e^{(K)} = a_e^T (L^+)^2 a_e, \quad \frac{\partial}{\partial b_e} n \text{ trace}(L^+) = -n G_e^{(K)}. \quad (11.27)$$

But a dynamic-hosting line screen should report the vector

$$\ell_e = \begin{bmatrix} G_e^{(2)} \\ G_e^{(K)} \\ -\min_{k \in \mathcal{K}} \partial h_k / \partial b_e \\ \partial S_g / \partial b_e \\ F_e \end{bmatrix}, \quad (11.28)$$

where F_e is a feasibility penalty. A line should be called a robustness improvement only after the selected margin improves and feasibility is acceptable. Otherwise it is only a graph-stiffness improvement.

For controller-aware planning, the line report should be extended to a safe reinforcement envelope:

$$\mathcal{R}_e = \begin{bmatrix} G_e^{(2)} \\ G_e^{(K)} \\ \zeta_{p,e}^{\text{static}} \\ \zeta_{p,e}^{\text{NEP}} \\ \rho_{\text{ctrl},e} \\ \rho_{\text{mem},e} \\ p_{e,\text{max}}^{\text{safe}} \end{bmatrix}. \quad (11.29)$$

Here $\rho_{\text{ctrl},e}$ is a magnitude-normalized controller-channel dominance score, $\rho_{\text{mem},e}$ is the memory-dominance score induced by $\Pi_s(s)$, and $p_{e,\text{max}}^{\text{safe}}$ is the local action budget before a protected margin is exhausted. The controller-distance term is $d_{k\ell}^{\text{ctrl}} = |s_k - \mu_\ell| / (|s_k| + |\mu_\ell|)$, where μ_ℓ is a condensed controller pole. A first-order version is

$$p_{e,\text{max}}^{\text{safe}} = \min_{k,\ell} \left\{ \frac{\zeta_k - \zeta_{\text{req}}}{[-\zeta_{p,e,k}^{\text{NEP}}]_+}, \frac{d_{k\ell}^{\text{ctrl}} - d_{\text{req}}}{[-d_{p,e,k\ell}^{\text{ctrl}}]_+} \right\}, \quad (11.30)$$

with the convention that a zero denominator gives no active local restriction. Finite upgrades must still be checked by continuation or full eigenvalue recomputation.

This envelope separates four planning classes:

1. Safe reinforcement: graph strength and protected damping both improve.
2. Ordinary tradeoff: graph strength improves, but damping ratio falls for the static stiffness-frequency reason.
3. Control-amplified reversal: the controller channel worsens a degradation already visible in the static screen.
4. Controller-induced reversal: the static screen accepts the line but the controller-aware model rejects it.

The fourth class is the most publishable because it says the reduced static model can choose the wrong reinforcement.

11.7 Braess, Braess-Like, and Control-Resonant Reversals

The literature uses Braess' paradox for cases where adding or increasing network capacity decreases system performance [38, 41]. In power systems, the performance loss may be larger flow on an already stressed line, loss of synchrony, or weaker linear stability [39, 40]. The present thesis separates three cases.

Definition 11.2 (Three reinforcement reversals). *Let b_e be an upgraded line.*

1. *A flow Braess reversal occurs when an upgrade increases loading on a protected line in the direction of its pre-existing flow.*
2. *A damping-margin reversal occurs when $\partial h_c / \partial b_e < 0$ or $\partial \zeta_c / \partial b_e < 0$ for the protected critical mode.*
3. *A control-resonant reversal occurs when the line action decreases the damping of a network-family pole, moves it closer to a condensed controller pole, and the self-energy term dominates the NEP sensitivity.*

Only the first case is a classical flow Braess statement. The second is a Braess-like spectral warning. The third is a stronger mechanism that belongs to the rational self-energy layer:

$$\frac{ds}{db_e} = - \frac{y^* (s \Sigma_{b_e}(s) + L_{b_e}) x}{y^* T_s(s) x}. \quad (11.31)$$

If the dominant negative contribution comes from the $s \Sigma_{b_e}(s)$ channel rather than from L_{b_e} alone, the reinforcement is not explained by scalar graph stiffness. It is a network-control resonance.

11.8 Internal Validation Gates from the Project

The project material preserved in `temp/` contains useful evidence, but it must be read as gates and diagnostics. The key validated and negative records are summarized in Table 11.1.

Table 11.1. Internal validation gates used to calibrate the theory. These are not universal claims; they define what the later empirical thesis must reproduce or falsify under registered controls.

Gate	Main record	Thesis use
Adaptive damping screen	In the <code>ieee39_rational_filter</code> output, 300 surrogate trials gave median relative error 4.68×10^{-3} for the diagonal screen, 3.21×10^{-4} for the second-order correction, and 3.69×10^{-4} for the reduced QEP.	Use diagonal damping as a fast screen and reduced QEP as a correction, not as a universal replacement for the full model.
Weak nodes and weak links	The same surrogate ranked bus 30 as the strongest local damping action ($\Delta\zeta = 1.12 \times 10^{-2}$) and line 25–37 as the strongest positive line action ($\Delta\zeta = 6.31 \times 10^{-3}$); other buses and lines were negative.	Weak elements are action- and margin-relative. A centrality list is not enough.
Braess-NEP phase	The <code>phase_braess_nep</code> output tested 460 line-mode perturbations: 348 margin reversals, 232 network-family reversals, and 40 resonant reversals. The clearest case was line 20–34 with self-energy-dominated $\Delta\zeta$ and a reported internal denominator metric of 0.959, which must be recomputed as a magnitude-normalized memory or controller-channel dominance index.	Call the effect control-resonant reinforcement risk after NEP validation, not a generic graph Braess theorem. The missing extraction is the strict subset with $d\zeta_{\text{static}}/dp \geq 0$ and $d\zeta_{\text{NEP}}/dp < 0$.
Inertia-control cross term	The registered full-ANDES gate found 0/13 measurable singleton GENROU-inertia by PLL-gain cross cases; the strongest cross-dominance index was 2.24×10^{-4} .	Keep cross terms in the local theory, but do not use them as the empirical headline.

11.9 Measurement Gate for Frequency Security

Dynamic hosting is also limited by what the protection and estimation layer can see. The estimator draft in the project compares frequency estimators under low-inertia IBR conditions and uses the trip-risk exposure

$$T_{\text{risk}}(e_{\text{th}}) = \sum_{m=0}^{N-1} \Delta t \mathbf{1}_{\{|e[m]| > e_{\text{th}}\}}, \quad (11.32)$$

where $e[m]$ is the estimator error relative to the reference frequency. This metric is included here for one reason: a low RMS estimator can still be unacceptable if its peak error or latency creates a long protection exposure. Thus a complete dynamic hosting study should pair spectral margins with measurement and protection margins. The theoretical hosting set (11.4) can be extended by adding $T_{\text{risk}} \leq T_{\text{max}}$ as a protection-aware constraint.

11.10 Planning Rule

The final planning rule is deliberately conservative.

1. Use static hosting constraints to reject voltage, thermal, and power quality violations.
2. Use $\mathcal{P}_{\text{host}}^{(1)}$ to screen SG-to-IBR conversions by damping headroom.
3. Rank virtual-inertia sites with separate RoCoF and damping-headroom weights; do not mix the objectives silently.
4. Rank line upgrades by $G_e^{(2)}$, $G_e^{(K)}$, and $\partial h_k / \partial b_e$, then reject actions that improve graph stiffness while degrading the protected dynamic margin.
5. If controller states are condensed, verify important candidates with the NEP sensitivity (11.31) or full eigenvalue recomputation.

This rule gives a defensible version of the practical advice: improve the graph first when the weak mechanism is graph-mediated, add damping when the weak mechanism is modal dissipation, add virtual inertia only with the correct frequency-security objective or co-designed damping, and call a reinforcement Braess-like only after the measured margin actually worsens.

Conclusions

This dissertation has developed an operator-valued theory of dynamic grid strength in converter-rich power systems and used it to define dynamic hosting capacity, virtual-inertia placement, weak elements, damping-aware network reinforcement, and a traceable source-to-claim ledger for the project artifacts. The hierarchy begins with the operational graph induced by the power-flow operating point, continues through commuting fixed damping, isolates the uniform-decay case $D = \gamma M$, passes through fixed non-commuting damping, continues to rational self-energy $\Pi(s)$ after controller-state condensation, introduces the Green function $T(s)^{-1}$ as the carrier of finite-action behavior, and then separates genuine nonlinear memory from small-signal NEP theory. The central conclusion is that a damping, hosting, or grid-strength claim is meaningful only relative to the operator, port/action family, margin, and validation gate that carry the mechanism.

12.1 Theoretical Conclusions

A first conclusion is that graph topology becomes dynamic only after the operating point is included. In the lossless retained model, the linearized stiffness is

$$L(z^\star) = BW(z^\star)B^T, \quad w_e = \kappa_e \cos \eta_e.$$

Thus line loading, angle cohesion, cuts, cycles, inertia placement, and local control fields enter the same graph-time operator. Static graph metrics are useful screens, but the protected object is the response of $T(s; z^\star) = s^2 M + sD + BWB^T + \Pi(s; z^\star)$.

The second conclusion is that $D = \gamma M$ is a special floor, not the full decouplable class. The exact fixed- D decoupling condition is $[\tilde{L}, \tilde{D}] = 0$. Under $D = \gamma M$, graph modes decouple and every underdamped mode has real part $-\gamma/2$. Under Rayleigh or other commuting damping, graph modes still decouple, but modal decay is not uniform. This distinction gives a precise foundation for classical graph-frequency intuition and also limits it.

The third conclusion is that fixed non-commuting damping breaks modal decoupling. The graph eigenvectors diagonalize $\tilde{L}$, but the modal damping matrix $\Gamma = Q^T \tilde{D} Q$ need not be diagonal. Off-diagonal damping creates coupled modal dynamics, and eliminating the other modes produces rational corrections. Thus, even a fixed- D system can require a frequency-dependent modal interpretation when one insists on reducing to a single retained mode.

The fourth conclusion is that controller-state condensation changes the reduced operator class. A Schur complement of the eliminated controller block gives

$$T(s) = s^2 M + sD_0 + L + \Pi(s), \quad \Pi(s) = \Pi_K(s) + s\Pi_D(s).$$

This rational correction has poles and residues inherited from the eliminated dynamics. A constant damping matrix cannot represent those features exactly. Static damping bridges are therefore local approximations whose validity must be tested, not assumed.

The fifth conclusion is that dynamic effective resistance closes the loop between graph metrics and finite actions. The Green function $G_T(s) = T(s)^{-1}$ defines

$$\mathcal{R}_e(s) = a_e^T G_T(s) a_e,$$

which reduces to classical effective resistance at the static graph limit and becomes a meromorphic port compliance away from zero. A finite line action satisfies $1 + \Delta b_e \mathcal{R}_e(s) = 0$, and general low-rank actions satisfy a small action determinant. Therefore local sensitivities are tangent approximations of exact finite-action laws, not standalone planning objects.

The sixth conclusion is that reduced models need certificates. A QEP screen, static self-energy approximation, data-identified $\hat{\Pi}(s)$, or condensed NEP should be accepted over a protected contour only when the relative perturbation bound

$$\sup_{\Gamma} \|\hat{T}(s)^{-1}(T(s) - \hat{T}(s))\|_2 < 1$$

or an equivalent certificate preserves the protected eigenvalue count. A margin derivative should be treated as a recommendation only when its sign is larger than its error bound.

The seventh conclusion is that fragility is structured. An unstructured pseudospectrum warns about non-normality, but a planning decision needs a physical perturbation family: line uncertainty, damping, inertia, controller gains, delays, operating-point variation, or identified-model error. The resulting fragility radius measures the smallest admissible perturbation that can move a pole to a protected boundary.

The eighth conclusion is that NEP theory and nonlinear stability are different modules. The Schur NEP preserves linearized memory around an operating point. The nonlinear DAE layer must still address equilibrium regularity, bifurcations, finite-amplitude stability, and regions of attraction. A small-signal damping margin is therefore a local gate, not a large-signal certificate.

The ninth conclusion is that hosting capacity in converter-rich systems is not only a voltage or thermal number. A static-feasible IBR penetration can still violate a damping-ratio margin, a protected resolvent-gain threshold, or a controller-resonance gate. The first-order spectral hosting screen is therefore a useful tangent screen, but it becomes an inner certificate only with curvature bounds and must be checked against the QEP or NEP when modal coupling or controller self-energy is important.

The tenth conclusion is that graph reinforcement and dynamic robustness are not synonymous. A line upgrade may improve algebraic connectivity or effective resistance while reducing damping headroom. The correct planning object is a vector containing graph gain, damping-margin derivative, protected-gain derivative, feasibility, and possible Braess-like or control-resonant reversal flags.

12.2 Consequences for Research Practice

The hierarchy imposes discipline on future simulation and planning studies. Before a graph metric is used as evidence, the study must state whether the operator is an operational graph at a fixed operating point, uniform proportional damping, commuting fixed damping, non-commuting fixed

damping, rational self-energy, or nonlinear memory. Before a weak bus or weak line is named, the margin and action family must be named. Before a reduced converter model is trusted, the spectral effect of removed states must be visible in either the retained matrix or the self-energy.

This thesis therefore separates three epistemic levels:

1. a certificate, which implies a margin under explicit assumptions;
2. a screen, which identifies candidates but does not imply the margin;
3. a gate, which tests whether the screen is reliable enough for the next claim.

The distinction is not editorial. It is necessary because the same graph quantity can be a certificate under proportional damping, a screen under fixed non-commuting damping, and an incomplete descriptor after controller condensation.

12.3 Open Theoretical Work

Several theoretical problems remain open.

1. Develop graph-time resonance atlases that relate operating-point geometry, graph-mode conversion, controller self-energy, and protected frequency-response peaks.
2. Turn the total derivative of $L = BW(z^*)B^T$ into a continuation tool for planning actions, so that line, dispatch, and converter changes are distinguished from frozen-stiffness perturbations.
3. Develop computable error bounds between the full Schur NEP and static or identified approximations of $\Pi(s)$ or its damping-channel projection $\Sigma(s)$ over specified spectral regions.
4. Turn the dynamic effective resistance law $1 + \Delta b_e \mathcal{R}_e(s) = 0$ into scalable algorithms for finite line screening and certified reinforcement budgets.
5. Define non-commutation measures tailored to inertia-normalized power networks, with explicit links to damping-ratio error.
6. Connect passivity or positive-real properties of $\Sigma(s)$ to small-signal damping margins in non-normal retained systems.
7. Replace scalar passivity summaries with directional passivity matrices or dynamic multipliers when MIMO channel geometry is important.
8. Develop a passivity-retuning protocol in which converter-controller gains are adjusted to improve a band-limited self-energy passivity index while also moving the protected Schur NEP pole in a damping-improving direction.
9. Compute structured fragility radii for topology, line loading, damping, inertia, controller, delay, and operating-point uncertainty families.
10. Characterize when grid-strength or Laplacian placement metrics are certificates, screens, or invalid descriptors for a given converter-control family.
11. Test whether cellular-sheaf Laplacians with bus/controller stalks and line restriction maps reveal weak nodes or weak links that scalar Laplacian sensitivity, fixed- D QEP screens, and Schur self-energy diagnostics miss.

12. Build theory-to-validation protocols that combine NEP sensitivity, pole-residue analysis, and full DAE eigenvalue recomputation.
13. Combine Loewner or contour-integral NEP extraction with continuation tracking so that local pole-motion maps can be followed through high-dimensional hosting and controller-retuning sweeps.
14. Extend the dynamic hosting screen beyond first order, including non-commuting damping, controller self-energy, nonlinear memory, structured fragility, and uncertainty in operating points.
15. Identify $\Pi(s)$ from black-box converter data with stability, causality, port, and passivity constraints, then propagate identification error to certified planning decisions.
16. Register line-reinforcement and virtual-inertia planning baselines so that Braess-like reversals and placement gains can be compared against meaningful controls.

12.4 Closing Statement

The transition from synchronous-machine dominated grids to converter-rich grids is often described as a loss of inertia. That description is incomplete. The deeper theoretical change is a change in the strength, damping, fragility, and memory operator. Uniform viscous damping, commuting fixed damping, non-commuting modal damping, rational controller self-energy, Green-function action calculus, structured fragility, and nonlinear memory are different mathematical objects. Static hosting, dynamic hosting, virtual inertia, controller retuning, and line reinforcement are likewise different planning questions unless they share the same port, operator, and protected margin. A doctoral theory of dynamic grid strength must keep those objects separate and then reconnect them through certificates. This dissertation provides that path: from static topology to operational graph-time resonance, from effective resistance to dynamic effective resistance, from fixed D to $\Pi(s)$, from $\Pi(s)$ to structured fragility, and from linear self-energy to nonlinear memory.

Integrated Claim Ledger and Algebraic Research Program

The previous chapters built the theory in layers. This chapter records how the project artifacts in **temp/** and the companion drafts map into that theory. Its purpose is traceability. A doctoral thesis cannot rely on the statement that a formula was “modified to taste.” It must state where the formula comes from, what model it assumes, what it proves, what it only screens, and what validation gate is still missing.

The central integrated object is

$$T(s; \rho) = s^2 M(\rho) + s D_0(\rho) + L(\rho) + \Pi(s; \rho), \quad \Pi(s; \rho) = \Pi_K(s; \rho) + s \Pi_D(s; \rho). \quad (\text{A.1})$$

Here L is the operating-point stiffness inherited from power flow, M is physical or virtual inertia, and $\Pi(s)$ is the condensed controller self-energy. The damping-channel specialization defines $\Sigma(s) = D_0 + \Pi_D(s)$ when Π_K is absent or explicitly absorbed into stiffness. The fixed- D theory is the special case $\Pi(s) = 0$ with direct damping $D_0 = D$. The uniform proportional damping floor is the smaller special case $D = \gamma M$, while the broader decoupled fixed- D class is $[\tilde{L}, \tilde{D}] = 0$. This single inclusion chain is the clean way to organize the project:

$$\begin{aligned} D = \gamma M &\subset \{D : [\tilde{L}, \tilde{D}] = 0\} \\ &\subset \{D : [\tilde{L}, \tilde{D}] \neq 0, D \text{ fixed}\} \\ &\subset \{\Pi(s) \text{ rational}\} \\ &\subset \{\text{nonlinear memory}\}. \end{aligned} \quad (\text{A.2})$$

Chapter visual promise

This chapter turns the project archive into a controlled research program. The reader should be able to trace each idea from source, to operator, to margin, to action, and finally to a validation gate.

Table A.1. Claim-status vocabulary used in this chapter.

Status	Meaning
Algebraic result	The statement follows from matrix identities or perturbation formulas under declared assumptions.
Screen	The calculation ranks candidates or flags risk, but does not by itself certify the protected margin.
Gate	The result is an internal validation or rejection test that constrains what can be claimed next.
Candidate	The mechanism is mathematically coherent but still requires a registered benchmark before it becomes an empirical contribution.

A.1 Source-to-Claim Ledger

Tables A.2 and A.3 convert the reviewed material into thesis claims. The language is deliberately conservative: an algebraic derivation is not a benchmark validation, and a validated internal gate is not an industry-wide theorem.

Table A.2. Ledger linking the main project sources to mathematical objects and claim status.

Source	Mathematical object	Incorporated thesis claim	Status
Handoff protocol	$\zeta_k \simeq \delta_k / (2\sqrt{\nu_k})$, adaptive reduced QEP, line sensitivity warnings	Graph-frequency metrics are screens; damping-margin rankings must use post-action $\zeta_{\min}$ or pole recomputation.	Calibrated protocol
Consolidated blueprint	Damping-region inequality $\delta_k \geq 2\zeta_{\star}\sqrt{\nu_k}$, reduced QEP selection, exact QEP sensitivity	Use the diagonal damping region when modes are separated, then promote only coupled low-damping clusters to a reduced QEP.	Theory plus surrogate gate
Unified attribution note	$T(s) = s^2M + sD_0 + L + \Pi(s)$, self-energy contribution, weak nodes and weak links	The final reduced controller object for IBR-rich models is rational in s ; its damping-channel projection is $\Sigma(s)$, not generally a constant D .	Core thesis frame
Hidden-resonance note	Distance d_{kj}^{ctrl} , self-energy dominance $\phi_{\Sigma}^{\text{abs}}$, hidden action margin	A stable pole may be close to a controller-resonant damping loss even when its present damping ratio is acceptable.	Candidate diagnostic
Line-shunt derivation	$\text{Sym}(n) = \mathcal{L}_{\text{all}} \oplus \mathcal{D}_{\text{sh}}$ and $\partial \tilde{L} / \partial M_i$	Virtual inertia has an existing-line plus shunt structural footprint in the inertia-normalized stiffness; it does not create new-line directions.	Algebraic result

Table A.3. Ledger linking internal drafts and validation records to incorporated claims.

Source	Mathematical object	Incorporated thesis claim	Status
Hosting-capacity draft PDF	Mass-normalized Laplacian, first-order spectral hosting screen, stochastic fragility, edge and node fragility	Dynamic hosting capacity should be a set constrained by damping, resolvent, and stochastic frequency-security margins, not only voltage limits.	Internal draft, integrated cautiously
Beyond-voltage draft PDF	Kirchhoff-consistent AC power flow, weak nodes, hidden dynamic margins	Voltage-safe operating points can still violate dynamic spectral margins; the power-flow Jacobian is the bridge from electrical operating point to L .	Internal draft, integrated cautiously
Frequency-estimator draft and ZIP	T_{risk} , latency, peak-error exposure	A dynamic hosting certificate should include what the protection and measurement layer can actually observe.	Protection gate
Braess-NEP phase	NEP line sensitivity with self-energy contribution	Some line reinforcements are control-resonant damping risks, not generic topological Braess theorems.	Internal NEP gate
Rational-filter output	Diagonal, second-order, reduced-QEP, and adaptive errors	The adaptive estimator is useful because diagonal screens are often accurate but fail under clustered or strongly coupled damping.	Surrogate validation
Coordination-lever gate	Equal-cost topology, damping, and inertia packages	Mixed lever packages can outperform the best single lever, but modal and resolvent margins need separate reporting.	Planning gate

A.2 Step-by-Step Analysis Stack

The analysis should be read as a sequence. Each step adds one layer of physics or mathematics.

1. **Start from the electrical operating point.** Bus voltages, current injections, and admittances satisfy $I = Y_{\text{bus}}V$. Complex power is $S_i = P_i + jQ_i = V_i I_i^*$. This gives the nonlinear AC power-flow map.
2. **Linearize active power with respect to angle.** At a feasible operating point, the block $J_{P\theta} = \partial P / \partial \theta$ gives the synchronizing stiffness. After the reference-angle direction is removed, write the positive stiffness as L . In the lossless small-angle case, L is a weighted Laplacian.
3. **Attach nodal dynamics.** Structure-preserving power-system models retain bus-level dynamics instead of hiding all loads inside a reduced network [51, 1, 52, 53]. For the electromechanical layer this gives

$$M\ddot{\theta} + D\dot{\theta} + L\theta = p.$$

4. **Remove the uniform angle mode.** Work on a subspace orthogonal to the rotational symmetry. If $R^T \mathbf{1} = 0$, define

$$M_r = R^T M R, \quad D_r = R^T D R, \quad L_r = R^T L R.$$

5. **Normalize inertia.** With

$$\tilde{L} = M_r^{-1/2} L_r M_r^{-1/2}, \quad \tilde{D} = M_r^{-1/2} D_r M_r^{-1/2},$$

the QEP becomes

$$(s^2 I + s\tilde{D} + \tilde{L})v = 0.$$

6. **Diagonalize stiffness, not damping.** Let $\tilde{L}Q = Q \text{diag}(\nu_k)$. The modal damping matrix is

$$\Gamma = Q^T \tilde{D} Q.$$

Only when Γ is diagonal do graph modes decouple exactly.

7. **Use the diagonal damping screen first.** The first screen is

$$\hat{\zeta}_k = \frac{\Gamma_{kk}}{2\sqrt{\nu_k}}, \quad h_k = \Gamma_{kk} - 2\zeta_* \sqrt{\nu_k}.$$

It is exact under proportional or commuting damping, and a first-order diagnostic otherwise.

8. **Promote dangerous clusters to a reduced QEP.** If off-diagonal modal damping is large or the graph frequencies are clustered, solve the QEP on the selected modal subspace.
9. **Condense controllers only after admitting that the object changes.** Eliminating controller states produces $\Pi(s)$, with $\Sigma(s)$ only as a declared damping-channel specialization, so the problem becomes the rational NEP (A.1).
10. **Plan with margins, not labels.** A bus, line, or controller is weak only relative to a margin and an action family: $\zeta_{\min}$, protected resolvent gain, control-pole distance, stochastic variance, or protection exposure.

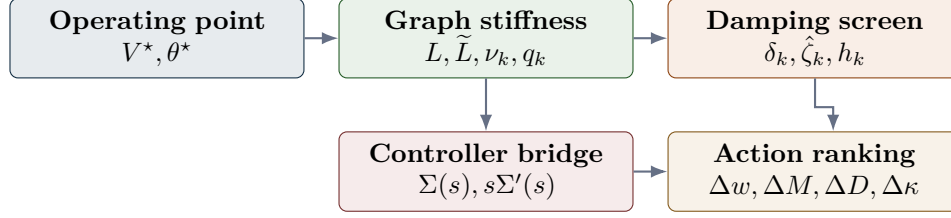


Figure A.1. Analysis stack used by the integrated research program. The diagonal damping screen is the fast route; the controller bridge is invoked when static damping cannot preserve the pole-motion mechanism.

A.3 Adaptive Reduced-QEP Algebra

The off-diagonal entries of Γ are the algebraic reason the scalar screen can fail. A compact coupling diagnostic is

$$\eta_{\text{off}} = \frac{\|\Gamma - \text{diag}(\Gamma)\|_F}{\|\Gamma\|_F}, \quad (\text{A.3})$$

and a mode-pair diagnostic is

$$c_{k\ell} = \frac{|\Gamma_{k\ell}|}{|\sqrt{\nu_k} - \sqrt{\nu_\ell}| + \varepsilon_{\text{gap}}}. \quad (\text{A.4})$$

Large $c_{k\ell}$ means that the damping coupling is not small relative to the frequency separation. This is a direct application of matrix perturbation logic [54, 45]: small denominators make apparently small off-diagonal terms important.

The adaptive estimator used by the project is therefore:

1. Compute (ν_k, q_k) of $\tilde{L}$ and $\Gamma = Q^T \tilde{D} Q$.
2. Rank all modes by $\hat{\zeta}_k = \Gamma_{kk} / (2\sqrt{\nu_k})$.
3. Select low- $\hat{\zeta}_k$ modes and augment the set with neighbors whose $c_{k\ell}$ exceeds a declared threshold.
4. Build $V = [q_k]_{k \in \mathcal{I}}$, $L_{\mathcal{I}} = V^T \tilde{L} V$, and $D_{\mathcal{I}} = V^T \tilde{D} V$.
5. Solve

$$\det(s^2 I_{\mathcal{I}} + s D_{\mathcal{I}} + L_{\mathcal{I}}) = 0.$$

6. Report the reduced-QEP damping margin and the gap to the diagonal screen. If the gap is large, the graph-only explanation is not enough.

The `ieee39_rational_filter` surrogate record supports this workflow: over 300 trials the diagonal screen had median relative error 4.68×10^{-3} , while the second-order correction and reduced-QEP variants reduced the median to approximately 3.2×10^{-4} and 3.7×10^{-4} , respectively. This does not prove universal accuracy. It does justify using the diagonal formula as a fast screen and the reduced QEP as a targeted correction.

A.4 Exact Lever Sensitivities

Let p be any scalar action: line weight, local inertia, local damping, converter gain, or hosting-penetration coordinate. For the fixed- D QEP

$$T_Q(s, p) = s^2 I + s \tilde{D}(p) + \tilde{L}(p),$$

with right and left eigenvectors v and y , a simple pole satisfies

$$\frac{ds}{dp} = -\frac{y^* \left(s \tilde{D}_p + \tilde{L}_p \right) v}{y^* \left(2sI + \tilde{D} \right) v}. \quad (\text{A.5})$$

This is the QEP version of the nonlinear eigenvalue sensitivity used for matrix polynomials and NEPs [5, 14, 55]. If $s = \sigma + j\omega$, $ds/dp = a + jb$, and $\zeta = -\sigma/|s|$, then

$$\frac{d\zeta}{dp} = \frac{\sigma\omega b - \omega^2 a}{|s|^3}. \quad (\text{A.6})$$

This equation is important because a line can move a pole left in real part and still reduce damping ratio if it increases oscillation frequency too much.

For a line $e = (i, j)$ in the inertia-normalized graph layer,

$$\frac{\partial \nu_k}{\partial w_e} = \left(\frac{q_k(i)}{\sqrt{M_i}} - \frac{q_k(j)}{\sqrt{M_j}} \right)^2. \quad (\text{A.7})$$

But damping requires more than $\partial \nu_k / \partial w_e$. With fixed modal labeling,

$$\frac{\partial \zeta_k}{\partial w_e} = \frac{1}{2\sqrt{\nu_k}} \frac{\partial \delta_k}{\partial w_e} - \frac{\delta_k}{4\nu_k^{3/2}} \frac{\partial \nu_k}{\partial w_e}, \quad (\text{A.8})$$

$$\frac{\partial \delta_k}{\partial w_e} = q_k^T \tilde{D}_{w_e} q_k + 2 \sum_{\ell \neq k} \frac{(q_\ell^T \tilde{L}_{w_e} q_k)(q_\ell^T \tilde{D} q_k)}{\nu_k - \nu_\ell}. \quad (\text{A.9})$$

The first term in (A.9) is direct damping change. The sum is modal rotation. It is zero under proportional damping and can be decisive under heterogeneous damping.

For the controller-condensed NEP (A.1),

$$\frac{ds}{dp} = -\frac{y^* (s^2 M_p + s D_{0,p} + L_p + \Pi_p(s)) x}{y^* (2sM + D_0 + \Pi_s(s)) x}. \quad (\text{A.10})$$

Here the subscript p denotes a total derivative after re-solving the operating point and updating the controller blocks. The additional term $\Pi_s(s)$, which reduces to $\Sigma(s) + s\Sigma'(s) - D_0$ in the damping-channel case, is the spectral memory of the condensed controller block. It is exactly the term that a static damping bridge removes.

A.4.1 Mechanism-Resolved Attribution

Equation (A.10) is not only a way to compute a pole derivative. It is also an attribution identity once the total derivative is logged by mechanism. For a declared operating-point continuation, write

$$T_p = T_p^{\text{net}} + T_p^{\text{op}} + T_p^{\text{ports}} + T_p^{\text{ctrl}} + T_p^{M,D}. \quad (\text{A.11})$$

The terms mean, respectively: direct network change, equilibrium or power-flow shift, change in controller port visibility and residues, movement of condensed controller poles, and direct mass-damping change. This split is not unique until the model assembly is fixed, but after that bookkeeping choice it is algebraic.

Let $s = \sigma + j\omega$ and define the real linear map

$$\mathcal{D}_\zeta(s)[\eta] = \frac{\sigma\omega \operatorname{Im} \eta - \omega^2 \operatorname{Re} \eta}{|s|^3}. \quad (\text{A.12})$$

For each channel $c \in \{\text{net}, \text{op}, \text{ports}, \text{ctrl}, \text{MD}\}$, set

$$g_c = \mathcal{D}_\zeta(s) \left[\frac{-y^* T_p^c x}{y^* T_s x} \right]. \quad (\text{A.13})$$

Then

$$\frac{d\zeta}{dp} = g_{\text{net}} + g_{\text{op}} + g_{\text{ports}} + g_{\text{ctrl}} + g_{M,D}. \quad (\text{A.14})$$

The denominator $y^* T_s x$ is shared by all channels. Therefore the rational memory term $\Pi_s(s)$ does more than add one extra numerator contribution: it can rotate or amplify every channel simultaneously. This is the causal language needed to turn a pole plot into a mechanism statement.

Definition A.1 (Strict controller-induced reinforcement reversal). *For a line or parameter action p , a strict controller-induced reinforcement reversal occurs when a network-strength score improves and the static reduction does not predict damping degradation, but the controller-aware NEP does:*

$$\frac{dG_{\text{graph}}}{dp} > 0, \quad \frac{d\zeta_{\text{static}}}{dp} \geq 0, \quad \frac{d\zeta_{\text{NEP}}}{dp} < 0. \quad (\text{A.15})$$

Equivalently, the static and controller-aware models disagree in sign:

$$\operatorname{sign} \left(\frac{d\zeta_{\text{static}}}{dp} \right) \neq \operatorname{sign} \left(\frac{d\zeta_{\text{NEP}}}{dp} \right). \quad (\text{A.16})$$

This definition is intentionally stronger than the statement $d\lambda_2/dp > 0$ and $d\zeta/dp < 0$. The latter can occur in an elementary QEP because increasing stiffness may raise modal frequency faster than decay. The stronger claim is that the usual static reduction can recommend the wrong line or parameter action.

Proposition A.1 (Controller-induced reversal certificate). *Assume a simple protected pole and the decomposition (A.11). Suppose the rational self-energy admits a local pole-residue expansion*

$$\Pi(s, p) = \sum_{\ell} \frac{Q_{\ell}(s, p)}{s - \alpha_{\ell}(p)} + \Pi_{\text{rem}}(s, p), \quad (\text{A.17})$$

and the contribution of all neglected or grouped terms to $d\zeta/dp$ is bounded above by $\bar{r}$. Let

$$g_0 = g_{\text{net}} + g_{\text{op}} + g_{M,D} \quad (\text{A.18})$$

and let g_{ℓ} be the damping-ratio contribution of one dominant controller pole α_{ℓ} :

$$g_{\ell} = \mathcal{D}_\zeta(s) \left[\frac{-y^* \Pi_{p,\ell}(s) x}{y^* T_s x} \right], \quad \Pi_{p,\ell}(s) = \frac{Q_{\ell,p}(s)}{s - \alpha_{\ell}} + \frac{Q_{\ell}(s) \alpha_{\ell,p}}{(s - \alpha_{\ell})^2}. \quad (\text{A.19})$$

If

$$g_0 \geq 0, \quad g_0 + g_{\ell} + \bar{r} < 0, \quad (\text{A.20})$$

then the first-order degradation is certified as controller-induced rather than merely controller-correlated.

Proof. Linearity of (A.10) and (A.12) gives the channel sum (A.14). The pole-residue derivative in (A.19) separates residue change from movement of the condensed pole. If the non-controller baseline is nonnegative and the dominant controller-pole term plus the worst admissible remainder is negative, then every compatible completion of the grouped remainder has negative total first-order damping-ratio derivative. The sign reversal is therefore forced by the rational controller channel. $\square$

A.5 Hidden Control-Resonance Margin

Let $\mu_j \in \text{spec}(A_{cc})$ be a condensed controller pole and let s_k be a retained network-family pole. Define

$$d_{kj}^{\text{ctrl}} = \frac{|s_k - \mu_j|}{|s_k| + |\mu_j|}, \quad \phi_{\Sigma}^{\text{abs}}(k) = \frac{|y_k^* s_k \Sigma'(s_k) x_k|}{|y_k^* 2s_k M x_k| + |y_k^* \Sigma(s_k) x_k| + |y_k^* s_k \Sigma'(s_k) x_k| + \varepsilon_{\text{reg}}}. \quad (\text{A.21})$$

Small d_{kj}^{ctrl} and large $\phi_{\Sigma}^{\text{abs}}(k)$ mean that a presently stable pole lies in the resolvent shadow of a controller pole. If A_{cc} is diagonalizable with eigenvector condition factor κ_c , then outside $\text{spec}(A_{cc})$

$$\|(sI - A_{cc})^{-1}\| \leq \frac{\kappa_c}{\Delta_c(s)}, \quad \Delta_c(s) = \min_j |s - \mu_j|, \quad (\text{A.22})$$

and

$$\|\Sigma'(s)\| \leq \frac{\|C\| \|B\| \kappa_c^2}{\Delta_c(s)^2}. \quad (\text{A.23})$$

Thus the warning is algebraic, not rhetorical: as a retained pole approaches a condensed controller pole, the self-energy slope can become large. $\phi_{\Sigma}^{\text{abs}}$ is a dominance index, not a strict fraction; the absolute denominator avoids artificial inflation or suppression caused by complex cancellation in $y^* T_s x$.

For an admissible action p and target margin ζ_* , define

$$m_p^{\zeta}(k) = \frac{\zeta_k - \zeta_*}{[-\partial \zeta_k / \partial p]_+}, \quad m_p^d(k, j) = \frac{d_{kj}^{\text{ctrl}} - d_*}{[-\partial d_{kj}^{\text{ctrl}} / \partial p]_+}. \quad (\text{A.24})$$

The hidden margin is

$$m_p^{\text{hid}}(k, j) = \min\{m_p^{\zeta}(k), m_p^d(k, j)\}. \quad (\text{A.25})$$

This quantity says how much of action p can be applied before either the damping threshold or the controller-distance threshold is locally exhausted. The `phase_braess_nep` record gives one internal mechanism gate: 460 line-mode perturbations produced 40 strict resonant cases, and the clearest line action moved a network-family mode toward a PLL-dominated condensed pole with a self-energy-dominated damping decrease. The correct thesis claim is therefore control-resonant reinforcement risk, not a universal Braess paradox. The next extraction must count the smaller but stronger subset satisfying (A.15); that subset is the empirical target for the controller-induced reversal paper.

A.5.1 Static Adequacy Gate and Resolvent Shadow

The practical question is when the fixed- D or static-QEP screen is sufficient. Let Ω_k be a disk or contour neighborhood of a protected pole s_k with radius r_k . A first-order staticization error indicator is

$$\varepsilon_{\text{static},k} = \kappa_k r_k \sup_{\xi \in \Omega_k} \|\Pi_s(\xi)\|_2, \quad \kappa_k = \frac{\|x_k\|_2 \|y_k\|_2}{|y_k^* T_s(s_k) x_k| + \varepsilon_{\text{reg}}}. \quad (\text{A.26})$$

If $\varepsilon_{\text{static},k} < \tau m_k^{\text{margin}}$ for a declared tolerance τ , then the static screen is acceptable for that mode-action pair. Otherwise the candidate lies in the resolvent shadow and must be checked with the Schur NEP or the full DAE model. This turns the hierarchy into a computational policy: use the cheap QEP broadly, escalate only shadowed modes, and reserve full-model recomputation for finalists.

Distance to a condensed pole is not enough when A_{cc} is non-normal. The screen should combine port-visible resolvent gain, modal projection, rational memory, and pole conditioning:

$$\chi_c(s_k) = \left\| C(s_k I - A_{cc})^{-1} (B_\theta + s_k B_\omega) \right\|_2, \quad (\text{A.27})$$

$$\chi_{k\ell}^{\text{proj}} = \frac{|y_k^* Q_\ell(s_k) x_k|}{|s_k - \alpha_\ell| \|y_k\|_2 \|x_k\|_2 + \varepsilon_{\text{reg}}}, \quad (\text{A.28})$$

$$\rho_{\text{mem},k} = \frac{|y_k^* \Pi_s(s_k) x_k|}{|y_k^* (2s_k M + D_0) x_k| + |y_k^* \Pi_s(s_k) x_k| + \varepsilon_{\text{reg}}}. \quad (\text{A.29})$$

The desired empirical result is not that every large value produces a reversal. It is that strict sign reversals concentrate where $\chi_{k\ell}^{\text{proj}}$, $\rho_{\text{mem},k}$, and κ_k are large. That statement can be tested with precision, recall, false-negative rate, and the share of strict reversals captured after reviewing only the top decile of line actions by resolvent-shadow score.

A.6 Stochastic Fragility and Protection Exposure

The hosting draft also contains a stochastic interpretation. In first-order state form

$$\dot{x} = Ax + B\xi, \quad z = Cx, \quad (\text{A.30})$$

with white input ξ , the stationary covariance X satisfies

$$AX + XA^T + BB^T = 0, \quad \mathbb{E} \|z\|^2 = \text{trace}(CXC^T). \quad (\text{A.31})$$

This is the same Gramian object used in network coherence analysis [56, 57]. In a homogeneous proportional limit, the graph contribution reduces to Kirchhoff-index-like quantities such as $\text{trace}(L^+)$, which explains why effective resistance can describe average vulnerability. In heterogeneous damping or rational self-energy models, however, the covariance must be computed from the actual linearization or an explicitly justified reduction.

The protection layer adds a separate constraint. Let $e[m]$ be a frequency estimator error and e_{th} a protection threshold. The estimator draft uses

$$T_{\text{risk}} = \sum_{m=0}^{N-1} \Delta t \mathbf{1}_{\{|e[m]| > e_{\text{th}}\}}, \quad (\text{A.32})$$

which measures how long a protection-relevant error is exceeded. A complete dynamic hosting set can therefore be written as

$$\mathcal{H}_{\text{dyn}}^{\text{full}} = \{ \rho \in \mathcal{H}_{\text{stat}} : \zeta_{\min}(\rho) \geq \zeta_\star, \ g_{\text{prot}}(\rho) \leq g_\star, \ \text{trace}(CX(\rho)C^T) \leq \sigma_\star^2, \ T_{\text{risk}}(\rho) \leq T_\star, \ d_{kj}^{\text{ctrl}}(\rho) \geq d_\star \}. \quad (\text{A.33})$$

The point is not that every study must include every constraint. The point is that each omitted constraint must be named.

A.7 Pole-Motion Cone and Planning Optimization

Collect line, inertia, damping, and controller actions in

$$u = [\Delta w \quad \Delta M \quad \Delta D \quad \Delta \kappa]^T. \quad (\text{A.34})$$

For each monitored pole s_k , NEP sensitivity gives

$$\Delta s_k \simeq J_k u, \quad J_k = [\partial s_k / \partial p_1 \quad \cdots \quad \partial s_k / \partial p_r]. \quad (\text{A.35})$$

Through (A.6), this induces

$$\Delta \zeta_k \simeq g_k^T u.$$

Feasible actions therefore form a cone of pole motions in the complex plane. Safe actions lie in the damping-improving half-space. Hidden-resonant actions may still improve a graph metric while moving a pole toward a dangerous controller resolvent region.

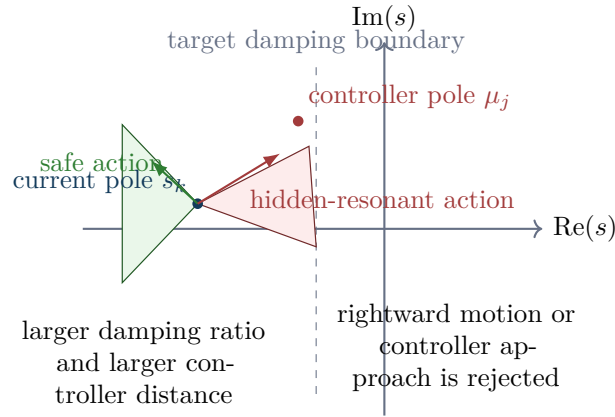


Figure A.2. Pole-motion cone induced by the sensitivity matrix J_k . A graph action is acceptable only if its pole-motion vector improves the protected margin and does not move the mode into a controller-resonant region.

A local planning problem follows:

$$\begin{aligned} & \max_{u,t} \quad t & (\text{A.36}) \\ \text{subject to} \quad & \zeta_k + g_k^T u \geq t, \quad k \in \mathcal{K}, \\ & d_{kj}^{\text{ctrl}} + h_{kj}^T u \geq d_\star, \quad (k,j) \in \mathcal{R}, \\ & c^T |u| \leq B, \quad u_{\min} \leq u \leq u_{\max}, \\ & \text{AC, thermal, voltage, and protection feasibility gates hold.} \end{aligned}$$

This problem states exactly what the project can do with the new analysis: rank SG-to-IBR conversions by dynamic headroom, identify where virtual damping or inertia is honest, find line upgrades that improve damping rather than only stiffness, detect control-resonant line risks, and compare mixed packages of topology, damping, inertia, and control under equal cost.

A.8 What This Analysis Enables

The framework is useful because it converts broad planning questions into declared algebraic tasks. Table A.4 summarizes the main actions.

Table A.4. Planning and research tasks enabled by the integrated analysis.

Task	Algebraic object	Output	Required gate
SG-to-IBR hosting direction	$h_k(\rho) = \delta_k - 2\zeta_\star \sqrt{\nu_k}$	Maximum conversion direction before damping headroom is exhausted	Full QEP or NEP recomputation
Virtual inertia placement	$\partial h_k / \partial M_i$, RoCoF and damping weights	Bus ranking that separates frequency support from damping support	Post-action $\zeta_{\min}$ and time-domain response
Line reinforcement without batteries	$\partial \nu_k / \partial w_e$, $\partial \zeta_k / \partial w_e$	Line list that improves margin, not only graph stiffness	Flow, voltage, and damping checks
Control-resonant risk audit	d_{kj}^{ctrl} , $\phi_\Sigma^{\text{abs}}(k)$, $s\Sigma'(s)$	Flag for stable poles that are close to controller-mediated damping loss	Schur NEP or full DAE eigenanalysis
Mixed lever package design	Pole-motion cone $J_k u$ and budget $c^T u \leq B$	Equal-cost packages of topology, damping, inertia, and control	Registered baseline comparison
Protection-aware hosting	T_{risk} , estimator latency, protected gain	Hosting set that includes what protection can observe	Estimator and relay exposure test
Self-energy passivity retuning	$\mu_\Sigma(\Omega) = \inf_{\omega \in \Omega} \lambda_{\min} H_\Sigma(\omega)$ and $ds_\star/d\rho$	Controller-gain update that improves both a passivity screen and the protected pole-motion direction	Schur NEP plus full-model retuning gate
Sheaf weak-element research	Cellular-sheaf Laplacian with bus/controller stalks and line restriction maps	Candidate weak-node or weak-link certificate that is richer than scalar Laplacian sensitivity	New benchmark showing information not recovered by L , QEP, or $\Sigma(s)$ alone

The last two rows are deliberately not ordinary planning claims. The passivity row is a near-term extension because it uses the existing Schur self-energy operator and adds a controller-respected dissipativity screen. The sheaf row is a longer-term research program. It may become a powerful structural language only if it produces a weak-element certificate that scalar graph theory and the present self-energy diagnostics cannot already reproduce.

A.9 Three Go–No-Go Studies

The next work should be staged sequentially. First, the sign-characteristic bridge must be grounded in the exact hypotheses of the matrix-polynomial literature. Second, the passivity-retuning result should be made operational because it extends the existing $\Sigma(s)$ framework and can fail cleanly. Third, the cellular-sheaf idea should be tested as a falsifiable structural extension, not adopted as a new language by default.

A.9.1 Sign-Characteristic Gate

The strongest possible theoretical headline is not that controller condensation makes a rational operator. That is Schur-complement mathematics. The stronger claim is that a planning action can move a network-family pole through the resolvent of a condensed controller and produce a sign collision that a constant- D model cannot predict.

This claim needs a gate before it becomes a theorem. The gate has four steps:

1. Read the sign-characteristic definitions in the Hermitian matrix-polynomial and self-adjoint meromorphic literature and list their assumptions: regularity, symmetry, leading coefficient, elementary divisors, and structure-preserving transformations.
2. State exactly which assumptions fail for $T(s) = s^2M + sD_0 + L + \Pi(s)$: rationality, dissipation, non-normality, and loss of a fixed symplectic or Hermitian product.
3. Build a two-bus SCR benchmark with a synchronous source and a grid-following converter. The reduced coefficients must come from the exact linearized algebraic-differential Jacobian; placeholders such as “or similar” are not allowed.
4. Track three quantities while varying SCR or PLL bandwidth: χ_k^{op} , ϕ_Σ^{abs} , and the full-model eigenvalues. The gate passes only if the same event appears as signature change or singularity, self-energy dominance, and damping loss.

If this gate passes, the first Transactions-style paper should be narrowly framed around controller-resolvent reinforcement reversals:

$$\frac{dG_{\text{graph}}}{dp} > 0, \quad \frac{d\zeta_{\text{static}}}{dp} \geq 0, \quad \frac{d\zeta_{\text{NEP}}}{dp} < 0,$$

with the degradation attributed to $\Pi_p(s)$ or $\Pi_s(s)$, not to the static graph term alone. That paper should not mix frequency-estimator latency, stochastic fragility, sheaves, and broad hosting optimization. The minimum package is a structured Schur derivation, total NEP sensitivity, a channel-resolved attribution, finite-difference validation, and ablation of L_p , Π_p , and Π_s . The empirical headline should be the rate of wrong-sign static recommendations,

$$\Pr \left[\text{sign}(\zeta_p^{\text{static}}) \neq \text{sign}(\zeta_p^{\text{full}}) \right],$$

plus the recall achieved by the resolvent-shadow filter when only a small fraction of line actions are escalated to the full model.

Table A.5. Minimal result package for the controller-induced reinforcement reversal paper.

Result	Claim	Evidence required
A	Exact Schur derivation and mechanism-resolved sensitivity attribution	Derive $\Pi(s)$, T_s , and the channel gains g_c .
B	Robust sufficient certificate for controller-induced sign reversal	Show $g_0 \geq 0$ and $g_0 + g_\ell + \bar{r} < 0$ under a registered remainder bound.
C	Static model wrong-sign rate	Report strict cases where $d\zeta_{\text{static}}/dp \geq 0$ but $d\zeta_{\text{full}}/dp < 0$.
D	Resolvent-shadow computational filter	Measure precision, recall, false negatives, and cases captured by the top 5–10% of candidate line actions.

A.9.2 Passivity Retuning Study

For a controller parameter κ , such as a PLL bandwidth or integral gain, the core object is

$$\mu_\Sigma(\Omega; \kappa) = \inf_{\omega \in \Omega} \lambda_{\min} H_\Sigma(\omega; \kappa).$$

The computational question is whether μ_Σ can be moved from negative to positive after a stressed operating condition or an $N - 1$ contingency, while the protected eigenvalue also moves toward the required damping region. The study is deliberately narrow:

1. Select a contingency, IBR penetration level, and weak-grid operating point; re-solve the power flow instead of freezing the pre-contingency linearization.
2. Sweep κ over the admissible controller range and compute $\mu_\Sigma(\Omega; \kappa)$ from the Schur self-energy.
3. Record the active frequency $\omega_\star$, active Hermitian eigenvector $q_\star$, and the envelope derivative $q_\star^H (\partial H_\Sigma / \partial \kappa) q_\star$.
4. Identify whether the margin is monotone, locally unimodal, or nonregular because active frequencies switch.
5. At the best candidate $\kappa_\star$, recompute the full DAE eigenvalues and time-domain response. The passivity screen is a gate; the final claim still needs the full-model check.
6. Compare against a static nodal- D screen. The negative control is important: if a scalar damping change predicts the same action, the rational self-energy did not add design information.

A successful result is not merely a plot where an eigenvalue moves left. It is a three-part certificate: the self-energy passivity margin improves, the protected pole moves in the desired direction, and the full model confirms the damping recovery. Failure is also useful. If μ_Σ is nonmonotone with

multiple active-frequency switches, the thesis has identified a controller interaction that cannot be summarized by one PLL retuning knob.

A.9.3 Cellular-Sheaf Weak-Link Test

The sheaf extension has one binary question:

Does a nonconstant sheaf Laplacian reveal a weak link that scalar $\partial\nu_k/\partial w_e$ does not?

The smallest useful construction is enough for this test. Let each converter bus carry a stalk $\mathcal{F}(v_i) \simeq \mathbb{R}^2$ representing a local PLL phase and frequency frame. Let each edge stalk $\mathcal{F}(e_{ij})$ represent the frame discrepancy that the line must reconcile. The restriction maps $\mathcal{F}_{v_i \rightarrow e_{ij}}$ and $\mathcal{F}_{v_j \rightarrow e_{ij}}$ are determined from the solved operating angles and the converter reference frames. With identity restrictions, this reduces to the constant sheaf and recovers the ordinary scalar graph intuition. With nontrivial frame rotations, the sheaf coboundary

$$\delta_{\mathcal{F}} : C^0(G; \mathcal{F}) \rightarrow C^1(G; \mathcal{F})$$

produces the sheaf Laplacian $L_{\mathcal{F}} = \delta_{\mathcal{F}}^* \delta_{\mathcal{F}}$ on node stalks.

The go-no-go protocol is:

1. Build $L_{\mathcal{F}}$ on the same 39-bus operating point used by the QEP and Schur studies. The solved angles, not the bare topology, define the restriction maps.
2. Compare the low spectrum of $L_{\mathcal{F}}$ with the scalar inertia-normalized Laplacian. The comparison should be made under the same loading and IBR placement assumptions.
3. Remove or down-weight each candidate line and measure the collapse of the lowest informative sheaf eigenvalues. This gives a sheaf weak-link ranking.
4. Compare that ranking against $\partial\nu_k/\partial w_e$ and the existing damping-risk vector. A material ranking difference is the first success condition.
5. Compute the first harmonic obstruction $\dim \ker \delta_{\mathcal{F}}^*$ on high-angle subgraphs. A nonzero value would indicate a frame-consistency obstruction, i.e., a synchronization frustration not visible in a scalar Laplacian screen.
6. Validate any sheaf-only weak link in the full model. A sheaf ranking that does not predict a changed damping, synchronization, or protection-relevant response is only a mathematical artifact.

The decision rule should be explicit. The sheaf program is a *GO* only if it produces a weak-link or weak-node ranking that differs materially from the scalar ranking in at least one sweep case, and the difference has a physical explanation in frame rotation, PLL alignment, or controller-coordinate compatibility. It is a *NO-GO* if the sheaf ranking reproduces the scalar ranking up to scale. In that case the sheaf remains a future mathematical extension, while the passivity-retuning paper remains the safer near-term contribution.

A.10 Mathematical Audit Gates Before Publication

The following gates prevent the thesis from overclaiming:

1. **Structured Schur gate.** A condensed model must state whether the rational correction is $\Pi(s)$, $\Pi_K(s)$, or $s\Pi_D(s)$. Calling the whole Schur correction “dynamic damping” is not allowed unless the stiffness channel has been eliminated or justified.
2. **Sign-convention gate.** The active-power Jacobian must be tied to one convention. This thesis uses exported electrical power, so $L = J_{P\theta}$ on retained angles.
3. **Invariant diagnostic gate.** Mode-family labels must use physical weights, port visibility, left-right participation, or projectors. Raw controller-state norm ratios are not admissible evidence.
4. **Curvature gate.** A first-order hosting polytope is a tangent screen. An inner hosting certificate needs a trust region and a Hessian or model error bound.
5. **Cluster gate.** Near repeated or clustered modes, use spectral projectors and eigenvalue conditioning rather than derivatives of a named mode.
6. **Resolvent gate.** Controller-pole distance is only a proxy. Non-normal A_{cc} requires resolvent gain or port-visible self-energy dominance.
7. **Passivity gate.** A Hermitian part of $\Sigma(j\omega)$ has a passivity interpretation only after the conjugate ports and supply rate are declared.
8. **Total-derivative gate.** A line, inertia, or controller action must be differentiated through the re-solved operating point whenever the claim is a planning claim rather than a frozen-linearization screen.

A.11 Validation Ladder

Every claim should climb the same ladder:

1. **Algebraic identity.** Verify formulas such as $\partial\tilde{L}/\partial M_i$ and the line-shunt decomposition to numerical precision.
2. **Surrogate QEP.** Compare diagonal, second-order, reduced-QEP, and full-QEP damping margins over controlled synthetic systems.
3. **IEEE topology QEP.** Use realistic network topology and operating points, while still keeping the reduced second-order abstraction explicit.
4. **Schur NEP.** Condense controller states and test whether $\Pi(s)$, its damping-channel projection $\Sigma(s)$, and resolvent dominance explain pole motion.
5. **Full ANDES eigenanalysis.** Recompute the complete DAE linearization after each selected action.
6. **Time-domain and protection gate.** Check RoCoF, nadir, oscillation decay, estimator exposure, and protection-relevant thresholds.

Negative gates are part of the thesis. The registered inertia-control cross term is mathematically real in the local expansion, but the full-ANDES singleton gate found no measurable effect in 13 tested cases. The correct use is therefore as a possible second-order diagnostic, not as a headline. The same discipline applies to hidden margins, line reinforcement, and virtual inertia: the theory names the mechanism; validation decides whether it is a planning claim in a given benchmark.

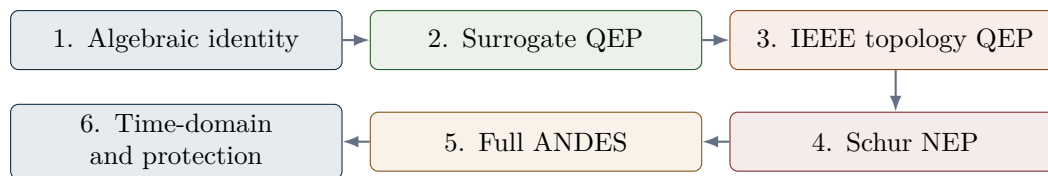


Figure A.3. Validation ladder for this research program. A result becomes more credible as it moves from algebraic identity to full linearization and time-domain/protection checks.

Reproducibility of the Theoretical Manuscript

This appendix records how to rebuild the thesis artifact. It intentionally does not define a simulation protocol. The present manuscript is the theoretical part of the dissertation.

B.1 Source Layout

The LaTeX source lives under:

```
reports/thesis/
```

The main file is:

```
reports/thesis/main.tex
```

The chapter files are:

```
reports/thesis/chapters/
```

The bibliography is:

```
reports/thesis/references.bib
```

B.2 Compile the Thesis

From the repository root, the manual build is:

```
cd reports/thesis
pdflatex -interaction=nonstopmode -halt-on-error main.tex
bibtex main
pdflatex -interaction=nonstopmode -halt-on-error main.tex
pdflatex -interaction=nonstopmode -halt-on-error main.tex
```

The final PDF is copied to:

```
outputs/pdf/thesis_draft.pdf
```

B.3 Evidence Boundary

This theoretical manuscript does not use plots or time-domain traces as evidence. Numerical studies, when added later, should be treated as a separate validation layer. A future validation artifact should record:

1. the full dynamic model and operating point;
2. the retained and condensed state partition;
3. the construction of $\Pi(s)$, its channel decomposition, and any $\Sigma(s)$ approximation;
4. full-model eigenvalue targets;
5. accepted error tolerances;
6. failure cases and rejected hypotheses.

This boundary protects the thesis from mixing mathematical derivation with uncalibrated simulation evidence.

Notation

This appendix records the principal notation used in the theoretical thesis.

s	Complex spectral parameter or eigenvalue.
x	Dynamic state vector in a first-order DAE or state-space model.
y	Algebraic variable vector in a DAE model.
A	Linearized first-order state matrix after algebraic elimination.
A_{ee}	Retained-state block of a partitioned state matrix.
A_{cc}	Condensed controller or auxiliary-state block.
A_{ec}, A_{ce}	Coupling blocks between retained and condensed states.
M	Diagonal inertia or mass matrix in a second-order retained model.
D	Fixed damping matrix. The case $D = \gamma M$ is the uniform damping-to-inertia case, not the full decouplable class.
γ_i	Local damping-to-inertia ratio D_i/M_i used in graph roughness and commutation diagnostics.
L	Synchronizing stiffness or active-power Jacobian Laplacian.
B	Oriented branch-bus incidence matrix of the retained network.
η	Branch angle-difference vector, $\eta = B^T \theta$.
κ_e	Nominal or parameter-dependent branch stiffness coefficient before the operating-point cosine factor.
f_e	Active power flow on branch e , $f_e = \kappa_e \sin \eta_e$ in the lossless branch model.
w_e	Operational branch stiffness $w_e = \kappa_e \cos \eta_e$.
W	Diagonal matrix of operational branch stiffnesses, so that $L = BWB^T$ on the angle subspace.
$\mathbf{1}$	Uniform angle-shift vector.
$\tilde{L}$	Inertia-normalized stiffness $M^{-1/2}LM^{-1/2}$.

$\tilde{D}$	Inertia-normalized damping $M^{-1/2}DM^{-1/2}$.
$\text{comm}(\tilde{L}, \tilde{D})$	Commutator $\tilde{L}\tilde{D} - \tilde{D}\tilde{L}$. For symmetric fixed operators, $[\tilde{L}, \tilde{D}] = 0$ is the exact modal decoupling condition.
Q	Orthogonal modal basis of $\tilde{L}$ on the nonzero angle subspace.
ν_k	Nonzero eigenvalue of $\tilde{L}$, interpreted as squared modal frequency in the undamped model.
ϕ_k	Generalized graph mode satisfying $L\phi_k = \nu_k M\phi_k$ and $\phi_k^T M\phi_\ell = \delta_{k\ell}$.
δ_k	Frozen-mode scalar damping $\phi_k^T D\phi_k$.
Λ	Diagonal matrix of nonzero $\tilde{L}$ eigenvalues.
Γ	Modal damping matrix $Q^T \tilde{D} Q$.
E	Off-diagonal part of Γ , responsible for modal damping coupling.
a_e	Incidence vector of line $e = (i, j)$, usually $a_e = e_i - e_j$.
b_e	Susceptance-like or stiffness weight of line e .
L^+	Moore–Penrose pseudoinverse of the Laplacian L .
K_f	Kirchhoff index or total effective resistance, $n\text{trace}(L^+)$.
$P(s)$	Quadratic eigenvalue operator $s^2 M + sD + L$.
$T(s)$	Nonlinear retained second-order operator.
$G_T(s)$	Dynamic Green function or retained compliance, $G_T(s) = T(s)^{-1}$ on the resolvent set.
$H_G(\omega)$	Graph-time transfer obtained by projecting the inertia-normalized operator into the graph-mode basis and evaluating at $s = j\omega$.
$\mathcal{R}_e(s)$	Dynamic effective resistance of line e , $a_e^T G_T(s) a_e$.
$S(s)$	First-order Schur complement operator for retained states.
$\Pi(s)$	General rational self-energy obtained after Schur condensation of controller or auxiliary states.
$\hat{\Pi}(s)$	Identified or approximate self-energy obtained from frequency scans, EMT data, PMU data, or black-box models.
$\Pi_K(s), \Pi_D(s)$	Stiffness-like and damping-like channels in $\Pi(s) = \Pi_K(s) + s\Pi_D(s)$.
$\Sigma(s)$	Damping-channel specialization $\Sigma(s) = D_0 + \Pi_D(s)$, used only after the channel decomposition is declared.
$\mathcal{P}_\Sigma(\omega)$	Weighted directional passivity matrix of the damping-channel self-energy.
U, K_u, V	Low-rank action factors in $T_u(s) = T(s) + U(s)K_u(s)V(s)^*$.

$\mathcal{G}_u(s)$	Action interaction matrix $V(s)^*T(s)^{-1}U(s)$.
η_Γ	Contour adequacy error $\sup_{s \in \Gamma} \ \hat{T}(s)^{-1}(T(s) - \hat{T}(s))\ _2$.
$\Lambda_\varepsilon^{\mathcal{A}}(T)$	Structured pseudospectrum under the declared physical action or uncertainty family $\mathcal{A}$.
r_{frag}	Structured dynamic fragility radius, the smallest normalized admissible perturbation that moves a root to a protected boundary.
r_{top}	Structured topology fragility radius over admissible line or edge-weight perturbations.
$r_e^{\mathbb{C}}, r_e^{\mathbb{R}}$	Complex and real frozen-line fragility radii for line e on a declared protected spectral boundary.
r_{forced}	Smallest admissible forcing or input amplitude that violates a declared protected response margin.
r_{ROA}	Nonlinear region-of-attraction fragility radius around the protected equilibrium.
D_0	Direct retained damping term inside the damping-channel specialization.
ζ	Damping ratio of a complex pole.
ζ_{req}	Required damping ratio.
S_ζ	Log damping-ratio margin.
g_{prot}	Protected resolvent gain over selected frequency, input, and output channels.
S_g	Log protected-resolvent margin.
ρ	Planning or hosting action vector, such as IBR conversion, line reinforcement, damping addition, or virtual-inertia placement.
$\mathcal{H}_{\text{stat}}$	Static hosting set defined by voltage, thermal, power-quality, protection, or operating constraints.
$\mathcal{H}_{\text{dyn}}$	Dynamic spectral hosting set after adding damping-ratio, protected-gain, and forbidden-root constraints.
$h_{\text{dyn}}(\alpha)$	Directional dynamic hosting capacity in penetration direction α .
h_k	Modal damping-region headroom $\delta_k - 2\zeta_{\text{req}}\sqrt{\nu_k}$.
$\mathcal{P}_{\text{host}}^{(1)}$	First-order spectral hosting polytope.
γ_i^v	Virtual-damping-to-virtual-inertia coefficient at bus i in a local fixed- D screen.
J_i^{VI}	Virtual-inertia placement score with declared frequency-security and damping-headroom weights.
$\mathcal{L}_{\text{all}}$	Linear space spanned by all edge Laplacian directions $b_{ij}b_{ij}^T$.
$\mathcal{L}_{\text{ex}}$	Linear space spanned by existing-line Laplacian directions.

$\mathcal{D}_{\text{sh}}$	Diagonal shunt space used in the line-shunt decomposition.
s_i	Modal substitutability score for replacing local inertia with existing-line actions.
T_{risk}	Frequency-estimation trip-risk exposure duration.
π_c	Fraction of eigenvector norm associated with condensed controller states.
$\mathcal{D}_i$	Fixed- D damping placement score at bus i .
$\mathcal{M}_i$	Fixed- D inertia-dilution exposure at bus i .
$G_e^{(2)}$	Algebraic-connectivity gain score for line e .
$G_e^{(K)}$	Effective-resistance gain score for line e .
H_i, H_e	Hidden damping-margin flags for bus- or line-related actions.
$r(a)$	Risk vector for a candidate planning action a .
$\mathcal{R}(a)$	Scalarized risk score $w^T r(a)$ after vector components have been reported.
GFL	Grid-following converter control family.
GFM	Grid-forming converter control family.

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